

Topologies and sheaves

SGA 3, Exposé IV

English draft

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About this draft

This is an unofficial English translation of Exposé IV, “Topologies and sheaves” from *Schémas en groupes* (SGA 3). The source is the recomposition edited by P. Gille and P. Polo, PDF `Exp4-13oct24.pdf` (74 pages), from <https://webusers.imj-prg.fr/~patrick.polo/SGA3/>. The seminar was directed by M. Demazure and A. Grothendieck (IHÉS, 1962–1964).

Statement numbers follow that PDF. A clear typographical slip is corrected in the English and marked with a `% typo:` comment; such slips are listed in the README. A slip that would change the mathematics is left as printed and marked `% typo?:`. Scholarly proofreading remains outstanding.

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Exposé IV. Topologies and sheaves

by M. Demazure^(*)

⁽⁰⁾This exposé is intended to acquaint the reader with the essentials of the language of topologies and of sheaves (without cohomology), particularly convenient in questions of passage to the quotient (among others).

The first three paragraphs develop the language of passage to the quotient. The fourth, which is the central part, is the exposition of the theory of sheaves, oriented principally towards the application to questions of quotients; the fifth is an application to passage to the quotient in groups and to principal homogeneous bundles. The last paragraph concerns more specifically the category of schemes, and defines various useful topologies on this category.

The reader will find it useful to refer to [AS], [MA], [D], and SGA 4; to [D] especially concerning the applications of topologies to descent theory, and to SGA 4 for questions of universes (particularly ill-treated in this exposé).

1. Universal effective epimorphisms

In the sequel of this exposé, one assumes fixed a category $\mathcal{C}$.

Definition 1.1. A morphism $u: T \rightarrow S$ is called an *epimorphism* if, for every object X , the corresponding map

$$X(S) = \text{Hom}(S, X) \rightarrow X(T) = \text{Hom}(T, X)$$

is injective⁽¹⁾. One says that u is a *universal epimorphism* if for every morphism $S' \rightarrow S$, the fiber product $T' = T \times_S S'$ exists, and $u': T' \rightarrow S'$ is an epimorphism.

Definition 1.2. A diagram

$$A \xrightarrow{u} B \begin{array}{c} \xrightarrow{v_1} \\ \xrightarrow{v_2} \end{array} C$$

of maps of sets is said to be *exact* if u is injective and if its image consists of the elements b of B such that $v_1(b) = v_2(b)$. A diagram of the same type in $\mathcal{C}$ is said to be *exact* if for every object X of $\mathcal{C}$, the corresponding diagram of sets

$$A(X) \longrightarrow B(X) \begin{array}{c} \longrightarrow \\ \longrightarrow \end{array} C(X)$$

is exact; one then also says that u makes A a *kernel* of the pair of arrows (v_1, v_2) . Dually, a diagram

$$C \begin{array}{c} \xrightarrow{v_1} \\ \xrightarrow{v_2} \end{array} B \xrightarrow{u} A$$

in $\mathcal{C}$ is said to be *exact* if it is exact as a diagram in the opposite category $\mathcal{C}^\circ$, i.e. if for every object X of $\mathcal{C}$, the corresponding diagram of sets

$$X(A) \longrightarrow X(B) \begin{array}{c} \longrightarrow \\ \longrightarrow \end{array} X(C)$$

^(*)This text develops the substance of two oral exposés by A. Grothendieck, completing the latter on several important points, which had been passed over in silence or barely touched upon.

⁽⁰⁾**N.D.E.** : Version of 13/10/2024

⁽¹⁾**N.D.E.** : that is, if u is right-cancellable.

is exact.⁽²⁾ One also says that u makes A a *cokernel* of the pair of arrows (v_1, v_2) .

Definition 1.3. A morphism $u: T \rightarrow S$ is called an *effective epimorphism* if the fiber square $T \times_S T$ exists, and if the diagram

$$T \times_S T \begin{array}{c} \xrightarrow{\text{pr}_1} \\ \xrightarrow{\text{pr}_2} \end{array} T \xrightarrow{u} S$$

is exact, i.e. if u makes S a cokernel of $(\text{pr}_1, \text{pr}_2)$. One says that u is a *universal effective epimorphism* if for every morphism $S' \rightarrow S$, the fiber product $T' = T \times_S S'$ exists, and the morphism $u': T' \rightarrow S'$ is an effective epimorphism.

One evidently has the implications:

$$\begin{array}{ccc} \text{universal effective} & & \text{effective} \\ \text{epimorphism} & \longrightarrow & \text{epimorphism} \\ \downarrow & & \downarrow \\ \text{universal} & & \\ \text{epimorphism} & \longrightarrow & \text{epimorphism}, \end{array}$$

but in general no other implication is valid.⁽³⁾

Definition 1.4.0.⁽⁴⁾One “recalls” that a morphism $u: T \rightarrow S$ is said to be *quarrable* if for every morphism $S' \rightarrow S$, the fiber product $T \times_S S'$ exists.

Lemma 1.4. Consider morphisms $U \xrightarrow{v} T \xrightarrow{u} S$. Then

- a) u, v epimorphisms $\Rightarrow uv$ epimorphism $\Rightarrow u$ epimorphism,
- b) u, v universal epimorphisms $\Rightarrow uv$ universal epimorphism and u quarrable $\Rightarrow u$ universal epimorphism.

Lemma 1.4 is trivial from the definitions. One concludes from it:

Corollary 1.5. Let $u: X \rightarrow Y$ and $u': X' \rightarrow Y'$ be universal epimorphisms, such that $Y \times Y'$ exists, then $X \times X'$ exists and $u \times u': X \times X' \rightarrow Y \times Y'$ is a universal epimorphism.

Let us also note:

Definition 1.6.0.⁽⁵⁾One says that an object S of $\mathcal{C}$ is *quarrable* if its product by every object of $\mathcal{C}$ exists. (If $\mathcal{C}$ has a final object e , this amounts to saying that the morphism $S \rightarrow e$ is quarrable, cf. 1.4.0.)

Lemma 1.6. Let $u: X \rightarrow Y$ be a morphism in $\mathcal{C}_{/S}$; for it to be an epimorphism (resp. a universal epimorphism, resp. an effective epimorphism, resp. a universal effective epimorphism), it suffices that the corresponding morphism in $\mathcal{C}$ be one, and this is also necessary if one assumes that S is a quarrable object of $\mathcal{C}$.

PROOF. Immediate, left to the reader. One uses the hypothesis “ S quarrable” in order to interpret the $\mathcal{C}$ -morphisms from an object Y of $\mathcal{C}_{/S}$ into an object Z of $\mathcal{C}$, as being the $\mathcal{C}_{/S}$ -morphisms from Y into $Z \times S$. $\square$

⁽²⁾**N.D.E.** : This implies, in particular, that u is an epimorphism.

⁽³⁾**N.D.E.** : For example, if $\mathcal{C} = (\text{Sch})$ is the category of schemes, one easily sees that every universal epimorphism is surjective. Let $T = \coprod_{p \text{ prime}} \text{Spec}(\mathbf{F}_p)$ and $S = \text{Spec}(\mathbf{Z})$; then the morphism $u: T \rightarrow S$ is an epimorphism which is *not universal*. On the other hand, one sees that $T \times_S T$ is identified with T , so that the two projections $T \times_S T \rightrightarrows T$ coincide; as id_T does not descend to a morphism $S \rightarrow T$, this shows that u is *not* an effective epimorphism.

⁽⁴⁾**N.D.E.** : One has added the numbering 1.4.0, for later references.

⁽⁵⁾**N.D.E.** : One has added the numbering 1.6.0, for later references.

Lemma 1.7. *With the notations of 1.4: u, v effective epimorphisms and v universal epimorphism $\Rightarrow uv$ effective epimorphism.*

To see this, one considers the diagram

$$\begin{array}{ccccc}
 S & \xleftarrow{u} & T & \xleftarrow{\quad} & T \times_S T \\
 & & \uparrow v & & \uparrow v \times_S v \\
 & & U & \xleftarrow{\quad} & U \times_S U \\
 & & \uparrow \uparrow & \nearrow & \\
 & & U \times_T U & &
 \end{array}$$

One notes that by hypothesis, row 1 and column 1 are exact, and that by virtue of 1.5 and 1.6, $v \times_S v$ is an epimorphism (v being a *universal* epimorphism). The conclusion follows from this by an evident diagram-chasing: if an element of $X(U)$ has the same images in $X(U \times_S U)$, it *a fortiori* has the same images in $X(U \times_T U)$, hence comes from an element of $X(T)$ since column 1 is exact. As row 1 is exact, it suffices to check that the element under consideration has the same images in $X(T \times_S T)$, and as $v \times_S v$ is an epimorphism, it suffices to check that the images in $X(U \times_S U)$ are the same, which is indeed the case.

Proposition 1.8. *Consider morphisms $U \xrightarrow{v} T \xrightarrow{u} S$. Then u, v universal effective epimorphisms $\Rightarrow uv$ universal effective epimorphism and u quarrable $\Rightarrow u$ universal effective epimorphism.*

The first implication follows at once from 1.7. For the second, one looks at the diagram (of “bisimplicial” type):

$$\begin{array}{ccccc}
 S & \xleftarrow{u} & T & \xleftarrow{\quad} & T \times_S T \\
 \uparrow uv & & \uparrow & & \uparrow \\
 U & \xleftarrow{\quad} & U \times_S T & \xleftarrow{\quad} & U \times_S T \times_S T \\
 \uparrow \uparrow & & \uparrow \uparrow & & \uparrow \uparrow \\
 U \times_S U & \xleftarrow{\quad} & U \times_S U \times_S T & \xleftarrow{\quad} & U \times_S U \times_S T \times_S T
 \end{array}$$

Columns 1, 2, 3 are exact by virtue of the hypothesis “ uv universal effective epimorphism”, row 2 is exact, for $U \times_S T \rightarrow U$ is an effective epimorphism (for it has a section over U), and the same holds for row 3 (same reason). An evident diagram-chasing then shows that row 1 is exact, i.e. u is an effective epimorphism. As the hypotheses made are invariant under an arbitrary base change $S' \rightarrow S$, it follows that u is even a universal effective epimorphism.

Corollary 1.9. *Let $u: X \rightarrow Y$ and $u': X' \rightarrow Y'$ be universal effective epimorphisms, such that $Y \times Y'$ exists; then $X \times X'$ exists and $u \times u': X \times X' \rightarrow Y \times Y'$ is a universal effective epimorphism.*

PROOF. As for 1.5, by the diagram

$$\begin{array}{ccccc}
 & & Y' & \xleftarrow{u'} & X' \\
 & & \uparrow & & \uparrow \\
 X & \xleftarrow{\quad} & X \times Y' & \xleftarrow{\quad} & X \times X' \\
 \downarrow u & & \downarrow & \nearrow u \times u' & \\
 Y & \xleftarrow{\quad} & Y \times Y' & &
 \end{array}$$

□

Corollary 1.10. *Consider a quarrable morphism $u: T \rightarrow S$, and a base change morphism $S' \rightarrow S$, which is a universal effective epimorphism. For u to be a universal effective epimorphism, it is necessary and sufficient that $u': T' = T \times_S S' \rightarrow S'$ be so:*

$$\begin{array}{ccc} T & \xleftarrow{v'} & T' \\ u \downarrow & & \downarrow u' \\ S & \xleftarrow{v} & S' \end{array} .$$

Only the “it suffices” requires a proof. Now if u' is a universal effective epimorphism, the same holds for vu' thanks to 1.8, and since $vu' = uv'$, one concludes by 1.8 that u is a universal effective epimorphism.

Remark 1.11. The same reasoning shows that in 1.10 one may replace “universal effective epimorphism” by “universal epimorphism” or “universal and effective epimorphism”, or simply by “epimorphism” (and in this last case, the hypothesis “ u quarrable” is evidently useless).

In the proof of 1.8 we have used the following result, which deserves to be spelled out:

Proposition 1.12. *Let $u: T \rightarrow S$ be a morphism which admits a section. Then u is an epimorphism, and if $T \times_S T$ exists, it is an effective epimorphism, and a universal effective epimorphism if moreover u is quarrable.*

The first assertion is contained in 1.4 a), and the third will follow at once from the second, which it will therefore suffice to establish. In fact one has a stronger conclusion: for every functor $F: \mathcal{C}^\circ \rightarrow (\text{Ens})$ (not necessarily representable), the diagram of sets

$$F(S) \longrightarrow F(T) \rightrightarrows F(T \times_S T)$$

is exact. This may be considered as a particular case of the formalism of Čech cohomology (in dimension 0!) which we content ourselves with recalling here. Let us simply assume that $T \times_S T$ exists; one then sets

$$H^0(T/S, F) = \text{Ker}(F(T) \rightrightarrows F(T \times_S T)).$$

One may regard $H^0(T/S, F)$ in an evident way as a contravariant functor in the argument T varying in $\mathcal{C}_S$, every S -morphism $T' \rightarrow T$ defining a map

$$(+)\quad H^0(T/S, F) \longrightarrow H^0(T'/S, F).$$

Let us fix T and T' in $\mathcal{C}_S$. A well-known computation shows that if there exists an S -morphism from T' into T , the corresponding map (+) is in fact independent of the choice of this morphism⁽⁶⁾, so that $H^0(T/S, F)$ may be regarded as a functor on the category associated with the set $\text{Ob } \mathcal{C}_S$ preordered by the relation of “domination” (T' dominates T if there exists an S -morphism from T' into T). In particular, if T and T' are isomorphic in this latter category, i.e. if each dominates the other, then (+) is an isomorphism of sets. This applies in particular to the case where T' is the final object of $\mathcal{C}_S$, i.e. essentially S itself; in any case T dominates $T' = S$, and the converse is true precisely if T/S has a section. This establishes 1.12 in the strengthened form announced.

Remark 1.13. For various applications, the notions introduced in the present exposé, and the results stated, are to be developed more generally relative to a *family* of morphisms

⁽⁶⁾**N.D.E.** : This is the following argument, communicated by M. Demazure. Let $f, g: T' \rightarrow T$, and let $\varphi: T' \rightarrow T \times_S T$ be the morphism with components f and g , whence $p_1 \circ \varphi = f$ and $p_2 \circ \varphi = g$. Then $F(\varphi): F(T \times_S T) \rightarrow F(T')$ satisfies $F(\varphi) \circ F(p_1) = F(f)$ and $F(\varphi) \circ F(p_2) = F(g)$. Now, for every $x \in H^0(T/S, F)$, one has $F(p_1)(x) = F(p_2)(x)$. Therefore, applying $F(\varphi)$ to both sides, one obtains $F(f)(x) = F(g)(x)$, which shows that f and g induce the same morphism.

$u_i: T_i \rightarrow S$ with the same target (instead of a single morphism $u: T \rightarrow S$). Thus, such a family will be said to be *epimorphic* if for every object X of $\mathcal{C}$, the corresponding map

$$X(S) \rightarrow \prod_i X(T_i)$$

is injective, and one likewise introduces the notion of effective epimorphic family and the “universal” variants of these notions. We shall admit if need be, in the sequel, that the results of the present exposé extend to this more general situation.

Remark 1.14. Every morphism which is at the same time a monomorphism and an effective epimorphism is an isomorphism. Indeed, in the notations of 1.3, the fact that $T \rightarrow S$ is a monomorphism implies that the two morphisms

$$\text{pr}_1, \text{pr}_2: T \times_S T \rightrightarrows T$$

are equal (and are isomorphisms). Now a diagram

$$C \begin{array}{c} \xrightarrow{v} \\ \xrightarrow{v} \end{array} B \xrightarrow{u} A$$

is exact only if u is an isomorphism, as follows immediately from the definition.⁽⁷⁾

2. Descent morphisms

Let us recall the following definitions:

Definition 2.1. Let $f: S' \rightarrow S$ be a morphism such that $S'' = S' \times_S S'$ exists, and let $u': X' \rightarrow S'$ be an object over S' . One calls *gluing datum* on X'/S' , relative to f , an S'' -isomorphism

$$c: X''_1 \xrightarrow{\sim} X''_2$$

where X''_i ($i = 1, 2$) denotes the inverse image (assumed to exist) of X'/S' by the projection $\text{pr}_i: S'' \rightarrow S'$. One says that the gluing datum c is a *descent datum* if it satisfies the “cocycle condition”

$$\text{pr}_{3,1}^*(c) = \text{pr}_{3,2}^*(c) \text{pr}_{2,1}^*(c)$$

where $\text{pr}_{i,j}$ ($1 \leq j < i \leq 3$) are the canonical projections of $S''' = S' \times_S S' \times_S S'$ into S'' (N.B. one now assumes that S''' also exists), where $\text{pr}_{i,j}^*(c)$ is the inverse image of c , considered as an S''' -morphism from X'''_j into X'''_i , and where for every integer k between 1 and 3, X'''_k denotes the inverse image (assumed to exist) of X'/S' by the projection of index k , $q_k: S''' \rightarrow S'$.

In the second part of the definition, one has therefore used identifications and abuses of notation in current use⁽⁸⁾, which experience proves to be harmless, but which it is obviously advisable to avoid in a rigorous exposé of the theory of descent, (which must precisely justify to a certain extent these current abuses of language). Such a formal exposé ([D]) has been written by Giraud, (with a view to justifying and making precise SGA 1, VII, which has never been written). For a detailed exposé of the results of faithfully flat descent, of which constant use will be made in the present Seminar, one may consult SGA 1, VIII.

Let $f: S' \rightarrow S$ still be a morphism such that $S'' = S' \times_S S'$ exists, and let X be an object over S such that $X' = X \times_S S'$ and $X'' = X \times_S S''$ exist; then the inverse images of X' by pr_i ($i = 1, 2$) exist and are canonically isomorphic, and consequently X'/S' is endowed with a canonical gluing datum relative to f . When S''' and $X''' = X \times_S S'''$ exist, it is even a descent datum. If Y is another object over S , satisfying the same conditions as X/S , then for

⁽⁷⁾**N.D.E.** : One will note that a monomorphism which is an epimorphism is not necessarily an isomorphism. For example, in $\mathcal{C} = (\text{Sch})$, the morphism $\text{Spec}(\mathbf{F}_p) \coprod \text{Spec}(\mathbf{Z}[1/p]) \rightarrow \text{Spec}(\mathbf{Z})$ is a monomorphism and a surjective epimorphism, but is not an isomorphism.

⁽⁸⁾**N.D.E.** : for example, one has identified, on the one hand, $\text{pr}_{2,1}^*(X''_1) = \text{pr}_{3,1}^*(X''_1)$ and, on the other hand, $\text{pr}_{2,1}^*(X''_2) = \text{pr}_{3,2}^*(X''_2)$.

every S -morphism $X \rightarrow Y$, the corresponding S' -morphism $X' \rightarrow Y'$ is “compatible with the canonical gluing data” on X', Y' . If in particular $S' \rightarrow S$ is a *quarrable* morphism, then

$$X \mapsto X' = X \times_S S'$$

is a functor from the category $\mathcal{C}_S$ into the “category of objects over S' endowed with a descent datum relative to f ”—a category whose definition is left to the reader, and which is a full subcategory of the “category of objects over S' endowed with a gluing datum relative to f ”.

This being said:

Definition 2.2. One says that a morphism $f: S' \rightarrow S$ is a *descent morphism* (resp. an *effective descent morphism*) if f is *quarrable* (i.e. for every $X \rightarrow S$, the fiber product $X \times_S S'$ exists), and if the preceding functor $X \mapsto X' = X \times_S S'$ from the category $\mathcal{C}_S$ of objects over S , into the category of objects over S' endowed with a descent datum relative to f , is fully faithful (resp. an equivalence of categories).

One will note that the first of these two notions can be expressed with the help of only the notion of gluing datum (hence without involving the triple fiber product S'''), f being a descent morphism if f is *quarrable* and $X \mapsto X'$ is a fully faithful functor from the category $\mathcal{C}_S$ into the category of objects over S' endowed with a gluing datum relative to f . When one makes this definition explicit, one finds that this means that for two objects X, Y over S , the following diagram of sets

$$(x) \quad \text{Hom}_S(X, Y) \longrightarrow \text{Hom}_{S'}(X', Y') \xrightleftharpoons[p_2]{p_1} \text{Hom}_{S''}(X'', Y'')$$

is exact, where $p_i(u')$ denotes the inverse image of $u' \in \text{Hom}_{S'}(X', Y')$ by the projection $\text{pr}_i: S'' \rightarrow S'$, for $i = 1, 2$; indeed, the kernel of the pair (p_1, p_2) is by definition nothing other than the set of S' -morphisms $X' \rightarrow Y'$ compatible with the canonical gluing data.

Let us note that, by definition of the inverse images Y', Y'' , one has canonical bijections

$$\text{Hom}_{S'}(X', Y') \simeq \text{Hom}_S(X', Y) \quad \text{and} \quad \text{Hom}_{S''}(X'', Y'') \simeq \text{Hom}_S(X'', Y),$$

so that the exactness of the diagram (x) is equivalent to that of

$$(xx) \quad \text{Hom}_S(X, Y) \longrightarrow \text{Hom}_S(X', Y) \rightrightarrows \text{Hom}_S(X'', Y),$$

obtained by applying $\text{Hom}_S(-, Y)$ to the diagram in $\mathcal{C}_S$:

$$(xxx) \quad X'' \rightrightarrows X' \longrightarrow X$$

which is deduced from

$$S'' \rightrightarrows S' \longrightarrow S$$

by the change of base $X \rightarrow S$. This proves, taking 1.6 into account, the first part of the following proposition:

Proposition 2.3. *Let $f: S' \rightarrow S$ be a morphism. If it is a universal effective epimorphism, it is a descent morphism, and the converse is true if S is a quarrable object of $\mathcal{C}$ (i.e. its product by every object of $\mathcal{C}$ exists).*

It remains to prove that if f is a descent morphism, it is a universal effective epimorphism, that is, that for every morphism $X \rightarrow S$, the diagram (xxx) is exact, i.e. for every object Z of $\mathcal{C}$, the transform of this diagram by $\text{Hom}(-, Z)$ is an exact diagram of sets. Now by hypothesis $Z \times S$ exists; let Y be the object of $\mathcal{C}_S$ which it defines; then the transform of diagram (xxx) by $\text{Hom}(-, Z)$ is isomorphic to the transform by $\text{Hom}_S(-, Y)$, now the latter is exact by the hypothesis on f .

One may therefore apply to universal effective epimorphisms the results on descent morphisms, such as the following:

Proposition 2.4. *Let $f: S' \rightarrow S$ be a descent morphism (for example a universal effective epimorphism). Then:*

a) *For every S -morphism $u: X \rightarrow Y$, u is an isomorphism (resp. a monomorphism) if and only if $u': X' \rightarrow Y'$ is.*

b) *Let X, Y be two subobjects of S , and X' and Y' the subobjects of S' , inverse images of the preceding ones. In order that X be majorized by Y (resp. be equal to Y), it is necessary and sufficient that the same hold for X', Y' .*

For (a), it follows from the definition that if u' is an isomorphism in the category of objects with gluing datum, then u is an isomorphism; now one sees at once that every isomorphism of objects over S' , compatible with gluing data, is an isomorphism of objects with gluing datum, i.e. its inverse is likewise compatible with the gluing data. For b), one is reduced to proving that if X' is majorized by Y' , i.e. if there exists an S' -morphism $X' \rightarrow Y'$, then the same holds for X, Y over S . Now since $Y' \rightarrow S'$, hence also $Y'' \rightarrow S''$, is a monomorphism, one sees that $X' \rightarrow Y'$ is automatically compatible with the gluing data, hence comes from an S -morphism $X \rightarrow Y$. Let us note that the proof holds more generally when one has two objects X, Y over S , with $Y \rightarrow S$ a monomorphism, and one asks whether the morphism $X \rightarrow S$ factors through Y : it suffices that $X' \rightarrow S'$ factor through Y' .

Corollary 2.5. *Let $f: S' \rightarrow S$ be a universal effective epimorphism and $g: S \rightarrow T$ a morphism such that $S \times_T S$ exists. Suppose that $S'' = S' \times_S S'$ is also a fiber product of S' by itself over T , i.e. $S' \times_S S' \xrightarrow{\sim} S' \times_T S'$. Then $g: S \rightarrow T$ is a monomorphism (and conversely, of course).*

Indeed, let us consider the cartesian diagram

$$\begin{array}{ccc} S' \times_S S' & \xrightarrow{\sim} & S' \times_T S' \\ \downarrow & & \downarrow f \\ S & \longrightarrow & S \times_T S, \end{array}$$

where the second horizontal arrow is the diagonal morphism. By virtue of 1.9 the second vertical arrow f is a universal effective epimorphism, by hypothesis the first horizontal arrow is an isomorphism, hence by virtue of 2.4 a) or b), at one's choice⁽⁹⁾, the same holds of $S \rightarrow S \times_T S$, which means precisely that $g: S \rightarrow T$ is a monomorphism.

Remark 2.6. The notions introduced in 2.1, in terms of morphisms between certain projective limits, are made explicit in an evident way in terms of the contravariant functors defined by the objects S, S', X' envisaged: subject to the existence of the fiber products envisaged in 2.1, there is a one-to-one correspondence between the gluing data (resp. descent data) on X'/S' relative to $S' \rightarrow S$, and the gluing data (resp. descent data) for the corresponding objects in $\widehat{\mathcal{C}} = \text{Hom}(\mathcal{C}^\circ, (\text{Ens}))$. This makes it possible, if one so desires, to extend these notions to the case where one makes no existence hypothesis on fiber products in $\mathcal{C}$.

Remark 2.7. The notions introduced in this number generalize to the case of families of morphisms. They can on the other hand be presented in a more intrinsic manner with the help of the notion of sieve (4.1). For these questions, the reader is referred to [D].

3. Universal effective equivalence relations

3.1. Equivalence relations: definitions

⁽⁹⁾**N.D.E.** : applied to $f: S' \times_T S' \rightarrow S \times_T S = Y$.

Definition 3.1.1. A *representable subfunctor* R of the functor $X \times X$, such that for every $S \in \text{Ob}(\mathcal{C})$, $R(S)$ is the graph of an equivalence relation in $X(S)$, is called a $\mathcal{C}$ -*equivalence relation* in $X \in \text{Ob}(\mathcal{C})$.

This definition applies in particular to the category $\widehat{\mathcal{C}}$. If one considers X as an object of $\widehat{\mathcal{C}}$, one sees then that a $\widehat{\mathcal{C}}$ -equivalence relation in X is nothing other than a subfunctor R of $X \times X$ such that $R(S)$ is the graph of an equivalence relation in $X(S)$ for every object S of $\mathcal{C}$.⁽¹⁰⁾

If R is a $\mathcal{C}$ -equivalence relation in X , one denotes by p_i the morphism from R into X induced by the projection $\text{pr}_i: X \times X \rightarrow X$. One therefore has a diagram

$$p_1, p_2: R \rightrightarrows X.$$

Definition 3.1.2. A morphism $u: X \rightarrow Z$ is said to be *compatible* with R if $up_1 = up_2$. An object of $\mathcal{C}$, cokernel of the pair (p_1, p_2) , is also called a *quotient-object* of X by R and denoted X/R .

One therefore has an exact diagram

$$R \xrightarrow{p_1, p_2} X \xrightarrow{p} X/R$$

and X/R represents the *covariant* functor

$$\text{Hom}_{\mathcal{C}}(X/R, Z) = \{\text{morphisms from } X \text{ into } Z \text{ compatible with } R\}.$$

As the quotient-objects have been chosen in $\mathcal{C}$, the quotient X/R is unique (when it exists).

These definitions generalize at once to the case of a $\widehat{\mathcal{C}}$ -equivalence relation in X , but one will observe that the identification of the objects of $\mathcal{C}$ with their images in $\widehat{\mathcal{C}}$ *does not commute with the formation of quotients*, that is, that the quotient X/R of X by R in $\mathcal{C}$ is not *a priori* a quotient of X by R in $\widehat{\mathcal{C}}$. One will therefore take care not to identify $\mathcal{C}$ rashly with its image in $\widehat{\mathcal{C}}$ in questions involving passages to the quotient.⁽¹¹⁾

In the sequel, *we shall say simply equivalence relation for $\widehat{\mathcal{C}}$ -equivalence relation*; we shall specify, when the case arises, whether it is a matter of $\mathcal{C}$ -equivalence relations.⁽¹²⁾

Definition 3.1.3. If X is an object of $\mathcal{C}$ over S , an equivalence relation R in X such that the structural morphism $X \rightarrow S$ is compatible with R is called an equivalence relation in X over S .

The canonical monomorphism $R \rightarrow X \times X$ then factors through the monomorphism

$$X \times_S X \rightarrow X \times X$$

⁽¹⁰⁾**N.D.E.** : The condition is obviously necessary. Conversely, if for every $S \in \text{Ob}(\mathcal{C})$, $R(S)$ is the graph of an equivalence relation, then this equivalence relation extends to $R(F)$ for every $F \in \text{Ob}(\widehat{\mathcal{C}})$, by declaring that two morphisms $\varphi, \psi: F \rightarrow R$ are equivalent if, for every $S \in \text{Ob}(\mathcal{C})$ and $x \in F(S)$, $\varphi(x)$ and $\psi(x)$ are equivalent in $X(S)$.

⁽¹¹⁾**N.D.E.** : Let us illustrate this remark by giving a glimpse of the sequel of this Exposé. Let G be a $\mathcal{C}$ -group, H a $\mathcal{C}$ -subgroup, R the equivalence relation in G defined by $G \times H \rightarrow G \times G$, $(g, h) \mapsto (g, gh)$ (cf. 3.2). The functor Q defined by $Q(S) = G(S)/H(S)$ is a quotient in $\widehat{\mathcal{C}}$ (by 4.4.9 applied to the coarsest topology, cf. 4.4.2), but it is not in general the quotient that one wishes. For example, for $\mathcal{C} = (\text{Sch})$, one has an exact sequence of (affine) group schemes:

$$1 \rightarrow \mu_2 \rightarrow \mathbf{G}_m \xrightarrow{p} \mathbf{G}_m \rightarrow 1$$

which identifies $\mathbf{G}_m$ with the quotient $\mathbf{G}_m/\mu_2$. Moreover, as p is a finite and locally free morphism, $\mathbf{G}_m$ is then the *quotient-sheaf* of $\mathbf{G}_m$ by μ_2 in the larger category of *sheaves for the (fppf) topology*, cf. 4.6.6 (ii) and 6.3.2. By contrast, the quotient Q in $\widehat{\mathcal{C}}$ is not isomorphic to $\mathbf{G}_m$ since, for example, $Q(\mathbf{Z}) = \{1\}$ whereas $\mathbf{G}_m(\mathbf{Z}) = \{\pm 1\}$. Hence Q is not an (fppf) sheaf, and a fortiori Q is not representable.

⁽¹²⁾**N.D.E.** : Take care that, even for a $\widehat{\mathcal{C}}$ -equivalence relation in X , one is interested in the existence of a quotient *in $\mathcal{C}$* .

and defines an equivalence relation in the object $X \rightarrow S$ of $\mathcal{C}/S$. When the quotient X/R exists, it is endowed with a canonical morphism into S and the corresponding object of $\mathcal{C}/S$ is a quotient of $X \in \text{Ob}(\mathcal{C}/S)$ by the preceding equivalence relation. Conversely, if S is a *quarrrable* object of $\mathcal{C}$ and if $Y \rightarrow S$ is a quotient of X by this equivalence relation (in $\mathcal{C}/S$), then Y is a quotient of X by R in $\mathcal{C}$. In any case, we shall never have to consider quotients in $\mathcal{C}/S$ which are not already quotients in $\mathcal{C}$.

Definition 3.1.4. If X (resp. X') is an object of $\mathcal{C}$ endowed with an equivalence relation R (resp. R'), a morphism

$$u: X \rightarrow X'$$

is said to be *compatible with R and R'* if the following equivalent conditions are satisfied:

- (i) for every $S \in \text{Ob}(\mathcal{C})$, two points of $X(S)$ congruent modulo $R(S)$ are transformed by u into two points of $X'(S)$ congruent modulo $R'(S)$;
- (ii) there exists a morphism $R \rightarrow R'$ (necessarily unique) making the following diagram commutative

$$\begin{array}{ccc} R & \longrightarrow & R' \\ \downarrow & & \downarrow \\ X \times X & \xrightarrow{u \times u} & X' \times X' \end{array}$$

By the universal property of X/R , there then exists (when the quotients X/R and X'/R' exist) a unique morphism v making the following diagram commutative

$$\begin{array}{ccc} X & \xrightarrow{p} & X/R \\ \downarrow u & & \downarrow v \\ X' & \xrightarrow{p'} & X'/R' \end{array}$$

Definition 3.1.5. A subobject (= a representable subfunctor) Y of X is said to be *stable* under the equivalence relation R if the following equivalent conditions are satisfied:

- (i) For every $S \in \text{Ob}(\mathcal{C})$, the subset $Y(S)$ of $X(S)$ is stable under $R(S)$.
- (ii) The two subobjects of R , inverse images of Y by p_1 and p_2 , are identical.

An important particular case of a stable subobject of X is the following: the quotient X/R exists and Y is the inverse image on X of a subobject of X/R .

Definition 3.1.6. Let R be an equivalence relation in X and $X' \rightarrow X$ a morphism. The equivalence relation R' in X' obtained by the cartesian diagram

$$\begin{array}{ccc} R' & \longrightarrow & R \\ \downarrow & & \downarrow \\ X' \times X' & \longrightarrow & X \times X \end{array}$$

is called the equivalence relation in X' , *inverse image* of the equivalence relation R in X . In particular, if X' is a subobject of X , one will say that it is the equivalence relation *induced* in X' by R ; one will denote it by $R_{X'}$.

The morphism $X' \rightarrow X$ is compatible with R' and R ; one therefore has, when the quotients exist, a morphism $X'/R' \rightarrow X/R$ (3.1.4). If X' is a subobject of X , we shall see later that in certain cases one can prove that $X'/R' \rightarrow X/R$ is a monomorphism, hence identifies X'/R' with a subobject of X/R . When this is the case, the inverse image of this subobject in X will be a subobject of X majorizing X' and stable under R : the *saturation* of X' for the equivalence relation R .

Proposition 3.1.7. *If the subobject Y of X is stable under R , one has two cartesian squares, for $i = 1, 2$:*

$$\begin{array}{ccc} R_Y & \longrightarrow & R \\ p_i \downarrow & & \downarrow p_i \\ Y & \longrightarrow & X. \end{array}$$

PROOF. Immediate. □

3.2. Equivalence relation defined by a group operating freely

Definition 3.2.1. Let X be an object of $\mathcal{C}$ and H a $\mathcal{C}$ -group operating on X (that is, equipped with a morphism of $\widehat{\mathcal{C}}$ -groups

$$H \longrightarrow \text{Aut}(X)).$$

One says that H operates *freely* on X if the following equivalent conditions are satisfied:

- (i) For every $S \in \text{Ob}(\mathcal{C})$, the group $H(S)$ operates freely on $X(S)$;
- (ii) The morphism of functors

$$H \times X \longrightarrow X \times X$$

defined set-theoretically by $(h, x) \mapsto (hx, x)$ is a monomorphism.

(In the preceding definition, one has supposed that the group H operated “on the left” on X . One evidently has an analogous notion in the case where the group operates “on the right”, that is, when one has been given a morphism of groups from the group H° opposite to H into $\text{Aut}(X)$.)

If H operates freely on X , the image of $H \times X$ by the morphism of (ii) is an equivalence relation in X , called the *equivalence relation defined by the action of H on X* . When the quotient of X by this equivalence relation exists, one denotes it by $H \backslash X$ (X/H when H operates on the right). It represents the following covariant functor: if Z is an object of $\mathcal{C}$, one has

$$\text{Hom}(H \backslash X, Z) = \{\text{morphisms from } X \text{ into } Z \text{ invariant under } H\}$$

where the morphism $f: X \longrightarrow Z$ is said to be invariant under H if for every $S \in \text{Ob}(\mathcal{C})$, the corresponding morphism $X(S) \longrightarrow Z(S)$ is invariant under the group $H(S)$.

Lemma 3.2.2. *Under the conditions of 3.2.1, let Y be a subobject of X . The following conditions are equivalent:*

- (i) Y is stable under the equivalence relation defined by H (3.1.5);
- (ii) For every $S \in \text{Ob}(\mathcal{C})$, the subset $Y(S)$ of $X(S)$ is stable under $H(S)$;
- (iii) There exists a morphism f , necessarily unique, making the following diagram commutative

$$\begin{array}{ccc} H \times Y & \xrightarrow{f} & Y \\ \downarrow & & \downarrow \\ H \times X & \longrightarrow & X. \end{array}$$

Under these conditions, f defines a morphism of $\widehat{\mathcal{C}}$ -groups

$$H \longrightarrow \text{Aut}(Y)$$

and the equivalence relation defined in Y by this operation of H is none other than the equivalence relation induced in Y by the equivalence relation defined in X by the action of H .

PROOF. The proof is immediate. One evidently has an analogous statement for a “right operation”. The operation of H on Y defined above will be called the operation *induced* in Y by the given operation of H on X . □

Let us now consider the following situation: H and G are two $\mathcal{C}$ -groups and one has been given a morphism of groups

$$u: H \longrightarrow G.$$

Then H operates on G by translations (one sets, set-theoretically, $hg = u(h)g$) and operates there *freely* if and only if u is a *monomorphism*. The quotient of G by this operation of H is denoted, when it exists, by $H \backslash G$. One likewise defines a right operation of H on G and a quotient G/H . These quotients are functorial with respect to the groups in question; precisely, one has the following lemma, stated for the right quotients:

Lemma 3.2.3. *Let $u: H \longrightarrow G$ and $u': H' \longrightarrow G'$ be two monomorphisms of $\mathcal{C}$ -groups. Suppose given a morphism of $\mathcal{C}$ -groups*

$$f: G \longrightarrow G'.$$

The following conditions are equivalent:

- (i) *f is compatible with the equivalence relations defined in G and G' by H and H' .*
- (ii) *For every $S \in \text{Ob}(\mathcal{C})$, one has $fu(H(S)) \subset u'(H'(S))$.*
- (iii) *There exists a morphism $g: H \longrightarrow H'$, necessarily unique and multiplicative, such that the following diagram is commutative*

$$\begin{array}{ccc} H & \xrightarrow{g} & H' \\ u \downarrow & & \downarrow u' \\ G & \xrightarrow{f} & G' \end{array}$$

Under these conditions, if the quotients G/H and G'/H' exist, there exists a unique morphism f making the following diagram commutative

$$\begin{array}{ccc} G & \xrightarrow{f} & G' \\ p \downarrow & & \downarrow p' \\ G/H & \xrightarrow{f} & G'/H' \end{array}$$

PROOF. The first part is proved by reduction to the set-theoretic case. The second follows immediately from (i). $\square$

One could carry over into the present situation the notions introduced above for general equivalence relations. Let us merely point out the following lemma, whose proof is immediate by reduction to the set-theoretic case:

Lemma 3.2.4. *Let $u: H \longrightarrow G$ be a monomorphism of $\mathcal{C}$ -groups and G' a $\mathcal{C}$ -subgroup of G . For the subobject G' of G to be stable under the equivalence relation defined by H , it is necessary and sufficient that u factor through the canonical monomorphism $G' \longrightarrow G$, and under this condition the operation of H on G' induced by the operation of H on G defined by u is none other than the operation deduced from the monomorphism $H \longrightarrow G'$ through which u factors.*

3.3. Universal effective equivalence relations

Definition 3.3.1. Let $f: X \longrightarrow Y$ be a morphism. One calls the *equivalence relation defined by f in X* , and one denotes by $\mathcal{R}(f)$, the $\hat{\mathcal{C}}$ -equivalence relation in X which is the image of the canonical monomorphism

$$X \times_Y X \longrightarrow X \times X.$$

Definition 3.3.2. Let R be an equivalence relation in X . One says that R is *effective* if

- (i) R is representable (i.e. is a $\mathcal{C}$ -equivalence relation);
- (ii) the quotient $Y = X/R$ exists in $\mathcal{C}^{(13)}$;
- (iii) the diagram

$$R \begin{array}{c} \xrightarrow{p_1} \\ \xrightarrow{p_2} \end{array} X \xrightarrow{p} Y$$

makes R the fiber square of X over Y , that is, R is the equivalence relation defined by p .

Scholium 3.3.2.1. ⁽¹⁴⁾ *If R is an effective equivalence relation in X , then p is an effective epimorphism (1.3). If $f: X \rightarrow Y$ is an effective epimorphism, then $\mathcal{R}(f)$ is an effective equivalence relation in X , one quotient of which is Y . There is therefore a one-to-one “Galois” correspondence between effective equivalence relations in X and effective quotients of X (i.e. equivalence classes of effective epimorphisms with source X).*

Definition 3.3.3. One says that the equivalence relation R in X is *universal effective* if the quotient $Y = X/R$ exists, and if, for every $Y' \rightarrow Y$, the fiber products $X' = X \times_Y Y'$ and $R' = R \times_Y Y'$ exist and R' is a fiber square of X' over Y' . It amounts to the same to say that R is *effective* and that $p: X \rightarrow X/R$ is a *universal effective epimorphism*.

Scholium 3.3.3.1. ⁽¹⁴⁾ *There is therefore, as above, a one-to-one correspondence between universal effective equivalence relations in X and universal effective quotients of X .*

Remark 3.3.3.2. ⁽¹⁴⁾ Suppose that $\mathcal{C}$ is the category of S -schemes, and denote by $\mathbf{A}^1$ the affine space of dimension 1 over S . Let $R \subset X \times_S X$ be a universal effective equivalence relation and $p: X \rightarrow Y$ the quotient. Then, for every open U of Y , $\mathcal{O}(U) = \text{Hom}_S(U, \mathbf{A}_S^1)$ is the set of elements ϕ of $\mathcal{O}(p^{-1}(U)) = \text{Hom}_S(p^{-1}(U), \mathbf{A}_S^1)$ such that $\phi \circ \text{pr}_1 = \phi \circ \text{pr}_2$. In particular, if R is given by the action of a group H operating freely on the right on X (cf. 3.2.1), then $\mathcal{O}(U)$ is the set of $\phi \in \mathcal{O}(p^{-1}(U))$ such that $\phi(xh) = \phi(x)$, for every $S' \rightarrow S$ and $x \in X(S')$, $h \in H(S')$.

Proposition 3.3.4. *Let R be a universal effective equivalence relation in X . Let $f: X \rightarrow Z$ be a morphism compatible with R , hence factoring through $g: X/R \rightarrow Z$. The following conditions are equivalent:*

- (i) g is a monomorphism;
- (ii) R is the equivalence relation defined by f .

PROOF. Indeed, (i) implies (ii) trivially, and the converse is none other than 2.5. □

Definition 3.3.5. Let H be a $\mathcal{C}$ -group operating freely on X . One says that H operates in an *effective* manner, or that the operation of H on X is *effective* (resp. *universal effective*), if the equivalence relation defined in X by the action of H is effective (resp. universal effective).

3.4. (M)-effectivity. — In practice, it is most often difficult to characterize the universal effective epimorphisms. One nevertheless often has at one’s disposal a certain number of morphisms of this type, for example in the theory of schemes, the faithfully flat quasi-compact morphisms. This leads to the developments below.

3.4.1. — Let us first state a certain number of conditions on a family (M) of morphisms of $\mathcal{C}$:

- (a) (M) is stable under base extension, i.e. every $u: T \rightarrow S$ which is an element of (M) is quarrable (cf. 1.4.0) and for every $S' \rightarrow S$, $u': T \times_S S' \rightarrow S'$ is an element of (M) .
- (b) The composite of two elements of (M) lies in (M) .

⁽¹³⁾N.D.E. : One has added “in $\mathcal{C}$ ”.

⁽¹⁴⁾N.D.E. One has added the numbering 3.3.2.1 (resp. 3.3.3.1) for later references. On the other hand, one has added remark 3.3.3.2.

- (c) *Every isomorphism is an element of (M) .*
- (d) *Every element of (M) is an effective epimorphism.*

Let us note that (a) and (b) imply:

(a') *The cartesian product of two elements of (M) lies in (M) : Let $u: X \rightarrow Y$ and $u': X' \rightarrow Y'$ be two S -morphisms which are elements of (M) . If $Y \times_S Y'$ exists, then $X \times_S X'$ exists and $u \times_S u'$ is an element of (M) .*

This follows from the diagram

$$\begin{array}{ccccc}
 & & Y' & \xleftarrow{u'} & X' \\
 & & \uparrow & & \uparrow \\
 X & \xleftarrow{\quad} & X \times_S Y' & \xleftarrow{\quad} & X \times_S X' \\
 \downarrow u & & \downarrow & \swarrow u \times_S u' & \\
 Y & \xleftarrow{\quad} & Y \times_S Y' & &
 \end{array}$$

Likewise (a) and (d) imply:

(d') *Every element of (M) is a universal effective epimorphism.*

3.4.2. — The family (M_0) of universal effective epimorphisms satisfies conditions (a) to (d) of 3.4.1. Indeed, (a), (c) and (d) are satisfied by definition, and (b) follows from 1.8. In what follows, we shall assume given a family (M) of morphisms of $\mathcal{C}$ satisfying these conditions: our results will therefore apply to the family (M_0) in particular.

Definition 3.4.3. One says that the equivalence relation R in X is of type (M) if it is representable and if $p_1 \in (M)$ (which by (b) and (c) implies that $p_2 \in (M)$).

One says that R is (M) -effective if it is effective and if the canonical morphism $X \rightarrow X/R$ is an element of (M) .

One says that the quotient Y of X is (M) -effective if the canonical morphism $X \rightarrow Y$ is an element of (M) .

From this definition there follow the following consequences:⁽¹⁵⁾

Proposition 3.4.3.1.

- (i) *An (M) -effective equivalence relation is of type (M) and universal effective.*
- (ii) *An (M) -effective quotient is universal effective (cf. 3.3.3).*
- (iii) *The maps $R \mapsto X/R$ and $p \mapsto \mathcal{R}(p)$ realize a one-to-one correspondence between (M) -effective equivalence relations in X and (M) -effective quotients of X .*
- (iv) *(M_0) -effective is equivalent to universal effective.*

PROOF. Let us prove point (i). Since R is (M) -effective, one has a cartesian square

$$\begin{array}{ccc}
 R & \xrightarrow{p_2} & X \\
 p_1 \downarrow & & \downarrow p \\
 X & \xrightarrow{p} & X/R,
 \end{array}$$

and $p \in (M)$. Then, by 3.4.1 (a), p_1 and p_2 belong to (M) , hence R is of type (M) .

Set $Y = X/R$, and let $Y' \rightarrow Y$ be an arbitrary morphism. By 3.4.1 (a), the fiber products $X' = X \times_Y Y'$ and $R' = R \times_Y Y'$ exist, and the morphisms $X' \rightarrow Y'$ and $p'_i: R' \rightarrow X'$ belong to (M) . Finally, since $R = X \times_Y X$, one obtains, by associativity of the fiber product:

$$R' = X \times_Y X \times_Y Y' = X' \times_{Y'} X'.$$

⁽¹⁵⁾**N.D.E.** : One has added the numbering 3.4.3.1, for later references, and one has set out in detail the proof of point (i).

This shows that R' is (M) -effective; hence, in particular, R is universal effective. This proves (i), and also (iv). Points (ii) and (iii) follow from this, taking definition 3.3.2 into account. $\square$

3.4.4. — Let H be an S -group whose structural morphism is an element of (M) . Then if H operates *freely* on the S -object X , it defines on it an equivalence relation of type (M) . Indeed, by (a) the fiber product $H \times_S X$ exists and $\text{pr}_2: H \times_S X \rightarrow X$ is an element of (M) . One will say that the operation of H is (M) -*effective* if the equivalence relation defined in X by this operation is (M) -effective.

Proposition 3.4.5 ((M)-effectivity and base change). *Let R be an (M) -effective equivalence relation in X over S . Set $Y = X/R$. Let $S' \rightarrow S$ be a base change such that $Y' = Y \times_S S'$ exists. Then $X' = X \times_S S'$ exists, $R' = R \times_S S'$ is an (M) -effective equivalence relation in X' over S' , and $X'/R' \simeq (X/R)'$.*

PROOF. Indeed, the canonical morphisms $X \rightarrow Y$ and $R \rightarrow Y$ are elements of (M) , hence by (a') X' and R' are representable. By associativity of the product, R' is the equivalence relation defined in X' by the canonical morphism $X' \rightarrow Y'$, which is an element of (M) , whence the conclusion. $\square$

Proposition 3.4.6 ((M)-effectivity and cartesian products). *Let R (resp. R') be an (M) -effective equivalence relation in X (resp. X') over S . If $(X/R) \times_S (X'/R')$ exists, then $X \times_S X'$ exists, $R \times_S R'$ is an (M) -effective equivalence relation in $X \times_S X'$ over S , and*

$$(X \times_S X')/(R \times_S R') \simeq (X/R) \times_S (X'/R').$$

PROOF. Set $Y = X/R$, $Y' = X'/R'$. By (a'), the fiber product $X \times_S X'$ exists and the canonical morphism

$$q: X \times_S X' \rightarrow Y \times_S Y'$$

is an element of (M) . Now the formula

$$(X \times X') \times_{(Y \times Y')} (X \times X') \simeq (X \times_Y X) \times (X' \times_{Y'} X')$$

(all unindexed products are taken over S) shows that $R \times_S R'$ is the equivalence relation defined by q in $X \times_S X'$, which completes the proof. $\square$

3.4.7. — ⁽¹⁶⁾Suppose that $\mathcal{C}$ possesses a final object e and let $f: G \rightarrow G'$ be a morphism of $\mathcal{C}$ -groups, such that $f \in (M)$. Then, by 3.4.1 (a), the kernel $\text{Ker}(f)$ is representable by $e \times_{G'} G$, and the morphism $\text{Ker}(f) \rightarrow e$ belongs to (M) .

On the other hand, the equivalence relation defined by f is the same as that defined by the action of $\text{Ker}(f)$ (say, on the right) on G , i.e. it is the image of the morphism $G \times \text{Ker}(f) \rightarrow G \times G$, defined set-theoretically by $(g, h) \mapsto (g, gh)$. Consequently, one deduces from 3.3.2.1 the following corollary:

Corollary 3.4.7.1. *Suppose that $\mathcal{C}$ possesses a final object e and let $f: G \rightarrow G'$ be a morphism of $\mathcal{C}$ -groups, such that $f \in (M)$. Then the action of $\text{Ker}(f)$ on G is (M) -effective and G' is the quotient $G/\text{Ker}(f)$.*

3.5. Construction of quotients by descent. — It frequently happens that one does not know how to construct a quotient directly, but that one knows how to do so after a suitable base change. The present number gives a useful criterion in this situation.

One has seen in paragraph 2.1 the definition of a descent datum on an object X' over S' relative to a morphism $S' \rightarrow S$.

⁽¹⁶⁾N.D.E. : One has added this paragraph.

Definition 3.5.1. One says that a descent datum on X' relative to $S' \rightarrow S$ is *effective*, if X' equipped with this descent datum is isomorphic to the inverse image on S' of an object X over S , equipped with its canonical descent datum.

If $S' \rightarrow S$ is a descent morphism (2.2), then the X of the definition is unique up to unique isomorphism. To say that $S' \rightarrow S$ is an effective descent morphism (2.2) is to say that it is a descent morphism and that every descent datum relative to this morphism is effective.

Let us now consider an equivalence relation R in an object X over S . Let X' (resp. X'' , resp. X''') be the inverse images of X on S' , $S'' = S' \times_S S'$ and $S''' = S' \times_S S' \times_S S'$, and let R' , R'' , R''' be the equivalence relations deduced from R by inverse image. Suppose that the equivalence relation R' in X' is (M) -effective, and consider the quotient $Y' = X'/R'$, which is an object over S' . Its two inverse images on S'' are isomorphic to X''/R'' by 3.4.5. The S' -object Y' is therefore equipped with a canonical gluing datum. Using likewise the uniqueness of X'''/R''' , one sees that it is even a descent datum. (Remark: one has implicitly supposed in this proof that all the fiber products written existed, which is the case in particular if $S' \rightarrow S$ is quarrable, for example a descent morphism.)

Proposition 3.5.2. *Let R be an equivalence relation in the object X over S . Let $S' \rightarrow S$ be a universal effective epimorphism. Suppose that every S -morphism whose inverse image on S' is in (M) is itself in (M) . The following conditions are equivalent:*

- (i) R is (M) -effective in X ;
- (ii) R' is (M) -effective in X' and the canonical descent datum on X'/R' is effective.

If this is so, the “descended” object of X'/R' is canonically isomorphic to X/R .

The fact that (i) entails (ii) is nothing other than the translation into the language of descent of 3.4.4. If one shows the converse, the last assertion of the proposition will be a consequence of the fact that a universal effective epimorphism is a descent morphism (2.3), hence that the “descended object” is unique (up to unique isomorphism).

Let us therefore prove (ii) $\Rightarrow$ (i). Let Y' be the quotient X'/R' and Y the descended object. As the canonical morphism $p': X' \rightarrow X'/R' = Y'$ is compatible by construction with the descent data (its inverse images on S'' coincide with the canonical morphism $X'' \rightarrow X''/R''$), it arises by inverse image on S' from an S -morphism $p: X \rightarrow Y$. As p' is an element of (M) , it follows from the hypothesis made on the morphism $S' \rightarrow S$ that p is likewise an element of (M) . As p' is compatible with the equivalence relation R' , p is compatible with R , again because a universal effective epimorphism is a descent morphism. One therefore has a morphism

$$R \longrightarrow X \times_Y X.$$

To prove that R is (M) -effective and that Y is isomorphic to X/R , it suffices to prove that this morphism is an isomorphism. Now it becomes one by extension of the base from S to S' , for R' is effective; it is therefore an isomorphism for the same reason as previously (2.4).

Let us remark that the hypothesis of the text is verified if one takes for (M) the family (M_0) of universal effective epimorphisms and if $\mathcal{C}$ possesses fiber products (1.10). One deduces from this the

Corollary 3.5.3. *Suppose that $\mathcal{C}$ possesses fiber products (over S would suffice). Let R be an equivalence relation in X over S and $S' \rightarrow S$ a universal effective epimorphism. The following conditions are equivalent:*

- (i) R is universally effective in X ,
- (ii) R' is universally effective in X' and the canonical descent datum on X'/R' is effective.

If this is so, the “descended” object of X'/R' is canonically isomorphic to X/R .

4. Topologies and sheaves

The notion of sieve, and the presentation of the notion of topology (4.2.1) adopted here, (more intrinsic and more convenient in many respects than that by covering families of [MA]), are due to J. Giraud [AS].

4.1. Sieves

Definition 4.1.1. One calls a sieve of the category $\mathcal{C}$ a subfunctor C of the final functor $\mathbf{e}: \mathcal{C}^\circ \rightarrow (\mathbf{Ens})$.

With every sieve C of $\mathcal{C}$ one associates the set $E(C)$ of the objects X of $\mathcal{C}$ such that $C(X) \neq \emptyset$, that is, such that the structural morphism $X \rightarrow \mathbf{e}$ factors through C . One therefore has the equivalences

$$(+) \quad \begin{cases} X \in E(C) & \iff C(X) = \mathbf{e}(X) = \{\emptyset\}. \\ X \notin E(C) & \iff C(X) = \emptyset. \end{cases}$$

The set $E = E(C)$ enjoys the following property:

(++) If $X \in E$ and if $\text{Hom}(Y, X) \neq \emptyset$, then $Y \in E$.

(Let us remark that if one equips the set $\text{Ob}(\mathcal{C})$ with its *natural preorder structure*, (Y dominating X if there exists an arrow from Y to X), the sets E satisfying (++) are the subsets of $\text{Ob}(\mathcal{C})$ which contain every majorant⁽¹⁷⁾ of one of their elements.)

Conversely, if E is a subset of $\text{Ob}(\mathcal{C})$ enjoying the property (++) , then E may be written in a unique manner in the form $E(C)$ and C is defined by the formulas (+). There is therefore a one-to-one correspondence between the sieves of $\mathcal{C}$ and the subsets of $\text{Ob}(\mathcal{C})$ satisfying the condition (++) . By *abuse of language*, we shall sometimes say that the set $E(C)$ is a sieve of $\mathcal{C}$.

⁽¹⁸⁾Let C and C' be two sieves of $\mathcal{C}$; as they are two subfunctors of the final functor $\mathbf{e}$, it amounts to the same to say that C dominates C' (for the domination relation in $\text{Ob}(\widehat{\mathcal{C}})$), or that C is a *subfunctor* of C' , or again that $E(C) \subset E(C')$; one will then say that C is *finer* than C' . One sees that one is dealing in this case with an order structure on the set of sieves of $\mathcal{C}$. Moreover, one has $E(C) \cap E(C') = E(C \times C')$ and hence the set of sieves of $\mathcal{C}$ is *filtered* for the order relation “being finer”.

Every subset E of $\text{Ob}(\widehat{\mathcal{C}})$, for example a subset of $\text{Ob}(\mathcal{C})$, defines a sieve $C(E)$: the set of the $X \in \text{Ob}(\mathcal{C})$ such that $F(X) \neq \emptyset$ for at least one $F \in E$ satisfies the condition (++) and defines the sieve sought.

This sieve may also be defined as the *image* of the family of morphisms $\{F \rightarrow \mathbf{e}, F \in E\}$ in the sense of the following definition:

Definition 4.1.2. Let $\{F_i \rightarrow F\}$ be a family of morphisms of $\widehat{\mathcal{C}}$ with the same target F . One calls the *image* of this family the subfunctor of F defined by

$$S \mapsto \bigcup_i \text{Im } F_i(S) \subset F(S).$$

Proposition 4.1.3. *The formation of the image commutes with extension of the base: in the preceding notations, let us denote by I the image of the family $\{F_i \rightarrow F\}$; for every morphism $G \rightarrow F$ of $\widehat{\mathcal{C}}$, the image of the family of morphisms $\{F_i \times_F G \rightarrow G\}$ is the subfunctor $I \times_F G$ of G .*

⁽¹⁷⁾**N.D.E.** : Here, “majorant” is taken in the sense of the preorder relation mentioned above, that is, Y majorizes X if there exists an arrow $Y \rightarrow X$. On the other hand, if X, Y are two subobjects of an object Z , one says (cf. 2.4) that Y majorizes X if $X \subset Y$. To avoid any ambiguity between these two terminologies, one has in the sequel replaced “majorant” by “dominating” in the first case, and by “containing”, in the second.

⁽¹⁸⁾**N.D.E.** : One has spelled out the sentence which follows.

Definition 4.1.4.0. ⁽¹⁹⁾Let C be a sieve of $\mathcal{C}$. If E is a subset of $\text{Ob}(\mathcal{C})$ such that $C(E) = C$, one says that E is a *base* of C . Every sieve C possesses a base, for example the set $E(C)$.

We propose to describe the set $\text{Hom}(C, F)$, where C is a sieve of $\mathcal{C}$ and F an object of $\widehat{\mathcal{C}}$, by means of a base $\{S_i\}$ of C . For each pair (i, j) , one has a diagram in $\widehat{\mathcal{C}}$:

$$\begin{array}{ccc} S_i \times S_j & \longrightarrow & S_i \\ \downarrow & & \downarrow \\ S_j & \longrightarrow & \mathbf{e}, \end{array}$$

whence a diagram of sets

$$\Gamma(F) = \text{Hom}(\mathbf{e}, F) \xrightarrow{\sigma} \prod_i \text{Hom}(S_i, F) \xrightleftharpoons[\tau_2]{\tau_1} \prod_{i,j} \text{Hom}(S_i \times S_j, F)$$

such that $\tau_1 \sigma = \tau_2 \sigma$. One therefore has a morphism

$$\text{Hom}(\mathbf{e}, F) \longrightarrow \text{Ker} \left(\prod_i \text{Hom}(S_i, F) \Rightarrow \prod_{i,j} \text{Hom}(S_i \times S_j, F) \right).$$

One verifies immediately:

Proposition 4.1.4. *One has an isomorphism, functorial in F ,*

$$\text{Hom}(C, F) \xrightarrow{\sim} \text{Ker} \left(\prod_i \text{Hom}(S_i, F) \Rightarrow \prod_{i,j} \text{Hom}(S_i \times S_j, F) \right),$$

such that the diagram

$$\begin{array}{ccc} \text{Hom}(\mathbf{e}, F) & \longrightarrow & \text{Ker} \left(\prod_i \text{Hom}(S_i, F) \Rightarrow \prod_{i,j} \text{Hom}(S_i \times S_j, F) \right) \\ & & \uparrow \wr \\ \text{Hom}(\mathbf{e}, F) & \longrightarrow & \text{Hom}(C, F), \end{array}$$

where the last line is induced by the canonical morphism $C \longrightarrow \mathbf{e}$, is commutative.

Corollary 4.1.5. *Suppose that the fiber products $S_i \times S_j$ are representable, for example that the S_i be quarrable. One then has, for every $F \in \text{Ob}(\widehat{\mathcal{C}})$, an isomorphism*

$$\text{Hom}(C, F) \xrightarrow{\sim} \text{Ker} \left[\prod_i F(S_i) \Rightarrow \prod_{i,j} F(S_i \times S_j) \right].$$

Remark 4.1.6. Let R be a sieve of $\mathcal{C}$; let us denote by $\mathcal{R}$ the full subcategory of $\mathcal{C}$ whose set of objects is $E(R)$ and by

$$i_R: \mathcal{R} \longrightarrow \mathcal{C}$$

the inclusion functor. One has an isomorphism, functorial in $F \in \text{Ob}(\widehat{\mathcal{C}})$,

$$\text{Hom}(R, F) \xrightarrow{\sim} \Gamma(F \circ i_R)$$

such that the diagram

$$\begin{array}{ccc} \text{Hom}(\mathbf{e}, F) & \longrightarrow & \text{Hom}(R, F) \\ \downarrow \wr & & \downarrow \wr \\ \Gamma(F) & \longrightarrow & \Gamma(F \circ i_R), \end{array}$$

⁽¹⁹⁾**N.D.E.** : One has added the numbering 4.1.4.0, in order to bring this definition into evidence.

where the second line is induced by the functor i_R , is commutative.

Definition 4.1.7. Let $\mathcal{C}$ be a category. One calls a sieve of the object S of $\mathcal{C}$ a sieve of the category $\mathcal{C}_{/S}$.

A sieve of S is therefore a sub- $\widehat{\mathcal{C}}$ -object of S . There corresponds to it canonically a subset of $\text{Ob}(\mathcal{C}_{/S})$ containing the source of every arrow whose target it contains. By *abuse of language*, such a set will also be called a sieve of S .

4.2. Topologies: definitions

Definition 4.2.1. Let $\mathcal{C}$ be a category. One calls a topology on $\mathcal{C}$ the giving, for each S of $\mathcal{C}$, of a set $J(S)$ of sieves of S , called *covering* sieves or *refinements* of S , this giving satisfying the following axioms:

- (T 1) For every refinement R of S and every morphism $T \rightarrow S$, the sieve $R \times_S T$ of T is *covering* (“stability under *base change*”).
- (T 2) If R, C are two sieves of S , if R is covering and if for every $T \in \text{Ob}(\mathcal{C})$ and every morphism $T \rightarrow R$, the sieve $C \times_S T$ of T is covering, then C is a refinement of S .⁽²⁰⁾
- (T 3) If $C \supset R$ are two sieves of S and if R is covering, then C is covering.
- (T 4) For every S , S is a refinement of S .

One may reformulate these axioms in the following manner. Suppose given a topology $S \mapsto J(S)$ on $\mathcal{C}$ and, for every object F of $\widehat{\mathcal{C}}$, let us denote by $J(F)$ the set of subfunctors R of F such that for every morphism $T \rightarrow F$ of $\widehat{\mathcal{C}}$, where T is *representable*, $R \times_F T$, which is a sieve of T , is *covering*. By virtue of (T 1), this notation is indeed compatible with the preceding one. One will likewise say that $R \in J(F)$ is a *refinement* of F . One verifies immediately that the preceding axioms entail the following properties:

- (T' 0) If $F \supset G$ are two objects of $\widehat{\mathcal{C}}$, and if for every $S \in \text{Ob}(\mathcal{C})$ and every morphism $S \rightarrow F$, $G \times_F S \in J(S)$, then $G \in J(F)$.
- (T' 1) If $G \in J(F)$, and if $H \rightarrow F$ is a morphism of $\widehat{\mathcal{C}}$, then $G \times_F H \in J(H)$.
- (T' 2) If $F \supset G \supset H$ are three objects of $\widehat{\mathcal{C}}$, if $G \in J(F)$ and $H \in J(G)$, then $H \in J(F)$.
- (T' 3) If $F \supset G \supset H$ are three objects of $\widehat{\mathcal{C}}$ and if $H \in J(F)$, then $G \in J(F)$.
- (T' 4) For every $F \in \text{Ob}(\widehat{\mathcal{C}})$, $F \in J(F)$.

Conversely, if one gives oneself for every $F \in \text{Ob}(\widehat{\mathcal{C}})$ a set $J(F)$ of subobjects of F satisfying the properties (T' 0) to (T' 4), the map $S \mapsto J(S)$ defines a topology on $\mathcal{C}$ and the two preceding constructions are inverse to one another.

From (T' 1), (T' 2) and (T' 3)⁽²¹⁾ there results the following property:

- (T' 5) If G and H are two subobjects of F and if $G, H \in J(F)$, then $G \cap H \in J(F)$.

The set $J(F)$, ordered by the relation $\supset$, is therefore *filtered*; this remark will be of use to us later.

4.2.2. — One will say that the topology defined by J is finer than the topology defined by J' if for every $S \in \text{Ob}(\mathcal{C})$, $J(S) \supset J'(S)$ (it amounts to the same to say that for every $F \in \text{Ob}(\widehat{\mathcal{C}})$, $J(F) \supset J'(F)$).

Every set of topologies on $\mathcal{C}$ possesses a greatest *lower* bound: let I be a set of indices, and for each $i \in I$, let $S \mapsto J_i(S)$ be a topology on $\mathcal{C}$. Set $J(S) = \bigcap_{i \in I} J_i(S)$; it is immediate

⁽²⁰⁾**N.D.E.** : that is: if C is covering “locally with respect to the covering sieve R ”, then C is covering.

⁽²¹⁾**N.D.E.** : (T' 1) and (T' 2) suffice: $G \cap H = G \times_F H$ belongs to $J(H)$, by (T' 1), hence to $J(F)$, by (T' 2).

that one has thus defined a topology on $\mathcal{C}$, and that it is indeed the greatest *lower* bound of the given set.

In particular, let us give ourselves, for each $S \in \text{Ob}(\mathcal{C})$, a set $E(S)$ of sieves of S . One calls the topology *generated* by these sets the *least fine* topology for which the elements of $E(S)$ are refinements of S for every S .

Definition 4.2.3. Let $\{F_i \rightarrow F\}$ be a family of morphisms of $\widehat{\mathcal{C}}$. Let $G \subset F$ be the image (4.1.2) of this family. The family is said to be *covering* if $G \in J(F)$. A morphism is said to be *covering* if the family reduced to this morphism is covering.

This definition applies in particular to an inclusion: a sieve C of S is covering if and only if the canonical morphism $C \rightarrow S$ is covering.

The axioms (T' 0) to (T' 5) entail for covering families the following properties:

(C 0) Let $\{F_i \rightarrow F\}$ be a family of morphisms of $\widehat{\mathcal{C}}$. If for every representable base change $S \rightarrow F$, the family $\{F_i \times_F S \rightarrow S\}$ is covering, then the original family is covering too.

(C 1) For every covering family $\{F_i \rightarrow F\}$ and every morphism $G \rightarrow F$, the family $\{F_i \times_F G \rightarrow G\}$ is covering (“stability under base change”).

(C 2) If $\{F_i \rightarrow F\}$ is a covering family and if, for each i , $\{F_{ij} \rightarrow F_i\}$ is a covering family, then the composite family $\{F_{ij} \rightarrow F\}$ is covering (“stability under composition”).

(C 3) If $\{G_j \rightarrow F\}$ is a covering family, and if $\{F_i \rightarrow F\}$ is a family of morphisms with target F such that for each j there exists an i such that $G_j \rightarrow F$ factors through $F_i \rightarrow F$, then $\{F_i \rightarrow F\}$ is covering (“saturation”).

(C 4) Every family consisting of a single isomorphism is covering.

Note that (C 2) and (C 3) also imply:

(C 5) If $\{F_i \rightarrow F\}$ is a family of morphisms with target F such that there exists a covering family $\{G_j \rightarrow F\}$ such that for every j the family $\{F_i \times_F G_j \rightarrow G_j\}$ is covering, then the family $\{F_i \rightarrow F\}$ is covering (“a locally covering family is covering”).

4.2.4. — Conversely, let $\mathcal{C}$ be a category possessing fibered products and, for each $S \in \text{Ob } \mathcal{C}$, let us give ourselves a set of families of morphisms of $\mathcal{C}$ with target S , called covering families, this datum satisfying axioms (C 1) to (C 4) (and hence also (C 5), which is a consequence of them). For every $S \in \text{Ob } \mathcal{C}$, let $J(S)$ be the set of sieves of S possessing a covering basis (or, what amounts to the same by (C 3), all of whose bases are covering). Then $S \mapsto J(S)$ defines a topology on $\mathcal{C}$. The two preceding constructions are inverse to one another.

In fact, in applications it is not very practical to consider all covering families, for one sometimes possesses fairly simple descriptions of a “sufficient” number of these families. This leads to the following definitions.

Definition 4.2.5.0. ⁽²²⁾ Let $\mathcal{C}$ be a category. Suppose that, for each $S \in \text{Ob } \mathcal{C}$, a set $P(S)$ of families of morphisms of $\mathcal{C}$ with target S is given. One calls the *topology generated by P* the least fine topology for which the given families are covering.

Definition 4.2.5. Let $\mathcal{C}$ be a category. One calls a *pretopology* on $\mathcal{C}$ the datum, for each $S \in \text{Ob } \mathcal{C}$, of a set $R(S)$ of families of morphisms $\{S_i \rightarrow S\}$ with target S , called covering for the pretopology in question, satisfying the following axioms:

(P 1) For every family $\{S_i \rightarrow S\} \in R(S)$ and every morphism $T \rightarrow S$, the fibered products $S_i \times_S T$ exist and $\{S_i \times_S T \rightarrow T\} \in R(T)$.

(P 2) If $\{S_i \rightarrow S\} \in R(S)$ and if for each i , $\{T_{ij} \rightarrow S_i\} \in R(S_i)$, then the composite family $\{T_{ij} \rightarrow S\}$ belongs to $R(S)$.

(P 3) Every family consisting of a single isomorphism is covering.

⁽²²⁾**N.D.E.** This definition (placed in the original after 4.2.5) has been put here, since it will be used in 6.2.1 in a slightly more general setting than that of 4.2.5.

Proposition 4.2.6. *For every S , let $J(S)$ be the set of sieves of S covering for the topology generated by the pretopology R . Let $J_R(S)$ be the subset of $J(S)$ consisting of the sieves defined by families in $R(S)$. Then $J_R(S)$ is cofinal in $J(S)$: every refinement of S contains a sieve defined by a family in $R(S)$.*

PROOF. For every S , let $J'(S)$ be the set of sieves of S containing a sieve in $J_R(S)$. One evidently has $J'(S) \subset J(S)$. To show that $J(S) = J'(S)$, it suffices to show that the $J'(S)$ form a topology on $\mathcal{C}$, that is, satisfy axioms (T 1) to (T 4). Now (T 1), (T 3), (T 4) are evidently satisfied. It remains to verify (T 2). ⁽²³⁾

Thus let U be an element of $J'(S)$ and C a sieve of S ; one assumes that for every $T \rightarrow U$, the sieve $C \times_S T$ is in $J'(T)$, and one must prove that $C \in J'(S)$. By definition of J' , U contains a refinement U' defined by a family $\{S_i \rightarrow S\} \in R(S)$. Since one has verified (T 3), it suffices to prove that $U' \cap C \in J'(S)$; one may therefore assume that $U = U'$. By hypothesis, for every i , $C \times_S S_i \in J'(S_i)$; there therefore exists, for each i , a covering family $\{T_{ij} \rightarrow S_i\} \in R(S_i)$ such that $T_{ij} \rightarrow S_i$ factors through $C \times_S S_i \rightarrow S_i$. The morphism $T_{ij} \rightarrow S$ therefore factors through $C \rightarrow S$, which shows that C contains the sieve defined by the composite family $\{T_{ij} \rightarrow S\}$, and one has finished by (P 2). $\square$

Axioms (P 1) to (P 3) are those of [MA]. In view of the practical interest of pretopologies, we shall interpret each important result with the aid of a pretopology defining the given topology.

Remark 4.2.7. One may introduce a slightly more general notion: for each S one gives a set of covering families satisfying (P 1), (P 3) and Proposition 4.2.6. This occurs in particular when the given families satisfy (P 1), (P 3) and (C 5). The reader may consult [D].

Definition 4.2.8. Let $\mathcal{C}$ be endowed with a topology, and let S be an object of $\mathcal{C}$. Let $P(S')$ be a relation involving an argument $S' \in \text{Ob } \mathcal{C}/_S$. One assumes that $\text{Hom}(S'', S') \neq \emptyset$ implies $P(S') \Rightarrow P(S'')$. One says that P is *true locally on S* for the topology considered if the following equivalent conditions are satisfied:

- (i) The set of $S' \rightarrow S$ such that $P(S')$ is true is a refinement of S .
- (ii) There exists a refinement of S such that $P(S')$ is true for every S' of this refinement.
- (iii) (If the given topology is defined by a pretopology.) There exists a covering family for this pretopology such that $P(S')$ is true for every S' of this family.

Example 4.2.9. Let $f : X \rightarrow Y$ be an S -morphism. One will say that f is locally an isomorphism if there exists a covering family $\{S_i \rightarrow S\}$ such that, for every i , $f \times_S S_i$ is an isomorphism. It amounts to the same to require that there exist a refinement R of S such that, for every $T \rightarrow R$, $X(T) \rightarrow Y(T)$ is an isomorphism.

One will see many other examples of “local” language in what follows.

4.3. Presheaves, sheaves, sheaf associated to a presheaf

Definition 4.3.1. Let $\mathcal{C}$ be a category. One calls a *presheaf of sets* on $\mathcal{C}$ any contravariant functor from $\mathcal{C}$ to the category of sets. The category $\hat{\mathcal{C}} = \text{Hom}(\mathcal{C}^{\text{op}}, (\text{Ens}))$ is called the *category of presheaves* on $\mathcal{C}$. If $\mathcal{C}$ is endowed with a topology, one says that the presheaf P is *separated* (resp. is a *sheaf*) if for every $S \in \text{Ob } \mathcal{C}$ and every $R \in J(S)$, the canonical map

$$(+) \quad P(S) = \text{Hom}(S, P) \rightarrow \text{Hom}(R, P)$$

⁽²³⁾**N.D.E.** In what follows, misprints in the original have been corrected.

is injective (resp. bijective). One calls the full subcategory of $\widehat{\mathcal{C}}$ whose objects are the sheaves the *category of sheaves*, and denotes it by $\widetilde{\mathcal{C}}$.⁽²⁴⁾

Proposition 4.3.2. *Let P be a separated presheaf (resp. a sheaf). For every functor $H \in \text{Ob } \widehat{\mathcal{C}}$ and every $R \in J(H)$, the canonical map*

$$(+)\quad \text{Hom}(H, P) \longrightarrow \text{Hom}(R, P)$$

is injective (resp. bijective).

PROOF. Indeed, let P be a separated presheaf, H a presheaf, $R \in J(H)$, and $u, v : H \rightarrow P$ such that $uj = vj$. For every $f : S \rightarrow H$, $S \in \text{Ob } \mathcal{C}$, $R \times_H S$ is a refinement of S and $ufj_S = vfj_S$:

$$\begin{array}{ccccc} & & R & \xrightarrow{j} & H & \xrightarrow{u,v} & P \\ & f_R \nearrow & & & \nearrow f & & \\ R \times_H S & \xrightarrow{j_S} & S & & & & \end{array}$$

Since P is separated, one deduces $uf = vf$. Since this is true for every representable S , one has $u = v$.

Suppose now that P is a sheaf. Let $g : R \rightarrow P$; let us show that it factors through H . For every $f : S \rightarrow H$, $S \in \text{Ob } \mathcal{C}$, $g \circ f_R : R \times_H S \rightarrow P$ factors uniquely through S , hence defines a morphism $h : S \rightarrow P$ which is evidently functorial in f , by uniqueness:

$$\begin{array}{ccccc} R \times_H S & \xrightarrow{f_R} & R & \xrightarrow{g} & P \\ \downarrow j_S & & \downarrow j & \nearrow h & \\ S & \xrightarrow{f} & H & & \end{array}$$

One has therefore defined, for every S , a map from $H(S)$ to $P(S)$, functorial in S , hence a morphism from H to P which does satisfy the required conditions. $\square$

Corollary 4.3.2.1.⁽²⁵⁾ *Let R, F be two sheaves. If R is a refinement of F , then $R = F$.*

PROOF. Indeed, suppose that R is a refinement of F and denote by j the inclusion $R \hookrightarrow F$. By 4.3.2, one has $\text{Hom}(F, R) = \text{End}(R)$, hence there exists $\pi : F \rightarrow R$ such that $\pi \circ j = \text{id}_R$. Similarly, one has $\text{End}(F) = \text{Hom}(R, F)$, and the equality $j \circ \pi \circ j = j$ implies $j \circ \pi = \text{id}_F$, so j is an isomorphism. $\square$

Proposition 4.3.3. ([AS], 1.3). *Let $\mathcal{C}$ be a category. Let P be a presheaf on $\mathcal{C}$; for every $S \in \text{Ob } \mathcal{C}$, denote by $J(S)$ the set of sieves R of S such that for every $T \rightarrow S$, the map*

$$(+)\quad \text{Hom}(T, P) \longrightarrow \text{Hom}(R \times_S T, P)$$

is injective (resp. bijective). Then the $J(S)$ define a topology on $\mathcal{C}$, i.e. satisfy axioms (T 1) to (T 4).

Corollary 4.3.4. *For every $S \in \text{Ob } \mathcal{C}$, let $K(S)$ be a family of sieves satisfying (T 1). Let P be a presheaf on $\mathcal{C}$. For it to be separated (resp. a sheaf) for the topology generated by the $K(S)$, it is necessary and sufficient that, for every $S \in \text{Ob } \mathcal{C}$ and every $R \in K(S)$, the canonical map*

$$(+)\quad \text{Hom}(S, P) \longrightarrow \text{Hom}(R, P)$$

⁽²⁴⁾**N.D.E.** One will note that if Q is a subpresheaf of a separated presheaf P , then Q is separated. Indeed, for every sieve R of S , the composite map $Q(S) \hookrightarrow P(S) \hookrightarrow P(R)$ is injective and factors through $Q(S) \rightarrow Q(R)$.

⁽²⁵⁾**N.D.E.** This corollary has been added.

be injective (resp. bijective).

Corollary 4.3.5. For each $S \in \text{Ob } \mathcal{C}$, let $R(S)$ be a set of families of morphisms of $\mathcal{C}$ with target S , satisfying (P 1) (for example defining a pretopology). Let P be a presheaf on $\mathcal{C}$. For P to be separated (resp. a sheaf) for the topology generated by R , it is necessary and sufficient that for every $S \in \text{Ob } \mathcal{C}$ and every family $\{S_i \longrightarrow S\} \in R(S)$ the map

$$P(S) \longrightarrow \prod_i P(S_i)$$

be injective (resp. the diagram

$$P(S) \longrightarrow \prod_i P(S_i) \rightrightarrows \prod_{i,j} P(S_i \times_S S_j)$$

be exact).

Definition 4.3.6. Let $\mathcal{C}$ be a category. One calls the *canonical topology* on $\mathcal{C}$ the finest topology for which all representable functors are sheaves.

Corollary 4.3.7. For a sieve R of S to be a refinement for the canonical topology, it is necessary and sufficient that, for every morphism $T \longrightarrow S$ of $\mathcal{C}$ and every $X \in \text{Ob } \mathcal{C}$, the canonical map

$$\text{Hom}(T, X) \longrightarrow \text{Hom}(R \times_S T, X)$$

be bijective.

Definition 4.3.8. A covering sieve for the canonical topology will be called a *universal effective epimorphic sieve*.

Corollary 4.3.9. A universal effective epimorphic family defines a universal effective epimorphic sieve. Conversely, every base-changeable family defining a universal effective epimorphic sieve is universal effective epimorphic.

Let us return to the case where $\mathcal{C}$ is endowed with an arbitrary topology and pass to the construction of the sheaf associated to a presheaf P . Let S be an object of $\mathcal{C}$. If $R \supset R'$ are two refinements of S , one has a diagram

$$\begin{array}{ccc} \text{Hom}(S, P) & \longrightarrow & \text{Hom}(R, P) \\ & \searrow & \downarrow \\ & & \text{Hom}(R', P) \end{array}$$

The ordered set $J(S)$ is filtered, as one has already remarked. Since S is an element of $J(S)$, one has an evident morphism

$$\text{Hom}(S, P) \longrightarrow \varinjlim_{R \in J(S)} \text{Hom}(R, P).$$

Definition 4.3.10.0. ⁽²⁶⁾ Set $\check{H}^0(S, P) = \varinjlim_{R \in J(S)} \text{Hom}(R, P)$. One verifies that $\check{H}^0(S, P)$ depends functorially on S , hence defines a functor LP by

$$(++)\quad \text{Hom}(S, LP) = \check{H}^0(S, P) = \varinjlim_{R \in J(S)} \text{Hom}(R, P).$$

⁽²⁶⁾**N.D.E.** The numbering 4.3.10.0 has been added for later references. Moreover, it follows from the definition that if $Q \longrightarrow P$ is a monomorphism, the same is true of $LP \longrightarrow LQ$; thus L “preserves monomorphisms” (see also 4.3.16 for a more general result: L “commutes with finite projective limits”).

By construction one has morphisms

$$\begin{aligned}\ell_P : P &\longrightarrow LP \\ z_R : \text{Hom}(R, P) &\longrightarrow \text{Hom}(S, LP).\end{aligned}$$

Lemma 4.3.10. (i) For every refinement R of S and every $u : R \longrightarrow P$, the diagram

$$\begin{array}{ccc} P & \xrightarrow{\ell_P} & LP \\ \uparrow u & & \uparrow z_R(u) \\ R & \xrightarrow{i_R} & S \end{array}$$

is commutative.

(ii) For every morphism $S \xrightarrow{v} LP$, there exists a refinement R of S and a morphism $u : R \longrightarrow P$ with $v = z_R(u)$.

(iii) Let Q be a functor and $u, v : Q \longrightarrow P$ such that $\ell_P u = \ell_P v$. Then the kernel of the pair (u, v) is a refinement of Q .

(iv) Let $u : R \longrightarrow P$ and $u' : R' \longrightarrow P$; for $z_R(u) = z_{R'}(u')$, it is necessary and sufficient that there exist a refinement $R'' \subset R \cap R'$ of S such that u and u' agree on R'' .

PROOF. (i): One must verify that $z_R(u)i_R = \ell_P u$. For this, it suffices to verify that the composites of these two morphisms with every morphism $T \xrightarrow{g} R$, where T is representable, are equal. Now consider $f = i_R g$ and the fibered product $R' = R \times_S T$:⁽²⁷⁾

$$\begin{array}{ccccc} P & \xrightarrow{\ell_P} & LP & & \\ \uparrow u & & \uparrow z_R(u) & & \\ R & \xrightarrow{i_R} & S & & \\ \uparrow p & \swarrow g & \uparrow f & & \\ R' & \xrightarrow{\sim} & T & & \end{array}$$

By definition of ℓ_P , $\ell_P u g = z_R(u) f$ (this is the particular case of what one is trying to prove in which i_R is an isomorphism), and $z_R(u) f = z_R(u) i_R g$.

(ii) and (iv) merely express the definition of $\text{Hom}(S, LP)$ as an inductive limit.

(iii): If K denotes the kernel of the pair (u, v) , for each morphism $f : S \longrightarrow Q$, where S is representable, $K \times_Q S$ is a subfunctor of the kernel of the pair of arrows $u f, v f : S \longrightarrow P$. By (T' 0), one is therefore reduced to proving the assertion in the case where $Q = S$ is representable. But in this case it follows from (ii) and (iv) that K contains a refinement of S , hence is a refinement of S . $\square$

Finally one verifies that $P \mapsto LP$ defines a functor

$$L : \widehat{\mathcal{C}} \longrightarrow \widehat{\mathcal{C}}$$

and $P \mapsto \ell_P$ a morphism of functors

$$\ell : \text{Id}_{\widehat{\mathcal{C}}} \longrightarrow L.$$

Let us now state the essential result:

Proposition 4.3.11. (i) If P is any presheaf, LP is separated and $\ell_P : P \longrightarrow LP$ is covering (4.2.3).

(ii) If P is a sheaf, $P \longrightarrow LP$ is an isomorphism.

⁽²⁷⁾**N.D.E.** Denote by p the projection $R' \longrightarrow R$. Since i_R is a monomorphism, one has $p = g i_{R'}$. Let g' be the section of $i_{R'}$ defined by g ; then $pg' = g$, hence $pg' i_{R'} = g i_{R'} = p$; this implies $g' i_{R'} = \text{id}_{R'}$, and thus $i_{R'} : R' \longrightarrow T$ is an isomorphism, with inverse g' .

(iii) For every presheaf P and every separated presheaf (resp. sheaf) F , the map

$$\mathrm{Hom}(\ell_P, F) : \mathrm{Hom}(LP, F) \longrightarrow \mathrm{Hom}(P, F)$$

is injective (resp. bijective).

(iv) If P is separated, $\ell_P : P \longrightarrow LP$ is a covering monomorphism (hence P is a refinement of LP), and LP is a sheaf.

PROOF. (i) First, $P \longrightarrow LP$ is covering; indeed, for every morphism $S \xrightarrow{v} LP$, there exists, by 4.3.10 (i) and (ii), a refinement R of S such that one has a commutative diagram:

$$\begin{array}{ccc} & P & \xrightarrow{\ell_P} LP \\ & \uparrow v' & \uparrow v \\ P \times_{LP} S & \xrightarrow{\quad} & S \\ \uparrow & \nearrow i_R & \\ R & & \end{array}$$

hence $P \times_{LP} S \longrightarrow S$ is covering. It follows, by (C 0), that $P \longrightarrow LP$ is covering.

If, on the other hand, two morphisms $S \xrightarrow{v_1, v_2} LP$ induce the same morphism from a refinement R of S , let us show that they are equal. There exist refinements R_i , $i = 1, 2$, and morphisms $u_i : R_i \longrightarrow P$ such that $z_{R_i}(u_i) = v_i$. Taking R sufficiently small, one may assume that $R_1 = R_2 = R$. It then follows from the commutative diagram of 4.3.10 (i) that $\ell_P u_1 = \ell_P u_2$. By *loc. cit.* (iii), u_1 and u_2 therefore agree on a refinement of R , hence a refinement of S , which implies $z_R(u_1) = z_R(u_2)$, by *loc. cit.* (iv).

(ii) is clear, for if P is a sheaf, $\mathrm{Hom}(S, P) \longrightarrow \mathrm{Hom}(R, P)$ is already an isomorphism for every refinement R of S .

(iii) Let u, v be two morphisms $LP \longrightarrow F$ such that $u\ell_P = v\ell_P$. To show that $u = v$, it suffices to see that $uf = vf$ for every $f : S \longrightarrow LP$, where S is representable. Now there exists a refinement R of S and a morphism $g : R \longrightarrow P$ with $f = z_R(g)$. Then uf and vf agree on R with $u\ell_P g = v\ell_P g$, hence agree on R . If F is separated, one therefore has $uf = vf$. Suppose now that F is a sheaf; one then has the commutative diagram

$$\begin{array}{ccc} P & \xrightarrow{\ell_P} & LP \\ \downarrow & \nearrow & \downarrow \\ F & \xrightarrow{\sim} & LF \end{array}$$

which shows that $\mathrm{Hom}(\ell_P, F)$ is surjective.

(iv) Let us show that if P is separated, $P \longrightarrow LP$ is a monomorphism. For this, it suffices to see that for every pair of morphisms $u, v : S \longrightarrow P$ (where S is representable) such that $\ell_P u = \ell_P v$, one has $u = v$. Now *loc. cit.* (iii) shows that u and v agree on a refinement of S , hence agree, since P is separated. This shows that $P \longrightarrow LP$ is a monomorphism; since it is covering by (i), one obtains that P is a refinement of LP .

Finally let us show that LP is a sheaf. Since one already knows by (i) that it is a separated presheaf, it suffices to see that for every $S \in \mathrm{Ob} \mathcal{C}$, every refinement R of S , and every morphism $h : R \longrightarrow LP$, there exists a morphism $u : S \longrightarrow LP$ with $u i_R = h$. Now $R' = P \times_{LP} R$ is a refinement of R , since P is a refinement of LP , hence R' is a refinement of S . Set $u = z_{R'}(h')$:

$$\begin{array}{ccccc} P & \xrightarrow{\ell_P} & LP & & \\ \uparrow h' & & \uparrow h & \nwarrow u & \\ R' & \xrightarrow{j} & R & \xrightarrow{i_R} & S \end{array}$$

One has $ui_{R'} = \ell_P h' = hj$, hence $ui_R j = hj$. Since R' is a refinement of R and LP is separated, 4.3.2 shows that $ui_R = h$. $\square$

Corollary 4.3.12. ⁽²⁸⁾ *Let F be a sheaf and R a sub- $\widehat{\mathcal{C}}$ -object of F . Then R is a separated presheaf, $\ell_R : R \rightarrow LR$ is a covering monomorphism, and one has a commutative diagram*

$$\begin{array}{ccc} R & \xrightarrow{i} & F \\ \ell_R \searrow & & \nearrow j \\ & LR & \end{array}$$

Consequently, R is a refinement of F if and only if j is an isomorphism.

PROOF. One has already noted that R is separated and that j is a monomorphism, cf. N.D.E. (24) and (26). By 4.3.11 (iv), ℓ_R is a covering monomorphism. Thus, if j is an isomorphism, R is a refinement of F . Conversely, if i is covering, j is too, hence is an isomorphism by 4.3.2.1. $\square$

Remark 4.3.13. If $J'(S)$ is a cofinal subset of $J(S)$, one has

$$\mathrm{Hom}(S, LP) = \varinjlim_{R \in J'(S)} \mathrm{Hom}(R, P).$$

In particular, let $S \mapsto R(S)$ be a pretopology generating the given topology. The functor L may be described with the aid of the covering families that are elements of $R(S)$. By making the formula above explicit, one recovers the construction of [MA].

Denote by i the inclusion functor $\widetilde{\mathcal{C}} \rightarrow \widehat{\mathcal{C}}$. The following theorem follows from Proposition 4.3.11:

Theorem 4.3.14. *There exists a unique functor $a : \widehat{\mathcal{C}} \rightarrow \widetilde{\mathcal{C}}$ such that the following diagram is commutative*

$$\begin{array}{ccc} \widehat{\mathcal{C}} & \xrightarrow{L} & \widehat{\mathcal{C}} \\ a \downarrow & & \downarrow L \\ \widetilde{\mathcal{C}} & \xrightarrow{i} & \widetilde{\mathcal{C}} \end{array}$$

i.e. for every presheaf P , $L(L(P))$ is a sheaf. The functors i, a are adjoint to one another: for every presheaf P and every sheaf F , one has an isomorphism functorial in P, F

$$\mathrm{Hom}_{\widehat{\mathcal{C}}}(P, i(F)) \simeq \mathrm{Hom}_{\widetilde{\mathcal{C}}}(a(P), F),$$

that is,

$$\mathrm{Hom}(P, F) \simeq \mathrm{Hom}(a(P), F).$$

Definition 4.3.15. The sheaf $a(P)$ is said to be *associated* to the presheaf P .

Remark 4.3.16. Since the functor L is constructed with the aid of projective limits and filtered inductive limits, it commutes with finite projective limits.⁽²⁹⁾ Moreover, if one identifies $L(P \times P)$ with $LP \times LP$, the morphism $\ell_{P \times P}$ is identified with $\ell_P \times \ell_P$. It follows, for example, that if P is a presheaf of groups, LP is also canonically endowed with the structure of a presheaf of groups and the canonical morphism $P \rightarrow LP$ is a morphism of groups. The same holds for the functor a , which shows that if P is a presheaf of groups and F a sheaf of groups, one has an isomorphism

$$\mathrm{Hom}_{\widehat{\mathcal{C}}\text{-gr.}}(P, i(F)) \simeq \mathrm{Hom}_{\widetilde{\mathcal{C}}\text{-gr.}}(a(P), F).$$

⁽²⁸⁾N.D.E. The statement of the corollary and its proof have been given in detail.

⁽²⁹⁾N.D.E. In particular, if K is the kernel of a pair of morphisms of presheaves $u, v : Q \rightrightarrows P$, then LK is the kernel of $Lu, Lv : LQ \rightrightarrows LP$ (this will be used in 4.4.5).

See [D] for more details.

4.3.17. — If $\mathcal{V}$ is any category, one calls a *presheaf on $\mathcal{C}$ with values in $\mathcal{V}$* a contravariant functor from $\mathcal{C}$ to $\mathcal{V}$. To define sheaves with values in $\mathcal{V}$, we must first recall the definition of the *projective limit* of a functor. If $\mathcal{R}, \mathcal{V}$ are two categories, and

$$F : \mathcal{R}^\circ \longrightarrow \mathcal{V}$$

is a contravariant functor from $\mathcal{R}$ to $\mathcal{V}$, one denotes by $\varprojlim F$ the object of $\widehat{\mathcal{V}}$ defined as follows:

$$(\varprojlim F)(X) = \text{Hom}_{\widehat{\mathcal{V}}}(X, \varprojlim F) = \varprojlim_{U \in \text{Ob } \mathcal{R}} \text{Hom}_{\mathcal{V}}(F(U), X) = \text{Hom}(c_X, F),$$

where X is a variable object of $\mathcal{V}$, where c_X denotes the contravariant functor from $\mathcal{R}$ to $\mathcal{V}$ which sends each object of $\mathcal{R}$ to X and each arrow of $\mathcal{R}$ to id_X , and where the last Hom is taken in the category $\text{Hom}(\mathcal{R}^\circ, \mathcal{V})$. If $\mathcal{R}$ possesses a final object $e_{\mathcal{R}}$, one has $\varprojlim F = F(e_{\mathcal{R}})$. If $\mathcal{V}$ is the category of sets, the functor $\varprojlim$ is identified with the functor Γ .

If S is an object of $\mathcal{C}$ and R a sieve of S , denote by $\mathcal{R}$ the full subcategory of $\mathcal{C}_{/S}$ whose set of objects is $E(R)$, and by $i_R : \mathcal{R} \longrightarrow \mathcal{C}_{/S}$ the canonical functor. If P is a presheaf on $\mathcal{C}$ with values in $\mathcal{V}$, it defines by restriction a functor $P_S : (\mathcal{C}_{/S})^\circ \longrightarrow \mathcal{V}$. The functor i_R induces a morphism of $\widehat{\mathcal{V}}$:

$$P(S) = P_S(S) = \varprojlim P_S \longrightarrow \varprojlim (P_S \circ i_R).$$

One denotes by $\varprojlim_R P$ the object $\varprojlim (P_S \circ i_R)$ of $\widehat{\mathcal{V}}$. By 4.1.6, Definition 4.3.1 generalizes to the

Definition 4.3.18. The presheaf P on $\mathcal{C}$ with values in $\mathcal{V}$ is said to be *separated* (resp. a *sheaf*) if, for every $S \in \text{Ob } \mathcal{C}$ and every $R \in J(S)$, the canonical morphism of $\widehat{\mathcal{V}}$

$$P(S) \longrightarrow \varprojlim_R P$$

is a monomorphism (resp. an isomorphism).

In the case where $\mathcal{V}$ is the category (Gr.) of groups (or any other category of sets endowed with algebraic structures defined by finite projective limits), one can see (cf. [D]) that there is an equivalence between the following notions: a presheaf on $\mathcal{C}$ with values in (Gr.) whose underlying presheaf of sets is a sheaf, and a group in the category of sheaves of sets. Taking these identifications into account, we shall always consider sheaves with values in a category of sets endowed with algebraic structures defined by finite projective limits as sheaves of sets endowed, in the category $\widetilde{\mathcal{C}}$, with the corresponding algebraic structure.

4.4. Exactness properties of the category of sheaves

Theorem 4.4.1. (i) *Arbitrary projective limits exist in $\widetilde{\mathcal{C}}$; “they are calculated in $\widehat{\mathcal{C}}$ ”, i.e. the inclusion functor $i : \widetilde{\mathcal{C}} \longrightarrow \widehat{\mathcal{C}}$ commutes with projective limits: if (X_α) is a projective system of sheaves, the presheaf*

$$\varprojlim i(X_\alpha) : S \longmapsto \varprojlim X_\alpha(S)$$

is a sheaf, and one has $i(\varprojlim X_\alpha) = \varprojlim i(X_\alpha)$.

(ii) *Arbitrary inductive limits exist in $\widetilde{\mathcal{C}}$: if (X_α) is an inductive system of sheaves, one has*

$$\varinjlim X_\alpha = a(\varinjlim i(X_\alpha)),$$

where $\varinjlim i(X_\alpha)$ is the inductive-limit presheaf of the $i(X_\alpha)$:

$$\varinjlim i(X_\alpha) : S \longmapsto \varinjlim X_\alpha(S).$$

(iii) The functor $a : \widehat{\mathcal{C}} \longrightarrow \widetilde{\mathcal{C}}$ commutes with arbitrary inductive limits and finite projective limits.

PROOF. Assertions (i) and (ii) follow formally from the adjunction formula (4.3.14), and⁽³⁰⁾ the first assertion of (iii) follows from (ii). Finally, the second assertion of (iii) has already been pointed out in 4.3.16. $\square$

Scholium 4.4.2. *This theorem allows one to use the following method to prove in $\widetilde{\mathcal{C}}$ an assertion concerning simultaneously arbitrary inductive limits and finite projective limits (for example: “every epimorphism is universal effective”, cf. below). One begins by proving the corresponding assertion in the category of sets, then extends it “argument by argument” to the category of presheaves. Next, one uses the preceding theorem to pass from the category of presheaves to the category of sheaves. One will see many examples of this method in what follows (4.4.3, 4.4.6, 4.4.9, etc.).*

Finally let us remark that the assertions concerning the category of presheaves are formally corollaries of the assertions concerning the category of sheaves. Indeed, it suffices to take as topology the least fine topology (“chaotic”), that is, the topology defined by $J(S) = \{S\}$ for every $S \in \text{Ob } \mathcal{C}$; every functor is indeed a sheaf for this topology.

Proposition 4.4.3. *Let $\mathcal{F} = \{F_i \longrightarrow F\}$ be a family of morphisms of sheaves. The following conditions are equivalent:*

- (i) $\mathcal{F}$ is an epimorphic family.
- (ii) $\mathcal{F}$ is a universal effective epimorphic family (1.13).
- (iii) $\mathcal{F}$ is covering (4.2.3).
- (iv) The image sheaf of $\mathcal{F}$ (that is, the sheaf associated to the image presheaf of $\mathcal{F}$ (4.1.2)) is F .

The equivalence of (iii) and (iv) follows from 4.3.12. The other equivalences will follow from the following lemmas.

Lemma 4.4.4. *Let $f : X \longrightarrow Y$ be a monomorphism of sheaves which is an epimorphism. Then f is an isomorphism.*

PROOF. The lemma is first clear in the category of sets. Let us next prove it in the category of presheaves. Consider the presheaf

$$V : S \longmapsto Y(S) \amalg^{X(S)} Y(S);$$

it is the amalgamated sum of Y and Y under X in the category of presheaves.⁽³¹⁾ Denote by i_1, i_2 the two “coordinate” morphisms $Y \longrightarrow V$. If $X \longrightarrow Y$ is an epimorphism in the category of presheaves, then $i_1 = i_2$. In this case, for each S , the map $X(S) \longrightarrow Y(S)$ is surjective; since it is also injective, it is a bijection, and thus $X \longrightarrow Y$ is an isomorphism.

Finally let us work in the category $\widetilde{\mathcal{C}}$ of sheaves. By 4.4.1 (ii), the amalgamated sum in $\widetilde{\mathcal{C}}$ of the sheaves Y, Y under X is the sheaf Z associated to the presheaf V . Consider the diagram of morphisms:

$$X \xrightarrow{f} Y \xrightarrow[i_2]{i_1} V \xrightarrow{\tau} Z = a(V).$$

One has $i_1 f = i_2 f$, hence $\tau i_1 f = \tau i_2 f$, and thus $\tau i_1 = \tau i_2$, since f is an epimorphism in $\widetilde{\mathcal{C}}$. By point (iii) of the lemma below, the presheaf V is separated, hence τ is a monomorphism (4.3.11 (iv)). Thus $i_1 = i_2$, and one has seen above that this implies that $f : X \longrightarrow Y$ is an isomorphism. This proves Lemma 4.4.4, once one has verified that V is separated. $\square$

⁽³⁰⁾**N.D.E.** The original has been modified here.

⁽³¹⁾**N.D.E.** The rest of the proof has been slightly modified.

(32) Let $R_\emptyset$ be the presheaf which associates the empty set to every $S \in \text{Ob } \mathcal{C}$; it is an initial object of $\widehat{\mathcal{C}}$.

Lemma 4.4.5. (i) Suppose that $R_\emptyset \notin J(S)$, for every $S \in \text{Ob } \mathcal{C}$. If (X_i) is a family of separated presheaves, the direct-sum presheaf $\coprod_i X_i$ is separated. (33)

(ii) Consider an equivalence relation in the category of presheaves:

$$X \xrightarrow[u]{u} Y$$

and let $w : Y \rightarrow Z$ be the quotient. If X is a sheaf and Y is separated, then Z is separated.

(iii) Consider an amalgamated sum in the category of presheaves, where u, u' are monomorphisms:

$$\begin{array}{ccc} & Y & \\ u \nearrow & & \searrow \\ X & & V \\ u' \searrow & & \nearrow \\ & Y' & \end{array} .$$

If Y, Y' are separated and X is a sheaf, then V is separated.

PROOF. (i) Set $X = \coprod_i X_i$. Let $S \in \text{Ob } \mathcal{C}$ and $\tau : R \hookrightarrow S$ be a refinement of S , and let x_1, x_2 be two elements of $X(S)$ such that $x_1 \circ \tau = x_2 \circ \tau$; there exist indices i, j such that $x_1 \in X_i(S)$, $x_2 \in X_j(S)$. Since $R \neq R_\emptyset$, there exists a morphism $\phi : T \rightarrow R$, with $T \in \text{Ob } \mathcal{C}$. Then $x_1 \circ \tau \circ \phi = x_2 \circ \tau \circ \phi$, and since $X(T)$ is the disjoint union of the $X_k(T)$, this implies $i = j$. Then, since X_i is separated and is a subobject of X , the map $X_i(S) \rightarrow X_i(R) \rightarrow X(R)$ is injective, hence $x_1 = x_2$. This proves that X is separated.

Let us prove (iii). Consider the morphisms $i : Y \rightarrow V$ and $j : Y' \rightarrow V$, and let K be the kernel of:

$$Y \times Y' \xrightarrow[p]{q} V ,$$

where $p = i \circ \text{pr}_1$, $q = j \circ \text{pr}_2$. Let $S \in \text{Ob } (\mathcal{C})$. Since u, u' are monomorphisms, one may identify $X(S)$ with its image in $Y(S)$ (resp. $Y'(S)$); denote by $Z(S)$ (resp. $Z'(S)$) the complement. Then $V(S)$ is identified with the disjoint union of $Z(S), Z'(S), X(S)$, and one easily sees that the maps $i(S) : Y(S) \rightarrow V(S)$, $j(S) : Y'(S) \rightarrow V(S)$ are injective, and that the map

$$(u \times u')(S) : X(S) \rightarrow K(S)$$

is bijective. Consequently i, j are monomorphisms, and $u \times u' : X \rightarrow K$ is an isomorphism.

Set $U = Y \amalg Y'$, and let $\tau : R \hookrightarrow S$ be a refinement of S ; one has a commutative diagram:

$$\begin{array}{ccc} U(S) & \xrightarrow{U(\tau)} & U(R) \\ \downarrow & & \downarrow \\ V(S) & \xrightarrow[V(\tau)]{} & V(R) \end{array} .$$

(32) **N.D.E.** Point (i) of Lemma 4.4.5 has been corrected, and the proofs of the three points have been given in detail.

(33) **N.D.E.** In general, the direct sum of two sheaves F, G is not a sheaf. Indeed, let $S_1, S_2 \in \text{Ob } \mathcal{C}$; one assumes that the direct sum $S = S_1 \amalg S_2$ exists in $\mathcal{C}$ and that the fibered product $S_1 \times_S S_2$ is an initial object $\emptyset$ of $\mathcal{C}$ (cf. I, 1.8). Let R be the sieve of S with basis $\{S_1, S_2\}$; then $(F \amalg G)(R)$ is the disjoint union of $F(S) \amalg G(S)$ and of $F(S_i) \times G(S_j)$ for $i \neq j$, hence $F \amalg G$ is not a sheaf in general. On the other hand, if $\mathcal{C}$ is the category with a single object S and id_S as its only morphism, endowed with the topology defined by $J(S) = \{R_\emptyset, S\}$, then the only separated presheaves are $R_\emptyset$ and $X = h_S$ (which is a sheaf), and $X \amalg X$ is not separated.

Let $v_1, v_2 \in V(S)$ have the same image in $V(R)$. By definition of V , v_1, v_2 lift to elements y_1, y_2 of $U(S)$; denote by z_1, z_2 their images in $U(R)$. Then z_1, z_2 have the same image in $V(R)$.

Since $i : Y \rightarrow V$, $j : Y' \rightarrow V$ are monomorphisms, the maps $Y(R) \rightarrow V(R)$, $Y'(R) \rightarrow V(R)$ are injective. Thus, since Y, Y' are separated, if y_1, y_2 both belong to $Y(S)$ or to $Y'(S)$, then $y_1 = y_2$. Otherwise one may assume that $y_1 \in Y(S)$, $y_2 \in Y'(S)$, hence $z_1 \in Y(R)$, $z_2 \in Y'(R)$. But then, since $i \circ z_1 = j \circ z_2$, the morphism $z_1 \boxtimes z_2 : R \rightarrow Y \times Y'$ factors through $K = X$. Since moreover $X(R) = X(S)$ (because X is a sheaf), there exists $x \in X(S)$ such that $u(x) \circ \tau = z_1 = y_1 \circ \tau$ and $u'(x) \circ \tau = z_2 = y_2 \circ \tau$, whence, since Y, Y' are separated, $u(x) = y_1$, $u'(x) = y_2$, and thus $v_1 = v_2$. This proves that V is separated.

Let us prove (ii). First remark that the morphism of presheaves

$$X \xrightarrow{u \boxtimes v} K,$$

where K denotes the kernel of the pair of morphisms $w \circ \text{pr}_i : Y \times Y \rightarrow Z$, is an isomorphism. Indeed, for every $T \in \text{Ob } \mathcal{C}$, $X(T)$ is an equivalence relation in $Y(T)$, so that the diagram

$$X(T) \xrightarrow{u \boxtimes v} Y(T) \times Y(T) \xrightarrow[w \circ \text{pr}_2]{w \circ \text{pr}_1} Z(T)$$

is exact in the category of sets.

Now let $S \in \text{Ob } \mathcal{C}$ and $g_1, g_2 : S \rightarrow Z$ be two morphisms which agree on a refinement $\tau : R \hookrightarrow S$ of S . Since $S \in \text{Ob } \mathcal{C}$, one has $Z(S) = Y(S)/X(S)$, by construction of Z , and hence there exist morphisms $f_1, f_2 : S \rightarrow Y$ such that $wf_i = g_i$. Then $wf_1\tau = wf_2\tau$, hence, by the preceding argument, there exists a morphism $\phi : R \rightarrow X$ such that $u\phi = f_1\tau$, $v\phi = f_2\tau$. Since X is a sheaf, there exists $\psi : S \rightarrow X$ such that $\phi = \psi\tau$, and one therefore has in $Y(R)$ the equalities:

$$u\psi\tau = f_1\tau, \quad v\psi\tau = f_2\tau.$$

Since $Y(S) \rightarrow Y(R)$ is injective (Y being separated), this implies $u\psi = f_1$, $v\psi = f_2$, whence $g_1 = g_2$. This proves that Z is separated. $\square$

Lemma 4.4.6. ⁽³⁴⁾ (i) Let $\{F_i \xrightarrow{f_i} F\}$ be a family of morphisms of presheaves, and let $G \subset F$ be the image presheaf. Then the following diagram in $\hat{\mathcal{C}}$ is exact:

$$(\coprod_i F_i) \times_G (\coprod_j F_j) \xrightarrow[\text{pr}_2]{\text{pr}_1} \coprod_i F_i \longrightarrow G$$

that is, for every presheaf H , the following diagram of sets is exact:

$$(*) \quad \text{Hom}(G, H) \longrightarrow \prod_i \text{Hom}(F_i, H) \xrightarrow[p]{q} \prod_{i,j} \text{Hom}(F_i \times_G F_j, H).$$

(ii) Every covering family of morphisms of sheaves is universal effective epimorphic.

PROOF. (i) Let H be a presheaf. The map f^* which associates to a morphism $\phi : G \rightarrow H$ the family of morphisms $\phi \circ f_i : F_i \rightarrow H$ is injective, for, for every $S \in \text{Ob } \mathcal{C}$, $\phi(S)$ is determined by the $(\phi \circ f_i)(S)$, since the family $f_i(S) : F_i(S) \rightarrow G(S)$ is surjective. It is clear that the image of f^* is contained in $\text{Ker}(p, q)$. Conversely, let $\phi_i : F_i \rightarrow H$ be a family of morphisms such that, for every i, j , the diagram below is commutative:

$$\begin{array}{ccc} F_i \times_G F_j & \longrightarrow & F_j \\ \downarrow & & \downarrow \phi_j \\ F_i & \xrightarrow{\phi_i} & H \end{array}$$

⁽³⁴⁾ **N.D.E.** The statement of the lemma and its proof have been given in detail.

Then, for every S , the map $\coprod_i F_i(S) \rightarrow H(S)$ factors uniquely through a map $\phi(S) : G(S) \rightarrow H(S)$, and this defines a morphism $\phi : G \rightarrow H$ such that $f^*(\phi) = (\phi_i)$. This proves the exactness of sequence (*), and point (i).

Let us prove (ii). Since the notion of covering family is stable under base extension, it suffices to show that every covering family is effective epimorphic. Thus let $\{F_i \rightarrow F\}$ be a covering family of morphisms of sheaves, and let G be the image presheaf of this family. Since the family is covering, the monomorphism $G \hookrightarrow F$ is too, hence, by 4.3.12, one has $a(G) = F$.

On the other hand, since $G \hookrightarrow F$ is a monomorphism, the fibered products $F_i \times_G F_j$, $F_i \times_F F_j$ are the same. Thus, by (i), the following diagram of sets is exact, for every presheaf H :

$$(**) \quad \text{Hom}(G, H) \longrightarrow \prod_i \text{Hom}(F_i, H) \xrightarrow[p]{q} \prod_{i,j} \text{Hom}(F_i \times_F F_j, H).$$

If H is moreover a sheaf, one has

$$\text{Hom}(G, H) = \text{Hom}(a(G), H) = \text{Hom}(F, H),$$

and then (**) shows that $\{F_i \rightarrow F\}$ is indeed an effective epimorphic family in the category $\tilde{\mathcal{C}}$ of sheaves. $\square$

Lemma 4.4.7. *Every family of morphisms of sheaves $\{F_i \rightarrow F\}$ factors into a covering family $\{F_i \rightarrow G\}$ and a monomorphism $G \rightarrow F$.*

PROOF. Indeed, it suffices to take for G the image sheaf of the given family. $\square$

PROOF OF PROPOSITION 4.4.3. One has seen in 4.4.6 that (iii) implies (ii), and evidently (ii) implies (i). Finally let $\{F_i \rightarrow F\}$ be an epimorphic family; by Lemma 4.4.7, it factors into a covering family followed by a monomorphism. But the latter, being dominated ⁽³⁵⁾ by an epimorphic family, is an epimorphism, hence an isomorphism by 4.4.4. $\square$

Remark 4.4.8. ⁽³⁶⁾ Since the image presheaf of the family $\mathcal{F}$ is separated, the construction of the associated sheaf shows that the conditions of Proposition 4.4.3 are also equivalent to the following:

- (v) For every $S \in \text{Ob } \mathcal{C}$, every $f \in F(S)$ is locally in the image of $\mathcal{F}$, that is:
- (vi) For every $S \in \text{Ob } \mathcal{C}$ and every $f \in F(S)$, the set of $S' \rightarrow S$ such that the image of f in $F(S')$ is in the image of one of the $F_i(S')$ is a refinement of S .
- (vii) (If the topology is defined by a pretopology R .) For every $S \in \text{Ob } \mathcal{C}$ and every $f \in F(S)$, there exists a family $\{S_j \rightarrow S\} \in R(S)$ such that for every j the image f_j of f in $F(S_j)$ is in the image of one of the $F_i(S_j)$.

Remark 4.4.8.bis. If the sheaf F is representable, the preceding conditions are also equivalent to:

- (viii) The set of $T \rightarrow F$ ($T \in \text{Ob } \mathcal{C}$) such that there exist an i and a commutative diagram

$$\begin{array}{ccc} F_i & \longrightarrow & F \\ \uparrow & \nearrow & \\ T & & \end{array}$$

is a refinement of F .

Indeed, if (viii) is satisfied, the image presheaf of the F_i is dominated by ⁽³⁷⁾ a refinement of F , which implies that the family is covering. Conversely, one applies (vi) to $\text{id}_F \in F(F)$.

⁽³⁵⁾**N.D.E.** We recall (cf. N.D.E. (17)) that a morphism $g : G \rightarrow H$ is said to be dominated by a family of morphisms $F_i \rightarrow H$ if this family factors through g .

⁽³⁶⁾**N.D.E.** To agree with later references, the numbering of the original, which had two nos. 4.4.7, has been corrected.

⁽³⁷⁾**N.D.E.** “majorizes” has been replaced by “is dominated by”, cf. N.D.E. (17).

This condition is expressed in figurative language as follows: locally on F , there exists an i such that $F_i \rightarrow F$ possesses a section. In particular a morphism $G \rightarrow F$, where G is a sheaf and F a representable sheaf, will be covering if and only if it possesses a section locally (on F).

(38) The following lemmas will be useful in VIB, 8.1 and 8.2.

Lemma 4.4.8.1. *Let $Q \subset P$ be presheaves of groups on $\mathcal{C}$, with Q normal in P . Suppose that P is separated and that Q is a sheaf. Then the presheaf of groups P/Q is separated. Consequently one has*

$$a(P/Q) = L(P/Q) = \varinjlim_{R \in J(S)} (P/Q)(R).$$

PROOF. It suffices to show the first assertion, for the second follows from it by 4.3.11 (iv) and (ii). Let $S \in \text{Ob } \mathcal{C}$, let R be a sieve of S , and let $y \in P(S)/Q(S)$ have the identity as its image in $(P/Q)(R)$. One must show that $y = 1$. Now, by hypothesis, the morphism $P(S) \rightarrow P(R)$ (resp. $Q(S) \rightarrow Q(R)$) is injective (resp. an isomorphism), and in the commutative diagram below the upper row is exact:

$$\begin{array}{ccccccc} 1 & \longrightarrow & Q(S) & \longrightarrow & P(S) & \longrightarrow & P(S)/Q(S) \longrightarrow 1 \\ & & \simeq \downarrow & & \downarrow & & \downarrow \\ 1 & \longrightarrow & Q(R) & \longrightarrow & P(R) & \longrightarrow & (P/Q)(R) \end{array}$$

The result will follow if one shows that the lower row is exact. Let $f : R \rightarrow P$ have the identity as its image in $(P/Q)(R)$, and let $\phi : T \rightarrow R$, with $T \in \text{Ob } \mathcal{C}$. Then $f \circ \phi$ is the identity of $(P/Q)(T) = P(T)/Q(T)$, i.e. $f \circ \phi \in Q(T)$. Thus $\phi \mapsto f \circ \phi$ is a functorial map $R(T) \rightarrow Q(T)$, hence defines a morphism of functors $\pi : R \rightarrow Q$ such that $\pi(\text{id}_R) = f$, whence $f \in Q(R)$. This proves the exactness of the lower row, and the lemma is proved. $\square$

Lemma 4.4.8.2. *Let $H \subset G$ be presheaves of groups on $\mathcal{C}$.*

- (i) *If H is normal in G , then $L(H)$ is normal in $L(G)$.*
- (ii) *If H is central in G , then $L(H)$ is central in $L(G)$.*

PROOF. Let $S \in \text{Ob } \mathcal{C}$; one must show (cf. I 2.3.6) that $L(H)(S)$ is normal (resp. central) in $L(G)(S)$. Let $h \in L(H)(S)$, $g \in L(G)(S)$; there exist a sieve R of S and elements $h' \in H(R)$, $g' \in G(R)$ such that $h = z_R(h')$, $g = z_R(g')$ (notation of 4.3.10). Since z_R is a morphism of groups, $ghg^{-1}h^{-1} = z_R(g'h'g'^{-1}h'^{-1})$. In case (i), one has $g'h'g'^{-1}h'^{-1} \in H(R)$, whence $ghg^{-1}h^{-1} \in LH(S)$; in case (ii), $g'h'g'^{-1}h'^{-1} = 1$, hence $ghg^{-1}h^{-1} = 1$. $\square$

Proposition 4.4.9. *Every equivalence relation in $\tilde{\mathcal{C}}$ is universal effective (3.3.3): let R be a $\tilde{\mathcal{C}}$ -equivalence relation in the sheaf X ; then the sheaf associated to the separated presheaf*

$$i(X)/i(R) : S \mapsto X(S)/R(S)$$

is a universal effective quotient of X by R .

PROOF. Let X/R be the quotient sheaf of X by R , which exists by 4.4.1 (ii): $X/R = a(i(X)/i(R))$. We must show that $X \rightarrow X/R$ is a universal effective epimorphism, and that the morphism $f : R \rightarrow X \times_{X/R} X$ is an isomorphism. The first assertion has already been proved (4.4.3). As for f , it comes, by application of the functor a , from the morphism $i(R) \rightarrow i(X) \times_{i(X)/i(R)} i(X)$ or, since $i(X)/i(R)$ is separated (4.4.5 (ii)), so that $i(X)/i(R) \rightarrow i(X/R)$ is a monomorphism, from the canonical morphism $i(R) \rightarrow i(X) \times_{i(X)/i(R)} i(X)$.

One is therefore reduced to proving the same assertion in the category of presheaves. But $i(X)/i(R)$ is the presheaf $S \mapsto X(S)/R(S)$, and one is reduced to proving the analogous assertion in the category of sets, where it is immediate. $\square$

(38) **N.D.E.** Lemmas 4.4.8.1 and 4.4.8.2 have been added.

Proposition 4.4.10. *Under the conditions of 4.4.9, let Y be a subsheaf of X . Denote by R_Y the equivalence relation induced in Y by R . Then the canonical morphism (3.1.6)*

$$Y/R_Y \longrightarrow X/R$$

is a monomorphism: it identifies Y/R_Y with a subsheaf of X/R , which is the image sheaf of the composite morphism

$$Y \longrightarrow X \longrightarrow X/R.$$

PROOF. The morphism of presheaves

$$i(Y)/i(R_Y) = i(Y)/i(R)_{i(Y)} \longrightarrow i(X)/i(R)$$

is a monomorphism. Since the functor a is left exact (4.3.16), it transforms monomorphisms into monomorphisms, and thus

$$Y/R_Y \longrightarrow X/R$$

is a monomorphism. The last assertion follows from the commutative diagram

$$\begin{array}{ccc} Y & \longrightarrow & X \\ \downarrow & & \downarrow \\ Y/R_Y & \longrightarrow & X/R \end{array},$$

and the fact that $Y \longrightarrow Y/R_Y$ is covering. $\square$

By virtue of this proposition, we shall always identify Y/R_Y with a subsheaf of X/R .

Proposition 4.4.11. *Let R be a $\tilde{C}$ -equivalence relation in the sheaf X . For every subsheaf Y of X stable under R , denote by Y' the quotient Y/R_Y , considered as a subsheaf of $X' = X/R$. Then $Y = Y' \times_{X'} X$, and the maps $Y \mapsto Y/R_Y$ and $Y' \mapsto Y' \times_{X'} X$ establish a bijective correspondence between the set of subsheaves Y of X stable under R and the set of subsheaves Y' of X' .*

PROOF. If Y' is a subsheaf of X' , then $Y' \times_{X'} X$ is a subsheaf of X stable under R . ⁽³⁹⁾ If Y' is obtained by passage to the quotient from a subsheaf Y of X , then Y is a subobject of $Y' \times_{X'} X$. It therefore suffices to show that if one has two subsheaves Y, Y_1 of X , stable under R , with Y_1 containing Y , and if the quotients $Y/R_Y, Y_1/R_{Y_1}$ are identical, then $Y = Y_1$. One is evidently reduced to proving the same assertion in the case where $Y_1 = X$. Then denoting by P (resp. Q) the presheaf $i(X)/i(R)$ (resp. $i(Y)/i(R_Y)$), the diagram

$$\begin{array}{ccc} X & \longrightarrow & P \\ \uparrow & & \uparrow \\ Y & \longrightarrow & Q \end{array}$$

is cartesian. Since one has a commutative diagram

$$\begin{array}{ccc} P & \hookrightarrow & a(P) \\ \uparrow & & \uparrow \\ Q & \hookrightarrow & a(Q) \end{array}$$

and since $Q \hookrightarrow a(Q)$ is covering (4.3.11), the monomorphism $Q \hookrightarrow P$ is covering, hence Q is a refinement of P . By base change, Y is a refinement of X . Since X, Y are sheaves, this implies (4.3.12) $Y = X$. $\square$

⁽³⁹⁾**N.D.E.** And one has $(Y' \times_{X'} X)/R = Y'$.

4.4.12. — In particular, if Y is a subsheaf of X , and if $Y' = Y/R_Y$, then the preceding correspondence defines a subsheaf $\bar{Y}$ of X , stable under R , containing Y , and minimal for these properties, which one calls the *saturation* of Y for the equivalence relation R .

4.5. The case of a topology less fine than the canonical topology

By 4.3.6 and 4.3.8, the following conditions are equivalent for a topology $\mathcal{T}$ on $\mathcal{C}$:

- (i) $\mathcal{T}$ is less fine than the canonical topology of $\mathcal{C}$.
- (ii) Every representable presheaf is a sheaf for $\mathcal{T}$.
- (iii) Every covering sieve for $\mathcal{T}$ is universal effective epimorphic.

If $\mathcal{T}$ is defined by a pretopology $S \mapsto R(S)$, these conditions are also equivalent to

- (iv) Every family belonging to $R(S)$ is universal effective epimorphic.

In the case where these conditions hold, the canonical functor $\mathcal{C} \rightarrow \hat{\mathcal{C}}$ factors through a functor $j_{\mathcal{C}} = j : \mathcal{C} \rightarrow \tilde{\mathcal{C}}$ (one will also write $j(S) = \tilde{S}$).⁽⁴⁰⁾

Proposition 4.5.1. *The functor $j : \mathcal{C} \rightarrow \tilde{\mathcal{C}}$ is fully faithful and commutes with arbitrary projective limits. In particular, it is left exact, and therefore preserves algebraic structures defined by finite projective limits.*

PROOF. This follows immediately by considering the commutative diagram

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow{\quad} & \hat{\mathcal{C}} \\ & \searrow j & \nearrow i \\ & \tilde{\mathcal{C}} & \end{array}$$

and from 4.4.1 (i). □

Before exhibiting other properties of the functor j , we must define the topology induced on a category $\mathcal{C}_{/S}$. No longer necessarily assuming that the given topology is less fine than the canonical topology, one does this as follows: if C is a sieve of T in $\mathcal{C}$ and if one has a morphism $T \rightarrow S$, then C naturally defines a sieve of T in $\mathcal{C}_{/S}$, denoted $J(T)$ (for the definition of a sieve of T depends only on the category $\mathcal{C}_{/T} = (\mathcal{C}_{/S})_{/T}$). If, for example, C is defined by the family $\{T_i \rightarrow T\}$, then its image in $\mathcal{C}_{/S}$ is defined by the same family considered as a family of morphisms of $\mathcal{C}_{/S}$. This said, the map $T \mapsto J(T)$ defines a topology on $\mathcal{C}_{/S}$, called the topology induced by the given topology. With the definitions of [AS], 2.3, this is the least fine topology on $\mathcal{C}_{/S}$ for which the canonical functor

$$i_S : \mathcal{C}_{/S} \rightarrow \mathcal{C}$$

is a comorphism.⁽⁴¹⁾ One will remark that the identifications

$$(\mathcal{C}_{/S})_{/T} = \mathcal{C}_{/T}$$

respect the topologies by definition.

Proposition 4.5.2. *Let S be a representable sheaf on $\mathcal{C}$ and $F \rightarrow S$ a morphism of $\hat{\mathcal{C}}$. For $S' \mapsto \text{Hom}_S(S', F)$ to be a separated presheaf (resp. a sheaf) on $\mathcal{C}_{/S}$, it is necessary and sufficient that F be a separated presheaf (resp. a sheaf) on $\mathcal{C}$.*

PROOF. For every functor P , one has (I 1.4.1)

$$\text{Hom}(P, F) = \coprod_{f \in \text{Hom}(P, S)} \text{Hom}_f(P, F).$$

⁽⁴⁰⁾**N.D.E.** And one will also write $h_S = \hat{S}$, cf. the first commutative diagram of 4.5.4.

⁽⁴¹⁾**N.D.E.** That is, such that for every object $T \rightarrow S$ of $\mathcal{C}_{/S}$, every covering sieve R' of $i_S(T \rightarrow S) = T$, considered as a sieve of $(T \rightarrow S) \in \text{Ob } \mathcal{C}_{/S}$, is covering.

$$\mathrm{Hom}_{\hat{C}}(S', F) \longrightarrow \mathrm{Hom}_{\hat{C}}(C', F)$$
$$\mathrm{Hom}_f(S', F) \longrightarrow \mathrm{Hom}_{f \circ \eta}(C', F).$$

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Corollary 4.5.4. *Suppose that the given topology on $\mathcal{C}$ is less fine than the canonical topology. For every $S \in \text{Ob } \mathcal{C}$, one has an equivalence of categories*

$$\tilde{\mathcal{C}}_{/\tilde{S}} \longrightarrow \widetilde{\mathcal{C}}_{/S}.$$

The following diagrams are commutative up to isomorphism (all unlabeled arrows are equivalences):

$$\begin{array}{ccccccc}
\mathcal{C}_{/S} & \xrightarrow{(jc)/s} & \widetilde{\mathcal{C}}_{/\widetilde{S}} & \xrightarrow{(ic)/\widetilde{f}_S} & \widehat{\mathcal{C}}_{/\widehat{S}} & \xrightarrow{(ac)/\widehat{f}_S} & \widetilde{\mathcal{C}}_{/\widetilde{S}} \\
\parallel & & \downarrow \simeq & & \downarrow \simeq & & \downarrow \simeq \\
\mathcal{C}_{/S} & \xrightarrow{j_{\mathcal{C}/S}} & \widetilde{\mathcal{C}}_{/S} & \xrightarrow{i_{\mathcal{C}/S}} & \widehat{\mathcal{C}}_{/S} & \xrightarrow{a_{\mathcal{C}/S}} & \widetilde{\mathcal{C}}_{/S} \\
(\widetilde{\mathcal{C}}_{/\widetilde{S}})_{/\widetilde{T}} & \longrightarrow & (\widetilde{\mathcal{C}}_{/S})_{/\widetilde{T}} & & & & \\
\downarrow \simeq & & & \searrow & & & \\
\widetilde{\mathcal{C}}_{/\widetilde{T}} & \longrightarrow & \widetilde{\mathcal{C}}_{/T} & \xrightarrow{\sim} & \widetilde{(\mathcal{C}_{/S})_{/T}} & &
\end{array}$$

PROOF. The commutativity of the first two squares follows from the definition of the equivalence $\widetilde{\mathcal{C}}_{/\widetilde{S}} \rightarrow \widetilde{\mathcal{C}}_{/S}$. To prove the commutativity of the last, one must see that the following square is commutative:

$$\begin{array}{ccc} \widehat{\mathcal{C}}_{/\widehat{S}} & \xrightarrow{L'} & \widehat{\mathcal{C}}_{/\widehat{S}}, \\ \downarrow & & \downarrow \\ \widehat{\mathcal{C}}_{/S} & \xrightarrow{L''} & \widehat{\mathcal{C}}_{/S} \end{array}$$

where L' is the restriction of the functor $L_{\mathcal{C}}$ to $\widehat{\mathcal{C}}_{/\widehat{S}}$, and L'' the functor $L_{\mathcal{C}/S}$; this is easily seen by returning to the definition of the functors L (given after 4.3.9). As for the second diagram, it is simply the restriction to the categories of sheaves of the corresponding diagram for the categories of presheaves (Exposé I, no. 1), which is commutative. $\square$

Scholium 4.5.5. *The various assertions of this number show that in the case where the given topology is less fine than the canonical topology, one may identify $\mathcal{C}$ with a full subcategory of $\widehat{\mathcal{C}}$, itself a full subcategory of $\widehat{\widehat{\mathcal{C}}}$, and that in this identification one may indulge in the customary abuses of language concerning $\mathcal{C} \longrightarrow \widehat{\widehat{\mathcal{C}}}$, justified by the preceding commutativities. Let us remark explicitly that the first diagram of 4.5.4 shows that one may use the functor α without special precautions.*

⁽⁴²⁾**N.D.E.** The original has been given in detail in what follows.

We shall see in the next number that the identification of $\mathcal{C}$ with a full subcategory of $\tilde{\mathcal{C}}$ (contrary to what happened for $\widehat{\mathcal{C}}$) commutes with the formation of certain inductive limits, and we shall then say how to use this fact.

From now on, unless expressly stated otherwise, we shall assume that the given topology is less fine than the canonical topology, and we shall systematically make the identifications set out above.

Proposition 4.5.6. *Let F, G be two sheaves over S , and $f : F \rightarrow G$ an S -morphism. The following conditions on f are equivalent:*

- (i) f is a monomorphism (resp. an epimorphism, resp. an isomorphism) in $\tilde{\mathcal{C}}$.
- (ii) f is a monomorphism (resp. an epimorphism, resp. an isomorphism) in $\tilde{\mathcal{C}}_S = \widetilde{\mathcal{C}_S}$.

PROOF. For monomorphisms and isomorphisms this is evident (it is a question of presheaves). For epimorphisms, this follows from the description of epimorphisms as covering morphisms (4.4.3) and the fact that, by definition of the induced topology, these are the same in $\tilde{\mathcal{C}}$ and $\tilde{\mathcal{C}}_S$. $\square$

Proposition 4.5.7. *Let $f : F \rightarrow G$ be a morphism of sheaves. The following conditions are equivalent:*

- (i) f is a monomorphism (resp. an epimorphism, resp. an isomorphism).
- (ii) For each $S \in \text{Ob } \mathcal{C}$, $f_S : F_S \rightarrow G_S$ is locally a monomorphism (resp. an epimorphism, resp. an isomorphism), that is:
- (iii) For each $S \in \text{Ob } \mathcal{C}$, the set of $T \rightarrow S$ such that $F_T \rightarrow G_T$ is a monomorphism (resp. an epimorphism, resp. an isomorphism) is a refinement of S .

If the given topology is defined by a pretopology R , these conditions are also equivalent to the following:

- (iv) For each $S \in \text{Ob } \mathcal{C}$, there exists a covering family $\{S_i \rightarrow S\} \in R(S)$ such that for every i , $F_{S_i} \rightarrow G_{S_i}$ is a monomorphism (resp. an epimorphism, resp. an isomorphism).

If the category $\mathcal{C}$ possesses a final object e , one may restrict oneself to taking $S = e$ in conditions (ii), (iii), (iv).

PROOF. One evidently has (ii) $\Leftrightarrow$ (iii) $\Leftrightarrow$ (iv). To prove the equivalence of (i) and (ii), as well as the supplement concerning the final object, one must show that (ii) implies (i) and that the notions considered are stable under base extension. Let us first prove the latter point. For monomorphisms and isomorphisms, this is evident (it is a question of presheaves). For epimorphisms, it follows from the fact that every epimorphism of sheaves is universal (4.4.3).

Finally let us show that (ii) implies (i). First suppose that $f_S : F_S \rightarrow G_S$ is locally a monomorphism (resp. an isomorphism). There then exists a covering sieve C of S such that for every $T \rightarrow C$, f_T is a monomorphism (resp. an isomorphism). Since a projective limit of monomorphisms (resp. isomorphisms) is one, $\text{Hom}(C, f)$ will be a monomorphism (resp. an isomorphism) (cf. 4.1.4). The commutative diagram

$$\begin{array}{ccc} \text{Hom}(C, F) & \xrightarrow{\text{Hom}(C, f)} & \text{Hom}(C, G) \\ \uparrow \simeq & & \uparrow \simeq \\ F(S) & \xrightarrow{f(S)} & G(S) \end{array}$$

then shows that $f(S)$ is injective (resp. bijective). Finally suppose that $f : F \rightarrow G$ is locally an epimorphism, and let $H \subset G$ be its image. For each $S \rightarrow G$, $H \times_G S$ is the image of $f \times_G S : F \times_G S \rightarrow S$. To show that f is an epimorphism, one must show that H is a refinement of G , that is, $H \times_G S$ is a refinement of S for each S . But since this is so after every base change $T \rightarrow S$ from a refinement of S (since $f \times_G S$ is locally covering), $H \times_G S$ is indeed a refinement of S (Axiom (T 2)). $\square$

Corollary 4.5.8. *Let F, G be two sheaves over S , and $f : F \rightarrow G$ an S -morphism. For f to be a monomorphism, resp. an epimorphism, resp. an isomorphism, it is necessary and sufficient that it be so locally on S .*

Remark 4.5.9. The proof of the proposition shows that it remains valid, for the part concerning monomorphisms (resp. isomorphisms), when one assumes only that F is a separated presheaf (resp. a sheaf) and G is any presheaf (resp. a separated presheaf).

Let us temporarily return to the case of an arbitrary topology and give a definition.

Definition 4.5.10. Let $G \rightarrow F$ be a morphism of $\widehat{\mathcal{C}}$. One says that G is a *relative sheaf over F* if for each F -functor H and each refinement R of H , the canonical map

$$(+) \quad \text{Hom}_F(H, G) \xrightarrow{\sim} \text{Hom}_F(R, G)$$

is bijective.

Proposition 4.5.2 generalizes at once:

Proposition 4.5.11. *If F is a sheaf, G is a relative sheaf over F if and only if it is a sheaf.*

Lemma 4.5.12. *In the situation $X \rightarrow T \rightarrow S$ (where X, T, S are three objects of $\widehat{\mathcal{C}}$), if X is a relative sheaf over T , then $U = \prod_{T/S} X$ is a relative sheaf over S .*

PROOF. Indeed, for every S -functor Y , one has

$$\text{Hom}_S(Y, U) = \text{Hom}_T(T \times_S Y, X).$$

If C is a sieve of Y , then $T \times_S C$ is a sieve of $T \times_S Y$; one concludes at once. $\square$

Corollary 4.5.13. *The presheaves $\underline{\text{Hom}}_{T/S}(X, Y)$, $\underline{\text{Isom}}_S(X, Y)$, etc., are sheaves when the arguments which occur in them are also sheaves.*

PROOF. Indeed, all these presheaves are constructed with the aid of fibered products and presheaves $\prod$ (I 1.7 and II 1). It therefore suffices to verify the result for a presheaf $\prod_{T/S} X$; in this case, the assertion follows from 4.5.11 and 4.5.12. $\square$

4.6. Description of the quotient of a sheaf by an equivalence relation

Recall that we assume that the given topology $\mathcal{T}$ is less fine than the canonical topology.

Proposition 4.6.1. *Let $R \xrightarrow{p_1, p_2} X$ be a $\widetilde{\mathcal{C}}$ -equivalence relation in the sheaf X . Let $F \in \text{Ob } \widehat{\mathcal{C}}$ be defined as follows: for each S of $\mathcal{C}$,*

$$F(S) = \left\{ \begin{array}{l} \text{sub-}S\text{-sheaves } Z \text{ of } X_S \text{ stable under } R \times S, \\ \text{whose quotient by } R_Z \text{ is } S, \text{ that is, such that} \\ \text{the diagram } R_Z \rightrightarrows Z \rightarrow S \text{ is exact} \end{array} \right\}.$$

(43) *Then, for every sheaf Y , $\text{Hom}(Y, F)$ is identified with the set:*

$$\left\{ \begin{array}{l} \text{sub-}Y\text{-sheaves of } X \times Y \text{ stable under } R \times Y \\ \text{and whose quotient is } Y \end{array} \right\}.$$

In particular, the subsheaf R of $X \times X$ corresponds to an element p of $\text{Hom}(X, F)$, and the diagram

$$R \xrightarrow[p_2]{p_1} X \xrightarrow{p} F$$

is exact, hence identifies F with the quotient sheaf X/R .

(43) **N.D.E.** $R \times S$ denotes the equivalence relation in $X \times S$ defined by $R \times S_{\text{diagonal}} \subset X \times X \times S \times S$, and R_Z is the equivalence relation which it induces in Z (cf. 3.1.6).

PROOF. Indeed, set $Q = X/R$. For every sheaf Y and every morphism $f \in \text{Hom}(Y, Q)$ corresponding to a section $s : Y \hookrightarrow Q \times Y$, consider the diagram

$$(*) \quad \begin{array}{ccccc} R \times Y & \rightrightarrows & X \times Y & \longrightarrow & Q \times Y \\ & & \uparrow & & \uparrow s \\ & & Z & \longrightarrow & Y \end{array}$$

Here the square is cartesian. It is immediate by 4.4.11 that Z is a sub- Y -sheaf of $X \times Y$, stable under $R \times Y$, whose quotient is Y , and that conversely, every Z of this type comes from a unique section of $Q \times Y$ over Y . Taking Y representable first, one deduces an isomorphism $Q \simeq F$. Taking arbitrary Y next, one deduces the announced form of $\text{Hom}(Y, F)$. Finally considering the canonical morphism $X \rightarrow Q$, one sees at once that it corresponds to the sub- X -sheaf R of $X \times X$, which completes the proof. $\square$

Corollary 4.6.2. *Let G be any subfunctor of F such that $\text{Hom}(X, G) \subset \text{Hom}(X, F)$ contains R . Then the canonical morphism $p : X \rightarrow F$ factors through G . Since p is covering (4.4.9 and 4.4.3), it follows that G is a refinement of F . In particular, every subsheaf G of F satisfying the preceding condition is equal to F (4.3.12).*

4.6.3. — We shall now concern ourselves with the case where X, R are representable. Let us first introduce some terminology. Besides conditions (a) to (d) introduced in 3.4.1, we shall use other conditions on a family (M) of morphisms of $\mathcal{C}$, which we state below, recalling conditions (a) to (c) already given, for completeness.

- (a) (M) is stable under base extension.
- (b) The composite of two elements of (M) is in (M).
- (c) Every isomorphism is an element of (M).
- (d $\mathcal{T}$) Every element of (M) is covering. ⁽⁴⁴⁾
- (e $\mathcal{T}$) Let $f : X \rightarrow Y$ be a morphism of $\mathcal{C}$. If there exists a refinement R of Y such that for every $Y' \rightarrow R$, $X \times_Y Y' \rightarrow Y'$ is an element of (M), then f is an element of (M).

Recall that (a) and (b) imply

- (a') The cartesian product of two elements of (M) is an element of (M).

On the other hand, (a) and (d $\mathcal{T}$) imply, by 4.3.9:

- (d') Every element of (M) is a universal effective epimorphism.

4.6.4. — The preceding conditions are satisfied by the family of covering morphisms, denoted (M $\mathcal{T}$), when $\mathcal{C}$ possesses fibered products. Indeed (cf. 4.2.3), (a) follows from (C 1), (b) from (C 2), (c) from (C 4), (d $\mathcal{T}$) from the definition, and (e $\mathcal{T}$) from (C 5). The results that we shall establish for a family satisfying these conditions will apply in particular to the family (M $\mathcal{T}$). In particular, one may take for $\mathcal{T}$ the canonical topology, and for (M) the family of universal effective epimorphisms.

Lemma 4.6.5. *Let (M) be a family of morphisms satisfying the preceding properties (a) to (e $\mathcal{T}$). Let R be a $\mathcal{C}$ -equivalence relation in $X \in \text{Ob } \mathcal{C}$, of type (M). Let $\tilde{X}$ be the sheaf defined by X , $\tilde{R}$ the $\tilde{\mathcal{C}}$ -equivalence relation in $\tilde{X}$ defined by R , and $\tilde{X}/\tilde{R}$ the quotient sheaf. For R to be (M)-effective, it is necessary and sufficient that $\tilde{X}/\tilde{R}$ be representable. If this is so, $\tilde{X}/\tilde{R}$ is represented by the quotient X/R .*

PROOF. First suppose that R is (M)-effective and write $Y = X/R$. The canonical morphism $p : X \rightarrow Y$ is an element of (M), hence covering by (d $\mathcal{T}$). The corresponding morphism

$$\tilde{p} : \tilde{X} \rightarrow \tilde{X}/\tilde{R}$$

is therefore a universal effective epimorphism of $\tilde{\mathcal{C}}$ (4.4.3), hence identifies $\tilde{X}/\tilde{R}$ with the quotient of $\tilde{X}$ by the equivalence relation R' defined in $\tilde{X}$ by $\tilde{p}$. Since the canonical functor

⁽⁴⁴⁾**N.D.E.** Recall that $\mathcal{T}$ denotes the given topology on $\mathcal{C}$, less fine than the canonical topology.

$\mathcal{C} \longrightarrow \tilde{\mathcal{C}}$ commutes with fibered products, R' is precisely $\tilde{R}$, since R is the equivalence relation defined by p .

Conversely, suppose that $\tilde{X}/\tilde{R}$ is represented by an object Y of $\mathcal{C}$. Let $p : X \longrightarrow Y$ be the morphism deduced from the canonical morphism $\tilde{X} \longrightarrow \tilde{X}/\tilde{R}$; it is a covering morphism by 4.4.3. It is clear, as before, that R is the equivalence relation defined by p . It remains only to show that $p \in (M)$. Now the cartesian square

$$\begin{array}{ccccc} R & \xrightarrow{\sim} & X \times_Y X & \longrightarrow & X \\ & \searrow p_2 & \downarrow \text{pr}_2 & & \downarrow p \\ & & X & \xrightarrow{p} & Y \end{array}$$

shows that p becomes p_2 , which is an element of (M) , after base change by the covering morphism p . One concludes by $(e_{\mathcal{T}})$. $\square$

Corollary 4.6.5.1. ⁽⁴⁵⁾ *Let (M) satisfy the preceding properties (a) to $(e_{\mathcal{T}})$, and let $f : G \longrightarrow G'$ be a morphism of $\mathcal{C}$ -groups such that $f \in (M)$. Suppose that $\text{Ker}(f)$ is representable (which is the case if $\mathcal{C}$ possesses a final object e). Then the equivalence relation in G defined by $H = \text{Ker}(f)$ is (M) -effective, and G' represents the quotient sheaf $\tilde{G}/\tilde{H}$ for the topology $\mathcal{T}$.*

PROOF. This follows from 4.6.5 and 3.4.7.1. $\square$

We are now in a position to state the principal result of this number.

Theorem 4.6.6. *Let (M) be a family of morphisms satisfying axioms (a) to $(e_{\mathcal{T}})$ of 4.6.3. Let R be a $\mathcal{C}$ -equivalence relation of type (M) (cf. 3.4.3) in the object X of $\mathcal{C}$. Consider the functor $F \in \text{Ob } \hat{\mathcal{C}}$ defined as follows:*

$$F(S) = \left\{ \begin{array}{l} \text{sub-}\mathcal{S}\text{-sheaves } Z \text{ of } X_S \text{ stable under } R \times S \\ \text{whose quotient by } R_Z \text{ is } S \end{array} \right\}.$$

Let F_0 be the subfunctor of F defined as follows: $F_0(S)$ consists of the representable $Z \in F(S)$, that is:

$$F_0(S) = \left\{ \begin{array}{l} \text{sub-}\mathcal{C}_{/S}\text{-objects } Z \text{ of } X_S \text{ stable under } R \times S, \\ \text{such that } R_Z \text{ is } (M)\text{-effective with quotient } S \\ \text{(i.e. such that } Z \longrightarrow S \text{ belongs to } (M) \text{ and } R_Z \simeq Z \times_S Z) \end{array} \right\}.$$

Then:

(i) The morphism $p \in \text{Hom}(X, F) = F(X)$ defined by the subobject R of $X \times X$ identifies F with the quotient sheaf of X by R .

(ii) The following conditions are equivalent:

- a) F is representable.
- b) F_0 is representable.
- c) R is (M) -effective.

Under these conditions, $F = F_0 = X/R$.

(iii) Let (N) be a family of morphisms stable under base change, such that for every covering family $\{S_i \longrightarrow S\}$ and every family $\{T_i \longrightarrow S_i\}$ of morphisms of (N) , every descent datum on the T_i relative to $\{S_i \longrightarrow S\}$ is effective. Suppose that X is base-changeable (cf. 1.6.0) and that the morphism $R \longrightarrow X \times X$ is an element of (N) . Then $F_0 = F$.

PROOF. (i) has already been proved (4.6.1).

(ii) One has seen the equivalence of a) and c), as well as the equality $F = X/R$. It remains to prove that b) or c) implies $F_0 = F$. First remark, as indeed asserted in the statement, that F_0 is a subfunctor of F ; indeed, for every $S \in \text{Ob } \mathcal{C}$ and every $Z \in F_0(S)$, the morphism

⁽⁴⁵⁾ **N.D.E.** This corollary has been added.

$Z \rightarrow S$ is base-changeable, hence $Z \times_S S'$ is an element of $F_0(S')$ for every $S' \rightarrow S$. Since $R \in F(X)$ belongs to $F_0(X)$, 4.6.2 shows that b) implies $F_0 = F$.

Now suppose that c) holds and let Q be an object of $\mathcal{C}$ representing X/R . Then the morphism $X \rightarrow Q$ is an element of (M), and for every $S \in \text{Ob } \mathcal{C}$ and every $Z \in F(S)$, diagram (*) of 4.6.1 shows that $Z = S \times_{(Q \times_S)} (X \times S)$ is representable, and $Z \rightarrow S$ belongs to (M), hence $Z \in F_0(S)$.

(iii) Let $f \in \text{Hom}(S, F)$ correspond to $Z \in F(S)$. One must show that f factors through F_0 , that is, that Z is representable. This is first clear if f factors through X , by virtue of:

Lemma 4.6.7. *Let $x_0 \in X(S)$. The image of x_0 in $F(S)$ corresponds to the subsheaf Z of X_S defined by the two cartesian squares*

$$\begin{array}{ccccc} X_S & \xrightarrow{\text{id}_{X_S} \times x_0} & X_S \times_S X_S & \longrightarrow & X \times X \\ \uparrow & & \uparrow & & \uparrow \\ Z & \longrightarrow & R_S & \longrightarrow & R \end{array}$$

PROOF. This lemma follows at once from the description of the morphism $X \rightarrow F$. $\square$

Let us return to the proof of the theorem. If f factors through X , then Z is representable, and, since $R \rightarrow X \times X$ is an element of (N), so is $Z \rightarrow X_S$. In general f does not necessarily factor through X ; but, since $X \rightarrow F$ is covering (4.4.3), there exists by 4.4.8 (vii) a covering family $\{S_i \rightarrow S\}$, and for each i a morphism $S_i \rightarrow X$ making the diagram

$$\begin{array}{ccc} X & \longrightarrow & F \\ \uparrow & & \uparrow f \\ S_i & \longrightarrow & S \end{array}$$

commutative. By the preceding argument, the morphism $f_i : S_i \rightarrow F$ defined by the preceding diagram belongs to $\text{Hom}(S_i, F_0)$ and corresponds to the subsheaf $Z \times_S S_i$ of X_{S_i} . The morphism $Z \times_S S_i \rightarrow X_{S_i}$ is an element of (N), and the family $X_{S_i} \rightarrow X_S$ is covering. It therefore remains only to establish:

Proposition 4.6.8. *Let $\{S_i \rightarrow S\}$ be a covering family, and Z a sheaf over S . Suppose that for each i , the S_i -functor $Z \times_S S_i$ is represented by an object T_i . Then the family of the T_i is endowed with a canonical descent datum relative to $S_i \rightarrow S$. For Z to be representable, it is necessary and sufficient that this datum be effective; if it is so, the “descended” object represents Z .*

PROOF. First remark that by 4.4.3, $S_i \rightarrow S$ is a universal effective epimorphic family in $\tilde{\mathcal{C}}$, hence a descent family in $\tilde{\mathcal{C}}$ (2.3). If Z is represented by the object T , then $T \times_S S_i$ (considered as a sheaf) is isomorphic to $Z \times_S S_i$, hence the descent datum on the T_i is effective, and the descended object (unique) is isomorphic to Z . Conversely, suppose that the canonical descent datum on the T_i is effective, and let T be the descended object. Since the family $\{S_i \rightarrow S\}$ is a descent family in $\tilde{\mathcal{C}}$, there exists an S -morphism $T \rightarrow Z$ which by base extension to each S_i gives back the canonical morphism $T_i \rightarrow Z \times_S S_i$. This morphism is locally an isomorphism; since T, Z are sheaves, it follows from 4.5.8 that it is an isomorphism. $\square$

$\square$

Corollary 4.6.9. *Let R be an (M) -effective equivalence relation in X . For every sheaf F , the map*

$$\text{Hom}(X/R, F) \longrightarrow \text{Hom}(X, F)$$

identifies the first set with the subset of the second consisting of morphisms compatible with R .

Corollary 4.6.10. *Let $\mathcal{T}'$ be a topology less fine than $\mathcal{T}$, for which the morphisms of (M) are covering. Under the conditions of 4.6.6 (iii), X/R is also the quotient sheaf of X by R in every topology intermediate between $\mathcal{T}'$ and the canonical topology.*

Remark 4.6.11. If, in the statement of 4.6.6 (iii), one assumes moreover that, under the hypotheses of the text, if one denotes by T the descended object, the morphism $T \rightarrow S$ is an element of (N) , then the inclusion morphisms $Z \hookrightarrow X_S$ are also elements of (N) , as follows at once from the construction of Z by descent.

Remark 4.6.12. The implications $c) \Rightarrow b) \Rightarrow a)$ and $c) \Rightarrow [F_0 = F = X/R]$ have been established without recourse to the “sufficient” part of Lemma 4.6.5, which is the only place where condition $(e_{\mathcal{T}})$ is used. They therefore remain valid if (M) satisfies only conditions (a) to $(d_{\mathcal{T}})$. An example of such a family (M) is that of base-changeable covering morphisms (compare with 4.6.4). In the case of the canonical topology, these are precisely the universal effective epimorphisms. One therefore has:

Corollary 4.6.13. *Let R be a universal effective equivalence relation in X . Then the object X/R of $\mathcal{C}$ is the quotient sheaf of X by R for the canonical topology. It represents the following functor: $(X/R)(S)$ is the set of sub- $\mathcal{C}_{/S}$ -objects Z of X_S stable under $R \times S$ and such that the induced equivalence relation is universal effective with quotient S .*

Similarly, for an arbitrary topology:

Corollary 4.6.14. *Let (M) be the family of base-changeable covering morphisms. If R is an (M) -effective equivalence relation in X , then the object X/R of $\mathcal{C}$ is the quotient sheaf of X by R , and represents the functor F_0 of 4.6.6.*

Scholium 4.6.15. *We may now add the following precisions to 4.5.5. Whereas in questions involving exclusively projective limits (fibered products, algebraic structures, etc.), one may, by the results of Exposé I and 4.5.5, identify $\mathcal{C}$ indifferently with a full subcategory of $\tilde{\mathcal{C}}$ or $\hat{\mathcal{C}}$, the same is not true in questions mixing projective and inductive limits. In all questions involving both projective limits and inductive limits, in particular passages to the quotient (example: group structure on the quotient of a group by an invariant subgroup), we shall consider the given category as embedded in the category of sheaves; thus if R is a $\mathcal{C}$ -equivalence relation in the object X of $\mathcal{C}$, X/R will denote the quotient sheaf of X by R (previously denoted by $j(X)/j(R)$), hence, in the case where this sheaf is representable, the object which it represents. The preceding results show that in the most important cases a quotient in $\mathcal{C}$ will also be a quotient in the category of sheaves; in any event we forbid ourselves the use of the notation X/R for a quotient in $\mathcal{C}$ which does not coincide with the quotient in $\tilde{\mathcal{C}}$ (for example, one which is not universal), thus modifying the definitions of no. 3.*

To study a problem of the above type, one therefore first works in the category of sheaves, where all the customary results are valid (cf. no. 4.4), then specializes the results obtained to the initial category, using the results of this number and, when available, criteria for effectiveness of descent. We shall see examples of this method in the following numbers.

4.7. Use of effectiveness criteria: isomorphism theorem

In this number we give an example of the use of effectiveness criteria. The initial data are a topology $\mathcal{T}$ on $\mathcal{C}$ (still less fine than the canonical topology), a family (M) of morphisms of $\mathcal{C}$ satisfying axioms (a) to $(e_{\mathcal{T}})$ of 4.6.3, and a family (N) of morphisms of $\mathcal{C}$ which may satisfy the following axioms:

- (a) (N) is stable under base extension.

(f_T) “Morphisms of (N) descend under the given topology”; that is: for every $S \in \text{Ob } \mathcal{C}$, every covering family $\{S_i \rightarrow S\}$ and every family $\{T_i \rightarrow S_i\}$ of morphisms of (N), every descent datum on the T_i relative to $\{S_i \rightarrow S\}$ is effective, and if one denotes by T the descended object, the morphism $T \rightarrow S$ is an element of (N).

Since every element of (M) is covering (condition 4.6.3 (d_T)), (f_T) implies the following axiom: ⁽⁴⁶⁾

(f_M) If $Y \rightarrow X$ is an element of (N) and $X \rightarrow X_1$ an element of (M), every descent datum on Y relative to $X \rightarrow X_1$ is effective; if one denotes by Y_1 the descended object, $Y_1 \rightarrow X_1$ is an element of (N).

Let us immediately point out an example of this situation, which will be treated later: $\mathcal{C}$ is the category of schemes, $\mathcal{T}$ the faithfully flat quasi-compact topology; (M) the family of faithfully flat quasi-compact morphisms, (N) the family of closed immersions or that of quasi-compact immersions. ⁽⁴⁷⁾

Recall the principal result of 4.6.6 (taking account of 4.6.11):

Proposition 4.7.1. *If X is a base-changeable object of $\mathcal{C}$, R an equivalence relation of type (M) in X such that $R \rightarrow X \times X$ is an element of (N), with (N) satisfying (a) and (f_T), then the quotient sheaf X/R is defined by*

$$(X/R)(S) = \left\{ \begin{array}{l} \text{sub-}S\text{-objects } Z \text{ of } X_S \text{ stable under } R \times S, \\ \text{such that } Z \rightarrow X_S \text{ belongs to (N), } Z \rightarrow S \text{ is covering} \\ \text{(or an element of (M)), and } R_Z \simeq Z \times_S Z \end{array} \right\}.$$

Moreover, one has:

Proposition 4.7.2. *Let $X \in \text{Ob } \mathcal{C}$, and R an (M)-effective equivalence relation in X . Let (N) be a family of morphisms satisfying (a) and (f_M).*

For every subobject Y of X stable under R and such that $Y \rightarrow X$ belongs to (N), the equivalence relation induced in Y by R is (M)-effective, and the quotient $Y/R_Y = Y'$ is a subobject of $X' = X/R$ such that $Y' \rightarrow X'$ belongs to (N).

The map $Y \mapsto Y'$ is a bijection between the set of subobjects Y of X stable under R , such that $Y \rightarrow X$ belongs to (N), and the set of subobjects Y' of X' such that $Y' \rightarrow X'$ belongs to (N). The inverse map is $Y' \mapsto Y' \times_{X'} X$.

PROOF. Since R is (M)-effective, the morphism $X \rightarrow X'$ belongs to (M). Let Y' be a subobject of X' such that the canonical morphism $Y' \rightarrow X'$ belongs to (N). Then the subobject $Y = Y' \times_{X'} X$ of X is stable under R , and the morphism $Y \rightarrow X$ (resp. $Y \rightarrow Y'$) belongs to (N) (resp. (M)), since (N) and (M) are stable under base change. Denote by R_Y the equivalence relation induced in Y by R . By 4.4.11, the quotient sheaf Y/R_Y is represented by Y' , and hence, by 4.6.5, R_Y is (M)-effective.

Conversely, let us show that every subobject Y of X stable under R , such that the structural morphism $Y \rightarrow X$ belongs to (N), is obtained in this way. Indeed, if Y is stable under R , its two images in $R = X \times_{X'} X$ are identical, and Y is endowed with a descent datum relative to $X \rightarrow X'$; the desired result follows, since the family (N) satisfies axiom (f_M). $\square$

Corollary 4.7.3. *Let $X \in \text{Ob } \mathcal{C}$, and R an (M)-effective equivalence relation in X ; suppose moreover that $R \rightarrow X \times X$ belongs to (N), where (N) satisfies (a) and (f_T). Then, for every Y as in 4.7.2, $R_Y \rightarrow Y \times Y$ also belongs to (N), and hence, by 4.7.1, one has:*

$$(Y/R_Y)(S) = \left\{ \begin{array}{l} \text{sub-}S\text{-objects } Z \text{ of } Y_S \text{ stable under } R_Y \times S, \\ \text{such that } Z \rightarrow Y_S \text{ belongs to (N) (then } Z \rightarrow X_S \text{ does too),} \\ Z \rightarrow S \text{ is covering, and } R_Z \simeq Z \times_S Z \end{array} \right\}.$$

⁽⁴⁶⁾**N.D.E.** Axiom (f_M) (which appeared before Proposition 4.7.2) has been placed here.

⁽⁴⁷⁾**N.D.E.** Cf. § 6.4, see also VIA, 5.3.1.

5. Passage to the quotient and algebraic structures

5.1. Principal homogeneous bundles

Definition 5.1.0. ⁽⁴⁸⁾ Recall (III 0.1) that an object X with a group of operators H (on the right) is said to be *formally principal homogeneous* under H ⁽⁴⁹⁾ if the canonical morphism (of functors)

$$X \times H \longrightarrow X \times X$$

defined by $(x, h) \mapsto (x, xh)$ is an isomorphism. It amounts to the same to say (cf. *loc. cit.*) that for every $S \in \text{Ob } \mathcal{C}$, $X(S)$ is formally principal homogeneous under $H(S)$, that is, empty or principal homogeneous under $H(S)$. In particular, if one lets H act on itself by translations (on the right), H becomes formally principal homogeneous under itself.

Definition 5.1.1. The object X with a group of operators H is said to be *trivial* if it is isomorphic (as an object with a group of operators H) to H on which H acts by translations.

Proposition 5.1.2. *Let X be formally principal homogeneous under H . One has an isomorphism*

$$\Gamma(X) \xrightarrow{\sim} \text{Isom}_{H\text{-obj.}}(H, X)$$

of principal homogeneous sets under $\Gamma(H)$.

PROOF. To every section x of X one associates the morphism from H to X defined set-theoretically by $h \mapsto xh$. The stated assertion is immediate, by reduction to the set-theoretic case. $\square$

Corollary 5.1.3. *One has an isomorphism of objects with operators H*

$$X \xrightarrow{\sim} \underline{\text{Isom}}_{H\text{-obj.}}(H, X).$$

Corollary 5.1.4. *For an object with a group of operators to be trivial, it is necessary and sufficient that it be formally principal homogeneous and that it possess a section.*

Definition 5.1.5. Let $\mathcal{C}$ be a category endowed with a topology. One says that the S -object X with an S -group of operators H is a *principal homogeneous bundle* under H if it is locally trivial, that is, if the following equivalent conditions hold: ⁽⁵⁰⁾

- (i) The set of $T \longrightarrow S$ such that (the functor) $X \times_S T$ is trivial under $H \times_S T$ is a refinement of S .
- (ii) There exists a covering family (for a pretopology defining the given topology) $\{S_i \longrightarrow S\}$ such that for each i , the S_i -functor $X \times_S S_i$ with an S_i -group-functor of operators $H \times_S S_i$ is trivial (= possesses a section over S_i).

Proposition 5.1.6. *Let $\mathcal{C}$ be a category endowed with a topology $\mathcal{T}$. Let (M) be a family of morphisms of $\mathcal{C}$ satisfying axioms (a) to (e $\mathcal{T}$) of 4.6.3. Let H be an S -group such that the structural morphism $H \longrightarrow S$ is an element of (M) , and P an S -object with an S -group of operators H . The following conditions are equivalent:*

- (i) P is a principal homogeneous bundle under H (Definition 5.1.5).
- (ii) P is formally principal homogeneous under H and the structural morphism $P \longrightarrow S$ is an element of (M) .

⁽⁴⁸⁾**N.D.E.** The numbering 5.1.0 has been introduced for later references.

⁽⁴⁹⁾**N.D.E.** One also says “pseudo-torsor”, cf. EGA IV₄, 16.5.15. Moreover, the more general notion of formally homogeneous object (not necessarily principal homogeneous) is defined in the addition 6.7.1 at the end of this exposé.

⁽⁵⁰⁾**N.D.E.** In this case one also says that X is an H -torsor.

- (iii) *There exists a morphism $S' \rightarrow S$ which is an element of (M) , such that by base extension from S to S' , P becomes trivial, that is, $P \times_S S'$ is trivial under $H \times_S S'$.*
- (iv) *H acts freely on P , in an (M) -effective manner, and the quotient P/H is isomorphic to S .*

PROOF. First remark that (ii) and (iv) are equivalent, taking account of the fact that in either case $P \rightarrow S$ is an element of (M) , hence base-changeable, which ensures the representability of the fibered products $H \times_S P$, $P \times_S P$. It is clear that (ii) implies (iii), for one may take P itself as S' , the hypothesis that P is formally principal homogeneous implying that $P \times_S P$ is trivial under $H \times_S P$ (5.1.4), for it has a section (the diagonal section). It is clear that (iii) implies (i), for $\{S' \rightarrow S\}$ is a covering family by axiom $(d_{\mathcal{T}})$. It therefore remains to show that (i) implies (ii). The morphism of sheaves $P \times_S H \rightarrow P \times_S P$ is locally an isomorphism, hence an isomorphism (4.5.8); thus P is formally principal homogeneous. The structural morphism $P \rightarrow S$ is locally isomorphic to the structural morphism $H \rightarrow S$, which is an element of (M) . It is therefore itself an element of (M) by $(e_{\mathcal{T}})$. $\square$

The equivalence between (i) and (iv) generalizes:

Proposition 5.1.7. *Under the same hypotheses on $\mathcal{C}$ and (M) , let H be an S -group, and X an S -object on which H acts (on the right). Suppose that the structural morphism $H \rightarrow S$ is an element of (M) . The following conditions are equivalent:*

- (i) *H acts freely on X , in an (M) -effective manner.*
- (ii) *There exists an S -morphism $p : X \rightarrow Y$ compatible with the equivalence relation defined in X by the action of H , such that the action of $H \times_S Y$ on X over Y deduced from it makes X a principal homogeneous bundle under H_Y over Y .*

Under these conditions p identifies Y with the quotient X/H .

PROOF. If $p : X \rightarrow Y$ is a morphism compatible with the action of H , then the action of $H \times_S Y$ on X over Y deduced from it defines in X the same equivalence relation as the action of H , by virtue of the formula

$$H_Y \times_Y X \xrightarrow{\sim} H \times_S X.$$

The proposition follows from this remark and the preceding equivalence (iv) $\Leftrightarrow$ (i). $\square$

Corollary 5.1.7.1. ⁽⁵¹⁾ *Let $\mathcal{C}$ be a category possessing a final object, stable under fibered products and endowed with a topology $\mathcal{T}$ less fine than the canonical topology. Let $f : G \rightarrow H$ be a morphism of $\mathcal{C}$ -groups, and $K = \text{Ker}(f)$. Suppose that f is covering for the topology $\mathcal{T}$.*

Then H represents the quotient sheaf G/K , and f is a K_H -torsor. (N.B. One will also say: “ G is a K -torsor over H ”.)

PROOF. Indeed, since f is covering, it is a universal effective epimorphism (4.4.3), hence by 3.3.3.1, H is the quotient of G by the equivalence relation $\mathcal{R}(f) = G \times_H G$. On the other hand, the morphism $G \times K \rightarrow G \times_H G$, $(g, k) \mapsto (g, gk)$, is an isomorphism of objects with a group of operators $K_G = G \times_H K_H$ (its inverse being given by $(g, g') \mapsto (g, g^{-1}g')$). Thus, on the one hand, $\mathcal{R}(f)$ is the equivalence relation defined by K ; on the other, since the morphism $f : G \rightarrow H$ is covering, f is a K_H -torsor by 5.1.6 (ii) (or directly by Definition 5.1.5 (ii)). $\square$

We may now specify Theorem 4.6.6 in the case of passage to the quotient by a group of operators:

Proposition 5.1.8. *Under the hypotheses of 5.1.7, denote by F_0 the functor over S defined as follows: for each $S' \rightarrow S$, $F_0(S')$ is the set of representable sub- S' -functors Z of $X \times_S S'$,*

⁽⁵¹⁾**N.D.E.** This particular case has been added as a corollary, and will be used several times in the following exposés.

stable under $H \times_S S'$ and principal homogeneous bundles under this S' -group for the induced action (3.2.2).

(i) The following conditions are equivalent:

a) The action of H on X is (M) -effective and free. ⁽⁵²⁾

b) F_0 is representable.

Under these conditions $F_0 = X/H$.

(ii) Let (N) be a family of morphisms stable under base change, such that for every covering family $\{S'_i \rightarrow S'\}$ and every family $\{T_i \rightarrow S'_i\}$ of morphisms of (N) , every descent datum on the T_i relative to $\{S'_i \rightarrow S'\}$ is effective. Suppose that the morphism $X \times_S H \rightarrow X \times_S X$ is an element of (N) , and that X is base-changeable. Then the element p of $\text{Hom}(X, F_0)$ corresponding to the subobject $X \times_S H$ of $X \times_S X$ identifies F_0 with the quotient sheaf X/H .

5.2. Group structures and passage to the quotient

In this number we concern ourselves with the algebraic structures that one can put on the quotient G/H of a group by a subgroup. We shall first work in the category of sheaves on $\mathcal{C}$ for an arbitrary topology. Taking the canonical topology and using 4.5.12, we shall obtain results for passage to the universal effective quotient in $\mathcal{C}$.

Proposition 5.2.1. *Let $u : H \rightarrow G$ be a monomorphism of sheaves of groups. There exists on the quotient sheaf G/H a unique structure of object with a group of operators G such that the canonical morphism*

$$p : G \rightarrow G/H$$

is a morphism of objects with a group of operators G . This structure is functorial in the pair (G, H) : if one has a commutative diagram

$$\begin{array}{ccc} H & \longrightarrow & G \\ \downarrow & & \downarrow f \\ H' & \longrightarrow & G' \end{array},$$

the morphism $\bar{f} : G/H \rightarrow G'/H'$ (3.2.3) is compatible with the morphism f on the groups of operators.

PROOF. Indeed, the sheaf G/H is the sheaf associated to the presheaf

$$i(G)/i(H) : S \mapsto G(S)/H(S);$$

since the functor a is left exact, it transforms objects with groups of operators into objects with a group of operators. Since the presheaf $i(G)/i(H)$ is endowed with a structure of object with a group of operators $i(G)$, $G/H = a(i(G)/i(H))$ is endowed with a structure of object with operators $a(i(G)) = G$. This structure evidently enjoys all the stated properties. $\square$

Corollary 5.2.2. *Let $u : H \rightarrow G$ be a monomorphism of $\mathcal{C}$ -groups. Suppose that the action of H on G is universal effective. There exists on the quotient object $G/H \in \text{Ob } \mathcal{C}$ a unique structure of object with a group of operators G such that $p : G \rightarrow G/H$ is a morphism of objects with operators. This structure is functorial in the pair (H, G) (H acting in a universal effective manner on G) in the preceding sense.*

Proposition 5.2.3. *Let $u : H \rightarrow G$ be a monomorphism of sheaves of groups identifying H with an invariant subsheaf of groups of G . There exists on the quotient sheaf G/H a unique structure of sheaf of groups such that the canonical morphism $p : G \rightarrow G/H$ is a morphism of groups. This structure is functorial in the pair (H, G) (H invariant).*

PROOF. The proof is similar to that of 5.2.1. $\square$

⁽⁵²⁾N.D.E. “and free” has been added.

Corollary 5.2.4. *Let $u : H \rightarrow G$ be a monomorphism of $\mathcal{C}$ -groups identifying H with an invariant subgroup of G . Suppose that the action of H on G is universal effective. There exists on the quotient object $G/H \in \text{Ob } \mathcal{C}$ a unique group structure such that the canonical morphism $G \rightarrow G/H$ is a morphism of groups. This structure is functorial in the pair (H, G) (H invariant, H acting in a universal effective manner).*

One can characterize the group structure of G/H in a more expressive manner:

Proposition 5.2.5. *Under the conditions of 5.2.4, let K be a $\mathcal{C}$ -group, and $f : G \rightarrow K$ a morphism. The following conditions are equivalent:*

- (i) *f is a morphism of groups compatible with the equivalence relation defined by H .*
- (ii) *f is a morphism of groups inducing the trivial morphism $H \rightarrow K$.*
- (iii) *f factors through a morphism of groups $G/H \rightarrow K$.*

In particular, one has an isomorphism functorial in the group K

$$\text{Hom}_{\mathcal{C}\text{-gr.}}(G/H, K) \xrightarrow{\sim} \{f \in \text{Hom}_{\mathcal{C}\text{-gr.}}(G, K) \mid f \circ u = e\}.$$

PROOF. The equivalence of (i) and (ii) is proved set-theoretically. One evidently has (iii) $\Rightarrow$ (ii). The equivalence of (iii) and (ii) follows from the formula

$$\text{Hom}(G/H, K) \simeq \text{Hom}(i(G)/i(H), K)$$

and the definition of the group structure of G/H . $\square$

Remark 5.2.6. In the preceding situation, if the kernel of f is exactly H , the morphism $G/H \rightarrow K$ through which f factors is a monomorphism. This follows at once from 3.3.4.

In the case of sheaves of groups, one can specify 4.4.11 by the

Proposition 5.2.7. *Let G be a sheaf of groups and H an invariant subsheaf of groups. For every subsheaf of groups K of G containing H , let K' be the quotient group K/H considered as a subgroup of $G' = G/H$.*

One has $K = K' \times_{G'} G$, and the maps $K \mapsto K/H$, $K' \mapsto K' \times_{G'} G$ establish a bijection between the set of subsheaves of groups of G containing H and the set of subsheaves of groups of G' . Under this correspondence, invariant subsheaves of groups of G, G' correspond.

PROOF. The first part follows easily from 4.4.11 and 3.2.4. It remains to see that K is invariant in G if and only if K' is invariant in G' . If K is invariant in G , then the presheaf $i(K)/i(H)$ is invariant in $i(G)/i(H)$. The same is true of the associated sheaves, by virtue of the customary argument. Conversely, if K' is invariant in G' , then the fibered product $K \times_{G'} G$ is invariant in G , as one sees immediately. $\square$

If now L is any subsheaf of groups of G , let $\bar{L}$ be the saturation of L for the equivalence relation defined by H ; one will also write $\bar{L} = L \cdot H$.

Proposition 5.2.8. *Under the preceding conditions, $L \cdot H$ is a subsheaf of groups of G containing H , and the image of L in G/H is identified with*

$$(L \cdot H)/H \simeq L/(H \cap L).$$

PROOF. Indeed, denote by L' the image sheaf of L in G/H . It is a subsheaf of groups of G/H corresponding to $L \cdot H$ under the correspondence of the preceding proposition. Since the morphism $L \rightarrow L'$ is covering, hence a universal effective epimorphism of sheaves, it follows from 4.4.9 that L' is identified with the quotient of L by the kernel of $L \rightarrow L'$, which is evidently precisely $H \cap L$. $\square$

Finally consider the following situation: one has a sheaf of groups G , a subsheaf of groups K , and a subsheaf of groups H of K , invariant in K . First let us define an action (on the right) of the sheaf of groups $H \backslash K (= K/H)$ on G/H . The group K acts by right translations on G . Since H is invariant in K , this action is compatible with the equivalence relation

defined by the action of H , and therefore defines an action of K on G/H , that is, a morphism from the group K° opposite to K to $\underline{\text{Aut}}(G/H)$. Since the latter is a sheaf (4.5.13) and this morphism is trivial on H , it factors through K/H and defines the desired action. Since the right and left actions of G on itself commute, the actions of $G, K/H$ on G/H commute.

Proposition 5.2.9. *Under the preceding conditions, K/H acts freely (on the right) on G/H , and one has a canonical isomorphism of sheaves with a group of operators G*

$$(G/H)/(K/H) \simeq G/K.$$

When K is invariant in G , in which case K/H is invariant in G/H (5.2.7), this isomorphism respects the group structures of the two members.

PROOF. One has an isomorphism of presheaves

$$(i(G)/i(H))/(i(K)/i(H)) \xleftarrow{\sim} i(G)/i(K),$$

which respects the structures of objects with a group of operators $i(G)$. The announced result is obtained by applying the functor a to this relation. $\square$

Corollary 5.2.10. *Let G be a $\mathcal{C}$ -group, K a sub- $\mathcal{C}$ -group of G , and H an invariant sub- $\mathcal{C}$ -group of K . Let (M) be a family of morphisms of $\mathcal{C}$ satisfying axioms (a) to $(e_{\mathcal{T}})$. Suppose that the action of H on G (resp. K) on the right is (M) -effective. Then K/H acts naturally and freely on the right on G/H ; this action commutes with that of G . The following conditions are equivalent:*

- (i) *The action of K on G is (M) -effective.*
- (ii) *The action of K/H on G/H is (M) -effective.*

Under these conditions one has an isomorphism of objects⁽⁵³⁾ with a group of operators G :

$$(G/H)/(K/H) \simeq G/K.$$

5.3. Use of effectiveness criteria: Noether's theorem

⁽⁵⁴⁾ Let $\mathcal{C}, \mathcal{T}$, and (M) be as usual. Let (N) be a family of morphisms satisfying axioms (a) and (f_M) of 4.7. Putting together 5.2.7 and 4.7.2, one obtains:

Proposition 5.3.1. *Let G be a $\mathcal{C}$ -group. Let H be an invariant sub- $\mathcal{C}$ -group of G , acting in an (M) -effective manner on G .*

For every sub- $\mathcal{C}$ -group K of G containing H and such that the morphism $K \rightarrow G$ belongs to (N) , H acts on K in an (M) -effective manner, and the quotient $K/H = K'$ is a sub- $\mathcal{C}$ -group of $G/H = G'$, such that the morphism $K' \rightarrow G'$ belongs to (N) .

The map $K \mapsto K'$ is a bijection between the set of sub- $\mathcal{C}$ -groups K of G containing H and such that $K \rightarrow G$ belongs to (N) , and the set of sub- $\mathcal{C}$ -groups K' of G' such that $K' \rightarrow G'$ belongs to (N) . The inverse map is $K' \mapsto K \times_{G'} G$. Under this correspondence, invariant subgroups of G, G' correspond.

Corollary 5.3.2. *If $H \rightarrow G$ is an element of (N) , then $\mathcal{C}$ possesses a final object e , and the unit section $e \rightarrow G/H$ is an element of (N) .⁽⁵⁵⁾*

6. Topologies in the category of schemes

6.1. The Zariski topology

⁽⁵³⁾N.D.E. Of $\mathcal{C}$.

⁽⁵⁴⁾N.D.E. In fact, the isomorphisms established in 5.2.8 to 5.2.10 would deserve to be called “Noether's isomorphism theorems”.

⁽⁵⁵⁾N.D.E. This follows from the proposition applied to $K = H$.

This is the topology generated by the following pretopology: a family of morphisms $\{S_i \rightarrow S\}$ is covering if each morphism is an open immersion and the union of the images of the S_i is all of S . It is denoted (Zar).

Definition 6.1.1. A sheaf for the Zariski topology is also called a *functor of local nature*: it is a contravariant functor from (Sch) to (Ens) such that for every scheme S and every covering of S by open sets S_i , one has an exact diagram:

$$F(S) \longrightarrow \prod_i F(S_i) \rightrightarrows \prod_{i,j} F(S_i \cap S_j).$$

In particular, a functor of local nature transforms direct sums into products. Since every representable functor is a sheaf, this topology is less fine than the canonical topology. As regards terminology, whenever we say “local”, “locally”, without further specification, this will refer to the Zariski topology, hence have the customary meaning.

6.2. A procedure for constructing topologies

Proposition 6.2.1. Let $\mathcal{C}$ be a category, $\mathcal{C}'$ a full subcategory, P a set of families of morphisms of $\mathcal{C}$ with the same target, stable under base change and composition (i.e. satisfying axioms (P 1) and (P 2) of 4.2.5), and P' a set of families of morphisms of $\mathcal{C}'$ containing the families consisting of a single identity isomorphism. Endow $\mathcal{C}$ with the topology generated by P, P' (cf. 4.2.5.0), and suppose that the following three conditions hold:

(a) If $\{S_i \rightarrow S\} \in P'$ (thus $S_i, S \in \text{Ob } \mathcal{C}'$), and if $T \rightarrow S$ is a morphism of $\mathcal{C}'$, then the fibered products $S_i \times_S T$ (in $\mathcal{C}$) exist, and the family $\{S_i \times_S T \rightarrow T\}$ belongs to P' (thus $S_i \times_S T \in \text{Ob } \mathcal{C}'$). (Remark: this condition implies that P' is stable under base change in $\mathcal{C}'$, but is not equivalent to it, for it assumes moreover that the inclusion functor from $\mathcal{C}'$ to $\mathcal{C}$ commutes with certain fibered products.)

(b) For every $S \in \text{Ob } \mathcal{C}$, there exists $\{S_i \rightarrow S\} \in P$, with $S_i \in \text{Ob } \mathcal{C}'$ for each i .

(c) In the following situation:

$$\begin{array}{ccc} & S_{ijk} & \\ & \downarrow (P') & \\ S_i & \xleftarrow{(P)} & S_{ij} \\ (P') \downarrow & & \\ S & & \end{array}$$

where $S, S_i, S_{ij}, S_{ijk} \in \text{Ob } \mathcal{C}'$; $\{S_i \rightarrow S\} \in P'$; $\{S_{ij} \rightarrow S_i\} \in P$ for each i ; $\{S_{ijk} \rightarrow S_{ij}\} \in P'$ for each ij , there exists a family $\{T_n \rightarrow S\} \in P'$ and, for each n , a multi-index (ijk) and a commutative diagram

$$\begin{array}{ccc} S_{ijk} & \xleftarrow{\quad} & T_n \\ & \searrow & \swarrow \\ & & S \end{array}$$

Then, for a sieve R of $S \in \text{Ob } \mathcal{C}$ to be covering, it is necessary and sufficient that there exist a composite family

$$\begin{array}{ccc} R & \xleftarrow{\quad} & S_{ij} \\ \downarrow & & \downarrow (P') \\ S & \xleftarrow{(P)} & S_i \end{array}$$

where $S_i, S_{ij} \in \text{Ob } \mathcal{C}'$, $\{S_i \rightarrow S\} \in P$, $\{S_{ij} \rightarrow S_i\} \in P'$ for each i , and that the resulting morphisms $S_{ij} \rightarrow S$ factor through R (in other words, that the sieve generated by this composite family be contained in R).

PROOF. Since families that are elements of P, P' are covering, a composite family of such families will also be covering (C 2); hence a sieve of the indicated form will be covering, since it contains a covering sieve.

Conversely, it suffices to see that the sieves of this form do form a topology, i.e. it suffices to verify axioms (T 1) to (T 4) of 4.2.1.

Axiom (T 4). Let $S \in \text{Ob } \mathcal{C}$. By (b), there exists a family $\{S_i \rightarrow S\} \in P$, with $S_i \in \text{Ob } \mathcal{C}'$. The families $\{S_i \xrightarrow{\text{id}_{S_i}} S_i\}$ are elements of P' by hypothesis. The sieve S of S is therefore of the desired form:

$$\begin{array}{ccc} S_i & \longrightarrow & S \\ (P') \downarrow \text{id} & & \downarrow \text{id}_S \\ S_i & \xrightarrow{(P)} & S \end{array} .$$

Axiom (T 3). Evident.

Axiom (T 2). Let R be a sieve of S of the desired form, and let C be a sieve⁽⁵⁶⁾ of S such that, for every $T \in \text{Ob } \mathcal{C}$ and every morphism $T \rightarrow S$ factoring through R , the sieve $C \times_S T$ of T is of the desired form. Then, since $S_{ij} \rightarrow S$ factors through R , the sieve $C_{ij} = C \times_S S_{ij}$ of S_{ij} :

$$\begin{array}{ccccc} & & S_{ij} & \longleftarrow & C_{ij} \\ & & \downarrow (P') & & \downarrow \\ & & S_i & & \\ & \swarrow & \downarrow (P) & & \downarrow \\ R & \longrightarrow & S & \longleftarrow & C \end{array}$$

is of the desired form; thus, for every ij , one has a diagram of the form:

$$\begin{array}{ccc} S_{ijkl} & & \\ (P') \downarrow & \searrow & \\ S_{ijk} & & \\ (P) \downarrow & & \\ S_{ij} & \longleftarrow & C_{ij} \end{array} .$$

One has therefore proved that there exists a composite family

$$S_{ijkl} \xrightarrow{(P')} S_{ijk} \xrightarrow{(P)} S_{ij} \xrightarrow{(P')} S_i \xrightarrow{(P)} S$$

belonging to $P \circ P' \circ P \circ P'$, factoring through C , and in which all objects other than S are in $\mathcal{C}'$. Applying condition (c) to each family $\{S_{ijkl} \rightarrow S_i\}$, one deduces that for each i there exists a family $\{T_{in} \rightarrow S_i\} \in P'$ such that $T_{in} \rightarrow S$ factors through one of the S_{ijkl} , hence

⁽⁵⁶⁾N.D.E. The original has been corrected here.

through C :

$$\begin{array}{ccc} T_{in} & \longrightarrow & S_{ijkl} \\ (P') \downarrow & & \downarrow \\ S_i & & \\ (P) \downarrow & & \downarrow \\ S & \longleftarrow & C \end{array}$$

The sieve C of S is therefore of the desired form, which completes the verification.

Axiom (T 1). Let R be a sieve of S of the given form, and $T \rightarrow S$ a morphism of $\mathcal{C}$. Let us show that the sieve $R \times_S T$ of T is of the desired form.

$$\begin{array}{ccccc} & S_{ij} & \longleftarrow & U_{ikj} & \\ & \downarrow (P') & & \downarrow (P') & \\ & S_i & \longleftarrow & T_i & \longleftarrow U_{ik} \\ & \downarrow (P) & & \downarrow (P) & \swarrow (P) \\ R & \longrightarrow & S & \longleftarrow & T \end{array}$$

Let $T_i = S_i \times_S T$. The family $\{T_i \rightarrow T\}$ belongs to P (by (P 1)). Applying (b), one constructs $\{U_{ik} \rightarrow T_i\} \in P$, with $U_{ik} \in \text{Ob } \mathcal{C}'$. By hypothesis (condition (P 2) on P), one has $\{U_{ik} \rightarrow T\} \in P$. By (a), $U_{ik} \times_{S_i} S_{ij} = U_{ikj}$ is an object of $\mathcal{C}'$, and for every ik , $\{U_{ikj} \rightarrow U_{ik}\} \in P'$. Then the commutative diagram below

$$\begin{array}{ccc} R & \longleftarrow & U_{ikj} \\ \downarrow & & \downarrow (P') \\ S & \longleftarrow & T \longleftarrow U_{ik} \end{array}$$

shows that the morphisms $U_{ikj} \rightarrow T$ factor through the sieve $T \times_S R$ of T , which is therefore of the desired form, and this completes the proof. $\square$

Corollary 6.2.2. *If $S \in \text{Ob } \mathcal{C}'$ and R is a sieve of S , R is covering if and only if there exists a family $\{T_i \rightarrow S\} \in P'$ factoring through R .*

PROOF. Indeed, such a sieve is covering. On the other hand, it suffices to apply (c), taking the family $\{S_i \rightarrow S\}$ to consist of the identity isomorphism of S , to deduce from the proposition that a covering sieve is of the indicated form. $\square$

Corollary 6.2.3. *For a presheaf F on $\mathcal{C}$ to be separated (resp. a sheaf), it is necessary and sufficient that the morphism*

$$F(S) \longrightarrow \prod_i F(S_i)$$

be injective (resp. that the diagram

$$F(S) \longrightarrow \prod_i F(S_i) \rightrightarrows \prod_{i,j} F(S_i \times_S S_j)$$

be exact) in the following two cases:

- (i) $\{S_i \rightarrow S\} \in P$,
- (ii) $S, S_i \in \text{Ob } \mathcal{C}'$; $\{S_i \rightarrow S\} \in P'$.

PROOF. Indeed, the conditions are necessary, for the families in question are covering. If C is the sieve of S which is the image of a family of morphisms $\{S_{ij} \xrightarrow{(P')} S_i \xrightarrow{(P)} S\}$, an easy diagram chase shows that the conditions of the corollary imply that $\text{Hom}(S, F) \xrightarrow{g} \text{Hom}(C, F)$

is injective (resp. bijective). But every refinement R of S contains a sieve C of the above type, and one has a commutative diagram

$$\begin{array}{ccc} \mathrm{Hom}(S, F) & \xrightarrow{f} & \mathrm{Hom}(R, F) \\ & \searrow g \quad \swarrow h & \\ & \mathrm{Hom}(C, F) & \end{array}$$

One knows that g is injective, hence f is too. Thus F is separated. But R is a refinement of C , hence h is also injective. If g is bijective, then f is too, hence F is a sheaf. $\square$

Remark 6.2.4. The preceding corollary does not follow from 4.3.5, since P' is not stable under base extension.

Remark 6.2.5. Condition (c) holds in particular in the case where

- (i) P' is stable under composition.
- (ii) If $\{S_j \rightarrow S\}$ is a family of morphisms of $\mathcal{C}'$, an element of P , there exists a subfamily of it which is an element of P' .

6.3. Application to the category of schemes

Take for $\mathcal{C}$ the category of schemes, for $\mathcal{C}'$ the full subcategory consisting of affine schemes, and for P the set of surjective families of open immersions. One will consider several sets P' :

P'_1 : finite surjective families consisting of flat morphisms.

P'_2 : finite surjective families consisting of flat morphisms of finite presentation which are quasi-finite. ⁽⁵⁷⁾

P'_3 : finite surjective families consisting of étale morphisms.

P'_4 : finite surjective families consisting of finite étale morphisms.

For each of these sets P'_i , except P'_4 , the conditions of Proposition 6.2.1 hold ((c) by 6.2.5, for, since an affine scheme is quasi-compact, every family of morphisms of $\mathcal{C}'$ which is an element of P contains a finite subfamily which is also in P , hence in P'_i for $i = 1, 2, 3$). The topology $\mathcal{T}_i$ generated by P, P'_i is denoted and called as follows: ⁽⁵⁷⁾

$\mathcal{T}_1 = (\mathrm{fpqc}) =$ faithfully flat quasi-compact topology.

$\mathcal{T}_2 = (\mathrm{fppf}) =$ faithfully flat (locally) finite presentation topology.

$\mathcal{T}_3 = (\mathrm{ét}) =$ étale topology.

$\mathcal{T}_4 = (\mathrm{étf}) =$ finite étale topology.

Since $P'_1 \supset P'_2 \supset P'_3 \supset P'_4$, one has

$$(\mathrm{fpqc}) \geq (\mathrm{fppf}) \geq (\mathrm{ét}) \geq (\mathrm{étf}) \geq (\mathrm{Zar}).$$

Proposition 6.3.1. (i) For the sieve R of S to be covering for $\mathcal{T}_i$, $1 \leq i \leq 3$, it is necessary and sufficient that there exist a covering (S_p) of S by affine open sets and, for each p , a family $\{S_{pq} \rightarrow S_p\}$ which is an element of P'_i , the S_{pq} being affine, such that each morphism $\{S_{pq} \rightarrow S\}$ factors through R . ⁽⁵⁸⁾

⁽⁵⁷⁾**N.D.E.** Let P'_2 be the set of finite surjective families consisting of flat morphisms of $\mathcal{C}'$ of finite presentation. By Proposition 6.3.1 below, the topology $\mathcal{T}_2$ generated by P, P'_2 coincides with that generated by P, P'_2 . This follows from the results of EGA IV₄, § 17.16 on quasi-sections; see the proof of 6.3.1. Let us cite here the following particular case of EGA IV₄, 17.16.2: let S be affine and $f : X \rightarrow S$ a surjective flat morphism locally of finite presentation; then there exists a faithfully flat morphism $S' \rightarrow S$ of finite presentation which is quasi-finite, with S' affine, and an S -morphism $S' \rightarrow X$.

⁽⁵⁸⁾**N.D.E.** By hypothesis, each family $\{S_{pq} \rightarrow S\} \in P'_i$ is finite, hence $S'_p = \coprod_q S_{pq}$ is affine, and the family can therefore be replaced by the morphism $S'_p \rightarrow S_p$, which still belongs to P'_i .

(ii) For a presheaf F on (Sch) to be a sheaf for (fpqc) (resp. (fppf) , (ét) , (étf)), it is necessary and sufficient that

- a) F be a sheaf for (Zar) , i.e. a functor of local nature.
- b) For every faithfully flat (resp. faithfully flat of finite presentation and quasi-finite, resp. surjective étale, resp. surjective finite étale) morphism $T \rightarrow S$, where T, S are affine, one have an exact diagram:

$$F(S) \rightarrow F(T) \rightrightarrows F(T \times_S T).$$

- (iii) The topologies $\mathcal{T}_i$, $i = 1, 2, 3, 4$, are less fine than the canonical topology.
- (iv) Every surjective family consisting of flat and open (resp. flat and locally of finite presentation, resp. étale, resp. finite étale) morphisms is covering for (fpqc) (resp. (fppf) , resp. (ét) , resp. (étf)).
- (v) Every finite surjective family consisting of flat quasi-compact morphisms is covering for (fpqc) . In particular, every faithfully flat quasi-compact morphism is covering for (fpqc) .

PROOF. (i) follows from 6.2.1, (ii) from 6.2.3, taking account of the fact that a sheaf for the Zariski topology transforms direct sums into products. Since every representable functor is a sheaf for (Zar) and satisfies condition (b) of (ii) by SGA 1, VIII 5.3, $\mathcal{T}_1$ is less fine than the canonical topology, which proves (iii).

Let us prove (iv). Let $\{S_i \rightarrow S\}$ be a family of morphisms as in the statement. Considering a covering of S by affine open sets, one is immediately reduced to the case where S is affine.⁽⁵⁹⁾ First let us treat the case where the morphisms $S_i \rightarrow S$ are flat and open (resp. étale). Let S_{ij} be a covering of S_i by affine open sets. Since the morphisms in question are open, the images T_{ij} of the S_{ij} form an open covering of S . Since S is affine, hence quasi-compact, it is covered by a finite number of open sets T_{ij} , with (i, j) ranging over a finite set F . Then $S' = \coprod_F S_{ij}$ is affine, and the morphism $S' \rightarrow S$ belongs to P'_1 , resp. P'_3 , hence is covering. Since it factors through the given family, the latter is covering.

In the case (étf) , each S_i is finite over S , hence affine; in the preceding argument, one may then take the covering $\{S_i\}$ of S , and one obtains that $S' \rightarrow S$ belongs to P'_4 .

Now consider the case where the morphisms $f_i : S_i \rightarrow S$ are flat and locally of finite presentation. For every $s \in S$, there exists, by (the proof of) EGA IV₄, 17.16.2, an affine subscheme $X(s)$ of some S_i , such that $s \in f_i(X(s))$ and the morphism $g_i : X(s) \rightarrow S$, the restriction of f_i , is flat, of finite presentation, and quasi-finite. Then $g_i(X(s))$ is an open neighborhood $U(s)$ of s (EGA IV₂, 2.4.6), and, since S is affine, it is covered by a finite number of such open sets $U(s_j)$, $j = 1, \dots, n$. Consequently $X' = \coprod_j X(s_j)$ is affine, and the morphism $X' \rightarrow S$ is surjective, flat, of finite presentation and quasi-finite, hence belongs to P'_2 .⁽⁶⁰⁾ This completes the proof of (iv).

Let us prove (v). Let $\{S_i \rightarrow S\}$ be a finite faithfully flat quasi-compact family. Let T_j be a covering of S by affine open sets. The $S_{ij} = T_j \times_S S_i$ are quasi-compact, and therefore possess finite affine open coverings T_{ijk} . Each morphism $T_{ijk} \rightarrow T_j$ is flat, and the family $\{T_{ijk} \rightarrow T_j\}$ is finite and surjective, hence covering for $\mathcal{T}_1$. The family $\{T_{ijk} \rightarrow S\}$ is therefore also covering, by composition. It factors through the given family, which is therefore also covering:

$$\begin{array}{ccccc} S_i & \longleftarrow & S_{ij} & \longleftarrow & T_{ijk} \\ \downarrow & & \downarrow & \swarrow & \\ S & \longleftarrow & T_j & & \end{array}$$

□

⁽⁵⁹⁾N.D.E. The rest has been simplified, taking advantage of the fact that one now assumes S affine.

⁽⁶⁰⁾N.D.E. This shows that, if one denotes by P'_2 the set of finite surjective families of flat morphisms of $\mathcal{C}'$ of finite presentation, the topology generated by P, P'_2 equals $\mathcal{T}_2$. On the other hand, with the notation at the beginning of the proof of (iv), if one takes a covering of S_i by affine open sets of finite presentation over S , one obtains that $S' \rightarrow S$ belongs to P'_2 .

Corollary 6.3.2. *Let (M_i) be the following family of morphisms:*

- (M_1) : *faithfully flat quasi-compact morphisms.*
- (M_2) : *faithfully flat morphisms locally of finite presentation.*
- (M_3) : *surjective étale morphisms.*
- (M_4) : *surjective finite étale morphisms.* ⁽⁶¹⁾ *The family (M_i) satisfies axioms (a), (b), (c), $(d_{\mathcal{T}_i})$, and $(e_{\mathcal{T}_i})$ of 4.6.3.*

PROOF. Indeed, for (a), (b), (c), this is classical (EGA and SGA 1, passim).⁽⁶²⁾ By 6.3.1 (iv) and (v), (M_i) satisfies $(d_{\mathcal{T}_i})$. It remains to see that (M_i) satisfies $(e_{\mathcal{T}_i})$; for this it suffices to see that (M_i) satisfies $(e_{\mathcal{T}_1})$, which implies the others. This follows from SGA 1, VIII (nos. 4 and 5). $\square$

Corollary 6.3.3. *If X is a scheme and R an equivalence relation in X of type (M_i) , R is (M_i) -effective if and only if the quotient sheaf of X by R for $\mathcal{T}_i$ is representable, and in this case it is represented by the quotient X/R .*

PROOF. Indeed, this is 4.6.5. $\square$

6.4. Effectiveness conditions

We now seek families (N) of morphisms satisfying axiom $(f_{\mathcal{T}})$ of 4.7. First remark that $(f_{\mathcal{T}_1})$ implies $(f_{\mathcal{T}_i})$, so that we may restrict ourselves to the case of the topology $(fpqc)$.

Lemma 6.4.1. *The following families of morphisms satisfy axiom $(f_{\mathcal{T}_1})$ of 4.7, that is, “descend under $(fpqc)$ ”:*

- (N) : *open immersions.*
- (N') : *closed immersions.*
- (N'') : *quasi-compact immersions.*

PROOF. By 6.3.1 (ii), it suffices to verify that the given families descend under the Zariski topology and under a faithfully flat quasi-compact morphism. The first assertion is clear; let us verify the second. For (N) , this is SGA 1, VIII 4.4; for (N') , it is *loc. cit.*, 1.9. For (N'') , one argues as in *loc. cit.*, 5.5, with the aid of the two preceding results. $\square$

Corollary 6.4.2. *The same result is valid for quasi-compact open immersions.*

These results allow one to apply to the present situation the general results of 4.7.1, 4.7.2, 5.1.8, 5.3.1, etc. Let us state one as an example, the first.

Corollary 6.4.3. $(= 4.7.1 + 4.6.10)$. *Let X be a scheme and R an equivalence relation in X . Suppose that $R \rightarrow X$ is faithfully flat and quasi-compact, and that $R \rightarrow X \times X$ is a closed immersion (resp. open, resp. quasi-compact, resp. quasi-compact open). Then the quotient sheaf X/R is the same for the topology $(fpqc)$ and for the canonical topology, and for each scheme S , one has*

$$(X/R)(S) = \left\{ \begin{array}{l} \text{closed (resp. open, resp. retrocompact,} \\ \text{resp. retrocompact open) subschemes } Z \text{ of } X_S \text{ stable under } R \times S, \\ \text{such that } Z \rightarrow S \text{ is faithfully flat quasi-compact} \\ \text{and the diagram } R_Z \rightrightarrows Z \rightarrow S \text{ is exact} \end{array} \right\}. \quad (63)$$

⁽⁶¹⁾**N.D.E.** The original has been corrected by adding for (M_3) , (M_4) the surjectivity hypothesis, which is automatically satisfied in the other cases.

⁽⁶²⁾**N.D.E.** Cf. EGA I, 6.6.4 for “quasi-compact”, EGA II, 6.1.5 for “finite”, EGA IV₁, 1.6.2 for “locally of finite presentation”, EGA IV₂, 2.2.13 for “faithfully flat”, and EGA IV₄, 17.3.3 for “étale”.

⁽⁶³⁾**N.D.E.** Recall that a subscheme Z of a scheme T is said to be retrocompact if the immersion $Z \hookrightarrow T$ is quasi-compact, cf. EGA 0_{III}, 9.1.1.

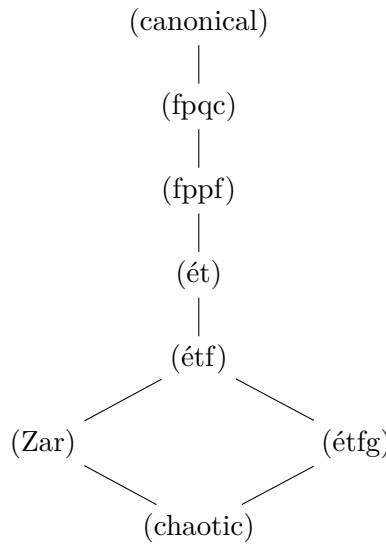
6.5. Principal homogeneous bundles

Let us simply point out the terminology:

Topology	Principal homogeneous bundles
(fpqc)	(without further qualification)
(ét)	quasi-isotrivial
(étf)	locally isotrivial
(Zar)	locally trivial.

6.6. Other topologies

One sometimes uses other topologies on the category of schemes. Let us point out one: the *global finite étale topology* (étfg), generated by the pretopology whose covering families are surjective families consisting of finite étale morphisms. It is not finer than the Zariski topology. The corresponding principal homogeneous bundles are called “isotrivial”.



(64)

6.7. Homogeneous spaces

⁽⁶⁵⁾ Let G be an S -group scheme, X an S -scheme with a group of operators G (on the left), and

$$\Phi : G \times_S X \longrightarrow X \times_S X$$

the morphism of S -schemes defined set-theoretically by $(g, x) \mapsto (gx, x)$. Recall (cf. 5.1.0 and III.0.1) that X is said to be a formally principal homogeneous space under G if the following equivalent conditions are satisfied:

- (i) For every $T \rightarrow S$, the set $X(T)$ is empty or principal homogeneous under $G(T)$.
- (ii) Φ is an isomorphism of S -functors.
- (iii) Φ is an isomorphism of S -schemes.

(The equivalence (i) $\Leftrightarrow$ (ii) is clear, and one has (ii) $\Leftrightarrow$ (iii), since $\mathcal{C} = (\text{Sch})_S$ is a full subcategory of $\widehat{\mathcal{C}}$.)

The definition of a formally homogeneous space (not necessarily principal homogeneous) is obtained by requiring that Φ be an epimorphism in the category of sheaves for an appropriate topology $\mathcal{T}$. Indeed, the condition that Φ be an epimorphism of S -functors is equivalent

⁽⁶⁴⁾**N.D.E.** Recall (cf. 4.4.2) that the chaotic topology is the least fine topology, defined by $J(S) = \{S\}$ for every $S \in \text{Ob } \mathcal{C}$.

⁽⁶⁵⁾**N.D.E.** The following numbers have been added.

to saying that for every $T \rightarrow S$, the set $X(T)$ is empty or homogeneous (not necessarily principal homogeneous) under $G(T)$, but this condition is too restrictive, as the following simple example shows. Let $S = \text{Spec } \mathbf{R}$, $G = \mathbf{G}_{m, \mathbf{R}}$, and $X = \mathbf{G}_{m, \mathbf{R}}$, on which G acts by $t \cdot x = t^2 x$. Then the morphism Φ is étale, finite, and surjective, hence an epimorphism in the category of sheaves for the topology (étf) (a fortiori, an epimorphism of S -schemes); on the other hand, the points $1, -1$ of $X(\mathbf{R})$ are not conjugate by an element of $G(\mathbf{R})$, so that the morphism $G(\mathbf{R}) \times X(\mathbf{R}) \rightarrow X(\mathbf{R}) \times X(\mathbf{R})$ is not surjective. ⁽⁶⁶⁾ One is therefore led to give the following definition:

Definition 6.7.1. Let G be an S -group, X an S -scheme with a group of operators G , and $\mathcal{T}$ a topology on $(\text{Sch})_S$ less fine than the canonical topology. One says that X is a *formally homogeneous space* under G (relative to the topology $\mathcal{T}$) if the following equivalent conditions are satisfied:

- (i) The morphism $\Phi : G \times_S X \rightarrow X \times_S X$ is an epimorphism in the category of sheaves for the topology $\mathcal{T}$.
- (ii) For every $T \rightarrow S$ and $x, y \in X(T)$, there exists a morphism $T' \rightarrow T$ covering for the topology $\mathcal{T}$, and $g \in G(T')$, such that $y_{T'} = g \cdot x_{T'}$.

Remark 6.7.2. Condition (i) implies in particular that Φ is a universal effective epimorphism in $(\text{Sch})_S$ (cf. 4.4.3). This implies, as one easily sees, that Φ is surjective (cf. 1.3, N.D.E. (3)).

Proposition and definition 6.7.3. ⁽⁶⁷⁾ Let G be an S -group, X an S -scheme with a group of operators G , and $\mathcal{T}$ a topology on $(\text{Sch})_S$ less fine than the canonical topology. The following conditions are equivalent:

- (i) X satisfies the following two hypotheses:
 - (1) The morphism $\Phi : G \times_S X \rightarrow X \times_S X$ is covering, i.e. X is a formally homogeneous G -space.
 - (2) The morphism $X \rightarrow S$ is also covering, that is, locally for the topology $\mathcal{T}$, it possesses a section (cf. RefIV.4.4.8bis.bis).
- (ii) “Locally on S for the topology $\mathcal{T}$ ”, X is isomorphic, as a scheme with a group of operators G , to the quotient sheaf (for $\mathcal{T}$) of G by a subgroup scheme H , that is, there exists a covering family $\{S_i \rightarrow S\}$ such that each $X \times_S S_i$ represents the quotient sheaf of $G \times_S S_i$ by some subgroup scheme H_i .

Under these conditions, one says that X is a *homogeneous G -space* (relative to the topology $\mathcal{T}$).

PROOF. Suppose that (ii) holds. Set $G_i = G \times_S S_i$, $X_i = X \times_S S_i$. Then X_i possesses a section over S_i , namely the composite of the unit section $\varepsilon_i : S_i \rightarrow G_i$ and the projection $\pi_i : G_i \rightarrow X_i = G_i/H_i$. Thus $X \rightarrow S$ is covering.

⁽⁶⁶⁾N.D.E. Evidently, this difficulty comes from the fact that if $\mathcal{C}'$ is a full subcategory of $\widehat{\mathcal{C}}$ containing $\mathcal{C}$, for example the category $\widetilde{\mathcal{C}}_{\mathcal{T}}$ of sheaves on $\mathcal{C}$ for a topology $\mathcal{T}$ less fine than the canonical topology, and if $f : X \rightarrow Y$ is a morphism in $\mathcal{C}$, then the implications:

$$\begin{aligned} f \text{ epimorphism of } \widehat{\mathcal{C}} &\Rightarrow f \text{ epimorphism of } \mathcal{C}' \\ &\Rightarrow f \text{ epimorphism of } \mathcal{C} \end{aligned}$$

are in general strict.

⁽⁶⁷⁾N.D.E. Cf. [Ray70], Def. VI.1.1.

On the other hand, π_i is covering, hence $\pi_i \times \pi_i$ is too (cf. 4.2.3 (C 1) and (C 2)), and one has a commutative diagram:

$$\begin{array}{ccc} G_i \times_{S_i} X_i & \xrightarrow{\Phi_i} & G_i \times_{S_i} X_i \\ \text{id} \times \pi_i \uparrow & & \uparrow \pi_i \times \pi_i \\ G_i \times_{S_i} G_i & \xrightarrow[\sim]{\Psi_i} & G_i \times_{S_i} G_i \end{array}$$

where Φ_i is deduced from Φ by the base change $S_i \rightarrow S$, and Ψ_i is the isomorphism defined set-theoretically by $(g, g') \mapsto (gg', g)$. Then $(\pi_i \times \pi_i) \circ \Psi_i$ is covering, hence Φ_i is too (4.2.3 (C 3)). This shows that Φ is “locally covering”, hence covering (4.2.3 (C 5)). This proves (ii) $\Rightarrow$ (i).

Conversely, suppose that (i) holds, and suppose moreover that the structural morphism $X \rightarrow S$ possesses a section σ . By EGA I, 5.3.13, σ is an immersion. Define $H = G \times_X S$ by the diagram below, both of whose squares are cartesian:

$$\begin{array}{ccccc} H & \longrightarrow & G & \xrightarrow{\text{id}_G \boxtimes \sigma} & G \times_S X \\ \downarrow & & \downarrow \pi & & \downarrow \Phi \\ S & \xrightarrow{\sigma} & X & \xrightarrow[\text{id}_X \boxtimes \sigma]{} & X \times_S X \end{array}$$

where $\pi, \text{id}_G \boxtimes \sigma$ and $\text{id}_X \boxtimes \sigma$ denote the morphisms defined set-theoretically, for $T \rightarrow S$ and $g \in G(T)$, $x \in X(T)$, by:

$$\begin{aligned} \pi(g) &= g \cdot \sigma_T, & (\text{id}_G \boxtimes \sigma)(g) &= (g, \sigma_T), \\ (\text{id}_X \boxtimes \sigma)(x) &= (x, \sigma_T). \end{aligned}$$

Then π is covering, and H is a subgroup scheme of G , representing the stabilizer $\underline{\text{Stab}}_G(\sigma)$ of σ (cf. I, 2.3.3), that is, for every $T \rightarrow S$, one has:

$$H(T) = \{g \in G(T) \mid g \cdot \sigma_T = \sigma_T\}.$$

Denote by G/H the *presheaf* $T \mapsto G(T)/H(T)$, and by $a(G/H)$ the associated sheaf for the topology $\mathcal{T}$. By the preceding argument, one obtains a commutative diagram of morphisms of presheaves with a group of operators G :

$$\begin{array}{ccc} G & \xrightarrow{\pi} & X \\ \downarrow & \nearrow \bar{\pi} & \\ G/H & & \end{array}$$

where $\bar{\pi}$ is a monomorphism. Since π is covering, $\bar{\pi}$ is too, and hence, by 4.3.12, $\bar{\pi}$ induces an isomorphism $a(G/H) \xrightarrow{\sim} X$. One has therefore proved that: *if X is a homogeneous G -space such that $X \rightarrow S$ admits a section σ , then X represents the quotient sheaf G/H , where $H = G \times_X S$ is the stabilizer of σ .*

In the general case, there exists by hypothesis a covering family $\{S_i \rightarrow S\}$ such that each morphism $X_i = X \times_S S_i \rightarrow S_i$ possesses a section σ_i . Set $G_i = G \times_S S_i$; then the morphism $\Phi_i : G_i \times_{S_i} X_i \rightarrow X_i \times_{S_i} X_i$ deduced from Φ by the base change $S_i \rightarrow S$ is still covering. Thus, by the preceding argument, $X_i \simeq G_i/H_i$, where H_i is the stabilizer in G_i of σ_i . This completes the proof of the implication (i) $\Rightarrow$ (ii). $\square$

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⁽⁶⁸⁾ **N.D.E.** This reference has been added.